Nonuniqueness of percolation on nonamenable transitive graphs

An account of the proof in the transitive case

Notes for a reading course*

Abstract

We present, for transitive graphs, a proof that Bernoulli bond percolation on every connected, locally finite, simple, nonamenable transitive graph has a phase with infinitely many infinite clusters, that is, $p_c < p_u$. The argument is the one of [Ope26], specialised and reorganised. Its core is the statement that the two-point connection operator is bounded on $\ell^2(V)$ at the critical parameter. Together with results of Hutchcroft this gives $p_c < p_{2\to 2} \leq p_u$, where $p_{2\to 2}$ is the largest parameter up to which the connection operator is bounded, a nonempty interval of simultaneous nonuniqueness, the triangle condition and mean-field critical behaviour. The proof of the operator bound compares two ways of measuring the connection kernel: its row sums and its norm on $\ell^2(V)$. A deformation that penalises connections through isolated open paths reduces the row sums at a quadratic rate while barely changing the norm, because the penalised paths are loop-erased random-walk paths and the random walk contracts $\ell^2(V)$. Making this quantitative needs near-mean-field a priori bounds on the susceptibility and the triangle diagram below p_c. These are derived from finiteness of all moments of order less than $1/2$ of the critical cluster size, which in turn is proved by contradiction with a construction that splices independent critical clusters. We decompose the argument into four steps, state each step precisely, and show how the steps combine before proving them.

*These notes rewrite, for transitive graphs, the proof given in the manuscript [Ope26], where the quasi-transitive case is treated.

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1 Introduction

In Bernoulli bond percolation with parameter $p \in [0,1]$ on an infinite connected graph, every edge is declared open with probability p, independently of all other edges, and one studies the connected components of the open subgraph, the *clusters*. Two parameters describe the large-scale picture. Infinite clusters exist for p above the *critical parameter* p_c, and there is exactly one infinite cluster for p above the *uniqueness threshold* p_u. On the hypercubic lattice $\mathbb{Z}^d$ there is never more than one infinite cluster [AKN87, BK89], and the argument of Burton and Keane extends this to every amenable transitive graph (see [LP16, Chapter 7]); there the two thresholds coincide. On the 3-regular tree, in contrast, $p_c = 1/2$ and $p_u = 1$: for every $p \in (1/2, 1)$ there are infinitely many infinite clusters, kept apart by the branching geometry of the tree. Benjamini and Schramm conjectured that the thresholds are separated on every quasi-transitive graph that shares the decisive feature of the tree, namely that the boundary of every finite vertex set is comparable to its size [BS96, Conjecture 6]. Such graphs are called nonamenable.

These notes present a proof of this conjecture for transitive graphs, following the manuscript [Ope26]. In informal terms, the main result reads as follows; the precise statement is Theorem 2.1.

> *Let G be a connected, locally finite, simple, nonamenable transitive graph. Then the matrix of connection probabilities at the critical parameter defines a bounded operator on $\ell^2(V)$. Consequently $p_c < p_{2\to 2} \le p_u$, so there is a nonempty interval of parameters with infinitely many infinite clusters, and percolation on G has mean-field critical behaviour.*

Here $p_{2\to 2}$ is the largest parameter up to which the connection probabilities define a bounded operator on $\ell^2(V)$. This threshold was introduced by Hutchcroft [Hut19, Hut20a], who conjectured that $p_c < p_{2\to 2}$ on every nonamenable quasi-transitive graph [Hut20a, Conjecture 1.3]. The inequality $p_{2\to 2} \le p_u$ is elementary: a unique infinite cluster forces all connection probabilities to be bounded away from zero, which is incompatible with boundedness on $\ell^2(V)$ (Theorem 4.10). The strict inequality $p_c < p_{2\to 2}$ is the substance of the result. It follows from boundedness at p_c itself, because a bounded connection operator remains bounded after a small increase of the parameter (Theorem 4.8).

Earlier results. Several classes of graphs were known to satisfy $p_c < p_u$. Benjamini and Schramm proved it for one-ended nonamenable transitive planar graphs [BS01], following Lalley's work on Fuchsian groups [Lal98]. Schonmann treated highly nonamenable graphs [Sch01], and Nachmias and Peres nonamenable Cayley graphs of large girth [NP12]. Gaboriau proved $p_c < p_u$ for unimodular transitive graphs carrying nonconstant harmonic Dirichlet functions [Gab05]. For Cayley graphs of a given nonamenable group, Pak and Smirnova-Nagnibeda [PSN00] and Thom [Tho15] showed that *some* generating set has $p_c < p_u$; the result presented here applies to *every* finite generating set (Theorem 2.2). In the operator framework, Hutchcroft proved $p_c < p_{2\to 2}$ for nonamenable Gromov hyperbolic graphs and for graphs with a quasi-transitive nonunimodular group of automorphisms [Hut19], building on [Hut20d]. Choi and Seo extended this to Cayley graphs of acylindrically hyperbolic groups [CS25]. The upper inequality $p_{2\to 2} \le p_u$ can be an equality [HP24].

Critical percolation on a nonamenable transitive graph has no infinite cluster: this was proved by Benjamini, Lyons, Peres and Schramm in the unimodular case [BLPS99a] and by Hutchcroft for all quasi-transitive graphs of exponential growth [Hut16]. We use it throughout, since it makes critical clusters finite. The behaviour of infinite clusters for $p_c < p < p_u$, including the simultaneous statements across parameters, was developed by Häggström, Peres and Schonmann [HP99, Sch99, HPS99].

Methods. The proof uses the classical toolbox of percolation: the Harris and van den Berg–Kesten inequalities [Har60, vdBK85], the tree-graph bounds of Aizenman and Newman [AN84], sharpness of the phase transition [AB87, Men86, DCT16], Russo's formula, and the mass-transport principle [BLPS99b, AL07]. Two further tools enter in an essential way. The first is the decision-tree covariance inequality of O'Donnell, Saks, Schramm and Servedio [OSSS05], used as in Hutchcroft's derivation of near-critical bounds from critical tails [Hut20c]. The second is a concentration estimate for exploration scores in the spirit of Hutchcroft's two-ghost inequalities [Hut20b, Hut21]. The genuinely new ingredients of [Ope26] are a construction that splices independent critical clusters together (Section 6), whose combinatorics is related to the finite modifications in Kozma's work on products of trees [Koz11], and a deformation of the connection kernel that penalises isolated open paths (Section 9).

About these notes. The manuscript [Ope26] treats quasi-transitive graphs, possibly with loops and multiple edges. We restrict to transitive simple graphs, which removes the averaging over vertex types, the reduction to simple graphs by subdivision and several case distinctions, and lets the main line of the argument stand out. The mathematics is that of [Ope26]; we have re-derived every step while rewriting it, specialised the statements to the transitive case, and reorganised the material around four steps whose precise statements are collected in Section 3. Where a proof deviates from [Ope26] beyond this specialisation, we say so. Classical results are cited rather than reproved; proofs are given for the less standard variants we need, some of them in Section A.

Outline. Section 2 fixes the setting and states the main results: Theorem 2.1, its consequence for Cayley graphs, and the mean-field consequences in Theorem 2.3. Section 3 is an overview of the whole proof. It reduces Theorem 2.1 to the boundedness of the critical connection operator (Theorem 3.1), explains the mechanism, states the four steps precisely, and proves Theorem 3.1 from them. Section 4 collects the tools. Sections 5 to 8 prove the a priori estimates on critical and near-critical clusters (Steps 1 and 2), and Sections 9 and 10 the properties of the deformation (Steps 3 and 4). Section 11 derives Theorem 2.3.

2 Setting and main results

2.1 Graphs and standing assumptions

Let $G = (V, E)$ be a graph with vertex set V and edge set E. We write $x \sim y$ if $\{x, y\} \in E$, dist for the graph distance, $B(x, r)$ for the closed ball of radius r around x, and $\mathrm{Aut}(G)$ for the group of graph automorphisms. For $A \subseteq V$ let $\partial_E A$ be the set of edges with exactly one endpoint in A, and let $\mathrm{N}[A]$ be the closed neighbourhood of A, consisting of A and all vertices adjacent to a vertex of A. The graph is *transitive* if $\mathrm{Aut}(G)$ acts transitively on V, and *nonamenable* if

$$\iota := \inf\Big\{ \frac{|\partial_E A|}{|A|} \colon A \subseteq V \text{ finite},\ A \neq \emptyset \Big\} > 0\,.$$

For graphs of bounded degree this is equivalent to positivity of the vertex-isoperimetric constant, because every vertex of the outer vertex boundary of A is incident to at least one and at most boundedly many edges of $\partial_E A$.

Throughout these notes G satisfies the following standing assumption.

(A1) $G = (V, E)$ is an infinite, connected, locally finite, simple graph, $\mathrm{Aut}(G)$ acts transitively on V, and G is nonamenable.

A transitive graph is regular; we write d for its degree and fix a root vertex $o \in V$. From Section 3 on, the main argument also uses the second assumption

(A2) the group $\mathrm{Aut}(G)$ is unimodular,

that is, $|\mathrm{Stab}(x)y| = |\mathrm{Stab}(y)x|$ for all $x, y \in V$, where $\mathrm{Stab}(x)y$ is the orbit of y under the stabiliser of x in $\mathrm{Aut}(G)$. Equivalently, the mass-transport principle holds (Section 4.2); every Cayley graph satisfies (A2). The nonunimodular case is covered by a theorem of Hutchcroft (Theorem 4.11).

2.2 Percolation, connection kernel and thresholds

Let $\mathbb{P}_p$ be the law of Bernoulli bond percolation with parameter $p \in [0,1]$ on G, that is, the product measure on $\{0,1\}^E$ under which each edge is *open* ($\omega(e) = 1$) with probability p, and let $\mathbb{E}_p$ be its expectation. We write $x \leftrightarrow y$ if x and y are joined by an open path; paths of length zero are allowed, so $x \leftrightarrow x$. The *cluster* C_x of x is the set of vertices y with $x \leftrightarrow y$, and $|C_x|$ is its number of vertices. For a vertex set $\Lambda \subseteq V$, a *domain*, we write $x \overset{\Lambda}{\longleftrightarrow} y$ if x and y are joined by an open path using only vertices of Λ, and C_x^Λ for the cluster of x in the subgraph induced by Λ. By convention $C_x^\Lambda = \emptyset$ if $x \notin \Lambda$, and every quantity that requires a terminal outside its domain is zero. Let

$$\theta(p) \coloneqq \mathbb{P}_p(|C_o| = \infty)\,, \qquad p_c \coloneqq \inf\{p \colon \theta(p) > 0\}\,, \qquad p_u \coloneqq \inf\{p \colon \mathbb{P}_p(N_\infty = 1) = 1\}\,,$$

where p ranges over $[0,1]$ and N_∞ is the number of infinite clusters. In the *standard monotone coupling* the edges carry independent labels U_e uniform on $[0,1]$, and an edge is open at parameter p if $U_e \le p$; this realises all $\mathbb{P}_p$ on one probability space.

The connection kernel. A *kernel* is a function $K \colon V \times V \to \mathbb{R}$. It is *invariant* if $K(gx, gy) = K(x,y)$ for all $g \in \mathrm{Aut}(G)$ and $x, y \in V$. We let a kernel act on finitely supported functions by $(Kf)(x) \coloneqq \sum_y K(x,y) f(y)$ and write $\|K\| \in [0,\infty]$ for its operator norm on $\ell^2(V)$, the Hilbert space of square-summable functions on V with counting measure; K is *bounded* if $\|K\| < \infty$. Products of nonnegative kernels, $(KL)(x,y) \coloneqq \sum_z K(x,z) L(z,y) \in [0,\infty]$, are always defined. We write Adj for the adjacency kernel, $\mathrm{Adj}(x,y) \coloneqq \mathbb{1}\{x \sim y\}$, and I for the identity. The *connection kernel* and the *operator threshold* are

$$T_p(x,y) \coloneqq \mathbb{P}_p(x \leftrightarrow y)\,, \qquad p_{2\to 2} \coloneqq \sup\{p \in [0,1] \colon \|T_p\| < \infty\}\,.$$

More generally $T_p^\Lambda(x,y) \coloneqq \mathbb{P}_p(x \overset{\Lambda}{\longleftrightarrow} y)$. The *susceptibility* and the *triangle diagram* are

$$\chi(p) \coloneqq \mathbb{E}_p|C_o| = \sum_{y \in V} T_p(o,y)\,, \qquad \nabla_p \coloneqq \sum_{y,z \in V} T_p(o,y)\, T_p(y,z)\, T_p(z,o)\,,$$

both with values in $[1,\infty]$. By transitivity they do not depend on the choice of o.

Constants. Unless indicated otherwise, $c, C \in (0,\infty)$ denote constants that depend only on G; their values may change from line to line. Dependence on further parameters is shown by subscripts, as in C_η.

2.3 Main results

Theorem 2.1 (Critical boundedness and nonuniqueness)**.** *Assume (A1). Then*

$$\sup_{0 \le p < p_c} \|T_p\| = \|T_{p_c}\| < \infty \qquad \text{and} \qquad p_c < p_{2\to 2} \le p_u\,. \tag{2.1}$$

Moreover there are $p_c < p_1 < p_2 < 1$ such that, in the standard monotone coupling, almost surely for every $p \in [p_1, p_2]$ the open subgraph at parameter p has infinitely many infinite clusters.

The strict inequality $p_c < p_u$ is the bond-percolation case of the conjecture of Benjamini and Schramm for transitive graphs.

Corollary 2.2 (Cayley graphs)**.** *Let Γ be a finitely generated nonamenable group and $S \subseteq \Gamma$ a finite symmetric generating set not containing the identity. Then the Cayley graph* $\mathrm{Cay}(\Gamma, S)$*, with vertex set Γ and edges $\{x, xs\}$ for $x \in \Gamma$ and $s \in S$, satisfies the conclusions of Theorems 2.1 and 2.3.*

Proof. The Cayley graph is simple, because the identity is not in S and edges are unordered pairs. It is connected because S generates Γ, and locally finite because S is finite. Left multiplications are automorphisms, so it is transitive. By Følner's criterion a finitely generated group is nonamenable if and only if one, equivalently every, of its Cayley graphs with respect to a finite generating set is nonamenable (see [LP16, Chapter 6]). Hence (A1) holds, and Theorems 2.1 and 2.3 apply. □

The operator bound also determines the critical behaviour. For a vertex v, the *extrinsic* and *intrinsic radius* of its cluster are

$$\mathrm{rad}_{\mathrm{ext}}(C_v) \coloneqq \sup_{x \in C_v} \mathrm{dist}(v, x)\,, \qquad \mathrm{rad}_{\mathrm{int}}(C_v) \coloneqq \sup_{x \in C_v} \mathrm{dist}_\omega(v, x)\,,$$

with values in $[0, \infty]$, where dist_ω is the graph distance in the open subgraph. We say that connection probabilities *decay exponentially* at p if there are $A < \infty$ and $a > 0$ with $T_p(x, y) \le A\mathrm{e}^{-a\,\mathrm{dist}(x,y)}$ for all $x, y \in V$, and we let p_{exp} be the supremum of all such p. We write $f \asymp g$ in a stated limit if $cg \le f \le Cg$ in a neighbourhood of that limit.

Theorem 2.3 (Mean-field behaviour and exponential connectivity)**.** *Assume (A1). Then*

$$\nabla_{p_c} \le \|T_{p_c}\|^3 < \infty\,, \tag{2.2}$$

and, as $p \uparrow p_c$, $p \downarrow p_c$ and $n \to \infty$ respectively,

$$\chi(p) \asymp (p_c - p)^{-1}\,, \qquad \theta(p) \asymp p - p_c\,, \tag{2.3}$$

$$\mathbb{P}_{p_c}(|C_o| \ge n) \asymp n^{-1/2}\,, \qquad \mathbb{P}_{p_c}(\mathrm{rad}_{\mathrm{int}}(C_o) \ge n) \asymp \mathbb{P}_{p_c}(\mathrm{rad}_{\mathrm{ext}}(C_o) \ge n) \asymp n^{-1}\,. \tag{2.4}$$

Moreover $p_c < p_{2\to 2} \le p_{\mathrm{exp}}$, and for every $p \in [0, p_{2\to 2})$ there are $A_p < \infty$ and $a_p > 0$ with

$$T_p(x, y) \le A_p\, \mathrm{e}^{-a_p\,\mathrm{dist}(x,y)} \qquad \text{for all } x, y \in V\,. \tag{2.5}$$

The bound (2.2) is the *triangle condition* of Aizenman and Newman [AN84]. Together with $p_c < p_{\mathrm{exp}}$ it verifies Conjectures 1.2 and 2.1 of [Hut20a] for transitive graphs. The asymptotics (2.3) and (2.4) are the mean-field values of the critical exponents. They follow from known conditional results, whose hypothesis is either the triangle condition or $p_c < p_{2\to 2}$; the new input is that these hypotheses hold for every graph satisfying (A1). The deduction is given in Section 11.

Remark 2.4 (Fixed parameters). For each fixed $p \in (p_c, p_u)$ there are almost surely infinitely many infinite clusters. Indeed, on a transitive graph N_∞ is almost surely constant and belongs to $\{0, 1, \infty\}$ [NS81] (see also [LP16, Chapter 7]); it is positive because $p > p_c$, and it is not one because $\mathbb{P}_p(N_\infty = 1) < 1$ for $p < p_u$ by the definition of p_u. Häggström, Peres and Schonmann proved the stronger statement that this holds simultaneously for all $p \in (p_c, p_u)$ in the standard coupling [HPS99]. The compact interval in Theorem 2.1 is obtained directly from the boundedness of a single connection operator (Theorem 4.12).

Remark 2.5 (Scope). The manuscript [Ope26] treats quasi-transitive graphs, in which $\mathrm{Aut}(G)$ has finitely many orbits, and also allows loops and multiple edges. A transitive graph with multiple edges does not reduce to a transitive simple graph: subdividing each edge, as in [Ope26], produces a simple graph with two vertex orbits. Such graphs are therefore not covered by these notes.

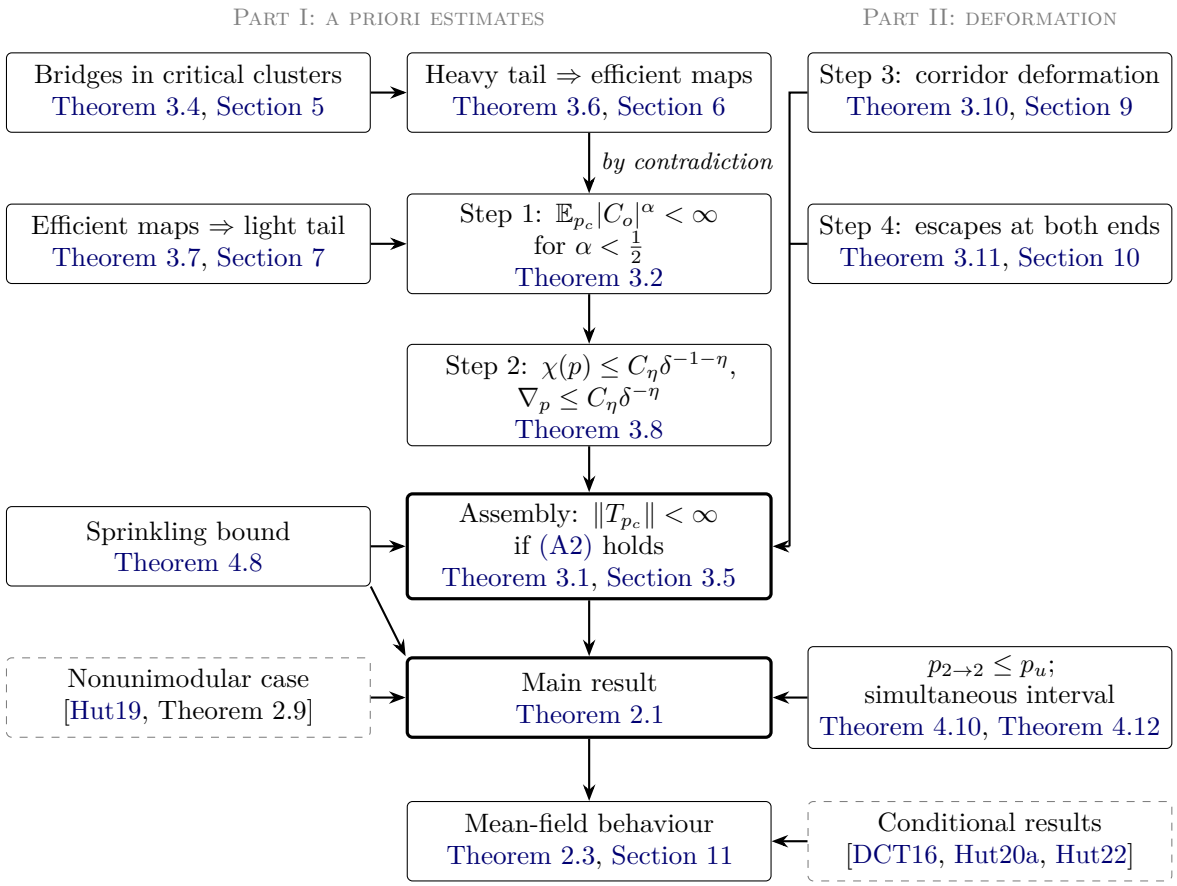


Figure 1: Logical structure of the proof. An arrow points from a statement to a statement whose proof uses it. Dashed boxes are results from the literature. Step 1 is proved by contradiction: if the critical cluster size had an infinite moment of order $\alpha < 1/2$, the amplification would produce efficient endpoint maps, which force all such moments to be finite. The assembly (Section 3.5) combines the four steps with the sprinkling bound.

3 Strategy of the proof

The proof of Theorem 2.1 has two layers. The first is a short reduction to the boundedness of the critical connection operator for graphs with a unimodular automorphism group (Section 3.1). The second proves this boundedness; it occupies Sections 4 to 10. We first explain the mechanism (Section 3.2). We then state the four steps of the proof precisely: two a priori estimates on critical and near-critical clusters (Steps 1 and 2, Section 3.3), and two properties of a deformation of the connection kernel (Steps 3 and 4, Section 3.4). Finally we show how the four steps combine (Section 3.5). Figure 1 summarises the logical structure.

3.1 Reduction to critical boundedness

The core of the argument is the following theorem.

Theorem 3.1 (Critical boundedness)**.** *Assume* (A1) *and* (A2)*. Then* $\|T_{p_c}\| < \infty$*.*

Everything else in Theorem 2.1 follows from Theorem 3.1 and the tools of Section 4.

Proof of Theorem 2.1 given Theorem 3.1. We first show that T_{p_c} is bounded and $p_c < p_{2\to 2}$. If $\mathrm{Aut}(G)$ is unimodular, $\|T_{p_c}\| < \infty$ by Theorem 3.1. Since $p_c < 1$ (Theorem 4.9), the sprinkling bound (Theorem 4.8) with a small $a > 0$ then shows that T_p is bounded at

$p = p_c + (1-p_c)a > p_c$, so $p_c < p_{2\to 2}$. If $\mathrm{Aut}(G)$ is not unimodular, then $p_c < p_{2\to 2}$ by a theorem of Hutchcroft (Theorem 4.11). Since $T_p \le T_{p'}$ entrywise for $p \le p'$, the kernel T_p is bounded for every $p < p_{2\to 2}$ (Theorem 4.1), in particular for $p = p_c$.

Theorem 4.10 gives $p_{2\to 2} \le p_u$, and Theorem 4.1 identifies $\|T_{p_c}\|$ with $\sup_{p<p_c}\|T_p\|$. This proves (2.1). Finally, choose $p_c < p_1 < p_2 < p_{2\to 2}$. Then T_{p_2} is bounded, and Theorem 4.12 yields the simultaneous statement on $[p_1, p_2]$. □

3.2 The mechanism: row sums versus norm

Row sums and norm. The connection kernel T_p is nonnegative, symmetric and invariant, and its row sums equal $\chi(p)$. By the Schur test (Theorem 4.1), $\|T_p\| \le \chi(p)$. Below p_c the susceptibility is finite by sharpness of the phase transition, so T_p is bounded. The row sums, however, explode as $p \uparrow p_c$: if χ stayed bounded, then $\chi(p_c) < \infty$ by monotone convergence, and the sprinkling bound (Theorem 4.8) would give $\chi(p) < \infty$ for some $p > p_c$, which is impossible because $\theta(p) > 0$. Theorem 3.1 therefore asserts that the norm of T_p stays bounded although its row sums do not. On a nonamenable graph this is possible because the ℓ^2 norm can be much smaller than the row sums. The basic example is the *lazy simple random walk*

$$P(x,y) := \frac{1}{2d}\,\mathbb{1}\{x \sim y\} + \frac{1}{2}\,\mathbb{1}\{x = y\}\,, \qquad \rho := \|P\| < 1\,,$$

whose row sums are one, while its norm is strictly smaller than one by nonamenability (Theorem 4.4). We use P and ρ throughout.

What unboundedness would mean. Suppose that $\|T_{p_c}\| = \infty$, and write $\delta := p_c - p$ for $p < p_c$. Adding an independent configuration of parameter a to one of parameter p and splitting open paths at the added edges gives the sprinkling bound (Theorem 4.8)

$$T_{p+(1-p)a} \le \sum_{j\ge 0} a^j\, T_p(\mathrm{Adj}\ T_p)^j \qquad \text{entrywise, and hence}$$
$$\|T_{p+(1-p)a}\| \le \frac{\|T_p\|}{1 - ad\|T_p\|} \qquad \text{if } ad\|T_p\| < 1\,.$$

With $a = \delta/(1-p)$, so that $p + (1-p)a = p_c$, this turns a small norm at p into boundedness at p_c. Hence

$$\|T_p\| \ge \frac{1-p}{d\,\delta} \qquad \text{for all } p < p_c\,. \tag{3.1}$$

Step 2 below shows that $\chi(p) \le C_\eta \delta^{-1-\eta}$ for every $\eta > 0$. Hence the ratio

$$R = R(p) := \frac{\chi(p)}{\|T_p\|} \in [1, \infty) \tag{3.2}$$

satisfies $R \le C_\eta \delta^{-\eta}$ for every $\eta > 0$: the norm of T_p would agree with its row sums up to a factor that is smaller than every power of $1/\delta$. We show that this is impossible by deforming T_p in a way that decreases its row sums by a large factor while decreasing its norm by at most a factor two. This contradicts $\|\cdot\| \le$ row sum.

Three structural facts. The comparison of norm and row sums rests on three properties of unimodular transitive graphs and of the connection kernel, which we use throughout. First, the *mass-transport principle* (Theorem 4.2): $\sum_y F(o,y) = \sum_x F(x,o)$ for every nonnegative function F on $V \times V$ that is invariant under the diagonal action of $\mathrm{Aut}(G)$; in particular every invariant kernel has equal row and column sums. Second, for $p < p_c$ the kernel T_p is *positive semidefinite* (Theorem 4.6). Third, the diagonal entry $\mathrm{tr}\, S := S(o,o)$ of a bounded invariant

operator is a *trace*: it is cyclic, and $|\mathrm{tr}(BA)| \le \|B\| \,\mathrm{tr}\, A$ for A positive semidefinite and B self-adjoint (Theorem 4.3). Two consequences show how the decay of the random walk on $\ell^2(V)$ enters. By the Cauchy–Schwarz inequality for T_p and the trace,

$$(P^m T_p)(a,b)^2 \le (P^m T_p P^m)(a,a)\, T_p(b,b) = \mathrm{tr}(P^{2m} T_p) \le \rho^{2m} \ :$$

a random walk of m steps smooths the connection kernel at rate ρ^m, however large $\|T_p\|$ is (Theorem 10.1). And sums of the form $\sum_v P^k(o,v)\, T_p^3(v,o) = \mathrm{tr}(P^k T_p^3)$ are at most $\rho^k \,\mathrm{tr}(T_p^3) = \rho^k \nabla_p$; this is how the triangle diagram enters the proof.

A deformation that separates the two. Fix an integer $k \ge 3$ and let w be a path of k steps of the lazy walk, with probability $\mu(w)$ under P. Erasing its loops produces a self-avoiding path γ_w from $u = w_0$ to $v = w_k$. On the *corridor event* Cor_w all edges of γ_w are open and all other edges at the interior vertices of γ_w are closed. Then γ_w is an isolated open path, a *corridor*. It joins the clusters of u and v outside its interior, and it is their only open link unless these two clusters coincide (Figure 2). The deformation multiplies the indicator of $\{x \leftrightarrow y\}$ by $\exp(-t\, c(w))$ for every corridor w all of whose edges are pivotal for $x \leftrightarrow y$, where $t \ge 0$ is the deformation time and the weight $c(w) = \mu(w)/\mathbb{P}_p(\mathrm{Cor}_w)$ is calibrated so that each corridor is penalised at rate equal to its random-walk probability. A penalised connection consists of a connection from x to u, a corridor, and a connection from v to y. Summed over corridors, the rate at which the deformed kernel T_t decreases is therefore at most $2\, T_p P^k T_p$, whose norm is at most $2\rho^k \|T_p\|^2$. The rate at which its row sum decreases is, roughly, the μ-average of the product of the cluster masses at u and v. Here no factor comparable to ρ^k appears, since the row sums of P^k equal one. Thus over a time interval of length $(kR)^C/\|T_p\|$ the norm changes by a negligible amount, while the row sum, if the masses at the ends of corridors are comparable to the full row sum, decreases at a rate proportional to its own square and must drop far below $\|T_p\|$.

The three costs. Turning the last sentence into a proof requires three corrections, and each has a price.

(i) *Distinct clusters.* The edges of a corridor are pivotal for $x \leftrightarrow y$ only if the clusters of u and v outside the corridor are distinct. By the van den Berg–Kesten inequality and the trace bound above, the resulting error is at most $C\chi(p)^2 \rho^k \nabla_p = C R^2 \nabla_p \rho^k \|T_p\|^2$ (Theorem 3.10(c)). It is negligible once k is a large multiple of $\log R + \log \nabla_p$.

(ii) *Removing the corridor.* The cluster masses at u and v are computed after deleting the interior of the corridor, which may cut off most of the mass. Step 4 shows that for a positive fraction of paths w, both endpoints are joined by short paths avoiding the corridor to vertices whose connections to the neighbourhood of the corridor are tiny. Its key input is the smoothing bound above. Transporting the mass along these short paths costs a factor of order $(kR)^{-C}$.

(iii) *Interaction of corridors.* Penalties of different corridors on the same connection interact, which costs a factor e^{-tc_k} with $c_k = \mathrm{e}^{C_0 k}$. This is harmless as long as $tc_k \ll 1$ up to the final time $t \approx (kR)^C/\|T_p\|$. By (3.1) this holds if $\mathrm{e}^{C_0 k}(kR)^C \ll 1/\delta$, in particular if k and $\log R$ are $o(\log(1/\delta))$.

The requirements (i) and (iii) can be met simultaneously if $\log R$ and $\log \nabla_p$ are $o(\log(1/\delta))$: then k can be chosen as a large multiple of $\log R + \log \nabla_p + \log\log(1/\delta)$, which is still $o(\log(1/\delta))$. A bound $R + \nabla_p \le C\delta^{-a}$ with a fixed $a > 0$ would also do, but only if a is smaller than a threshold determined by constants of the construction (such as C_0 and the constants of Step 4) that we do not control explicitly. This is why Step 2 proves the a priori bounds with an arbitrarily small power of $1/\delta$.

Where the a priori bounds come from. Mean-field theory predicts $\mathbb{P}_{p_c}(|C_o| \geq n) \asymp n^{-1/2}$; Theorem 2.3 confirms this a posteriori. Step 1 proves the weaker statement that $\mathbb{E}_{p_c}|C_o|^\alpha < \infty$ for every $\alpha < 1/2$. Step 2 converts it into the near-critical bounds. A decision-tree inequality transfers the critical tail to $\chi(p) \leq \delta^{-1-o(1)}$. A differential inequality for the triangle diagram gives $\nabla_p \leq \delta^{-o(1)}$; in it, the critical tail bounds the probability that two adjacent distinct clusters are both large, and a concentration estimate bounds the number of edges between them.

Critical moments by contradiction. Step 1 is proved by contradiction. If $\mathbb{E}_{p_c}|C_o|^\alpha = \infty$ for some $\alpha < 1/2$, then at infinitely many scales n the probability that $|C_o| \in [n, 2n)$ is at least $n^{-\beta}$ for some $\beta < 1/2$: there are more large critical clusters than mean-field theory allows. We exploit them by splicing independent small critical clusters onto a large one. This produces *local endpoint maps*, random rules that select a vertex of the cluster of the root, whose selection kernel has a very small norm on $\ell^2(V)$ compared with their cost, measured by the logarithms of the inverse success probability and of the cluster size. The multiplicity of the splicing is controlled by a bound on the number of bridges in pieces cut out of critical clusters. Running such an efficient map slightly below p_c shows that χ grows at least as fast as a solution of $\chi' = \chi^{2-\gamma}$ for every $\gamma > 0$. This gives $\chi(p) \leq \delta^{-1-o(1)}$ and hence a near-mean-field tail bound for the critical cluster size, which contradicts the assumption. The susceptibility bound obtained in this way is conditional on the heavy tail, which is used to construct the maps; Step 2 proves it unconditionally from Step 1.

3.3 Part I: a priori estimates (Steps 1 and 2)

Theorem 3.2 (Step 1: critical moments)**.** *Assume (A1) and (A2). Then $\mathbb{E}_{p_c}|C_o|^\alpha < \infty$ for every $\alpha \in (0, 1/2)$.*

The proof of Theorem 3.2 combines three propositions. The first concerns the geometry of critical clusters. A *bridge* of a finite connected graph H is an edge whose removal disconnects H. For vertices x, y of H let $d^H_{\mathrm{br}}(x, y)$ be the number of bridges of H that separate x from y; this is a pseudometric, the *bridge distance*, and $\mathrm{diam}_{\mathrm{br}}(H) \coloneqq \max_{x,y} d^H_{\mathrm{br}}(x, y)$ is the *bridge diameter*. Throughout, a vertex set B of an open cluster is regarded as a graph with the open edges joining its vertices.

Definition 3.3 (Good clusters)**.** Let $\varepsilon \in (0, 1/2)$ and $\sigma \coloneqq 1/2 + \varepsilon$. A finite open cluster C with $|C| \leq M$ is *M-good* if the following holds for every $A \subseteq C$ and every connected component B of the open subgraph induced by $C \setminus A$: if the set $\mathcal{K}$ of open edges between B and A is not empty, then $\mathrm{diam}_{\mathrm{br}}(B) \leq |\mathcal{K}| M^\sigma$.

Proposition 3.4 (Step 1a: bridges in critical clusters)**.** *Assume (A1) and let $\varepsilon \in (0, 1/2)$. For all sufficiently large M,*

$$\mathbb{P}_{p_c}\big(|C_o| \leq M \text{ and } C_o \text{ is not } M\text{-good}\big) \leq \exp(-M^\varepsilon)\,.$$

Moreover, the event $\{|C_o| \leq M,\ C_o \text{ is } M\text{-good}\}$ is determined by the states of the edges with an endpoint in $B(o, M)$.

In a critical branching process of total size M, every edge is a bridge and the height is of order $M^{1/2}$; heuristically, critical clusters on a nonamenable graph behave alike. Theorem 3.4 is a robust form of this heuristic: pieces cut out of a critical cluster have bridge diameter at most $M^{1/2+\varepsilon}$ per cut edge, outside an event of stretched-exponentially small probability. It is proved in Section 5 by an exploration-martingale bound on the number of pivotal edges of a marked cluster. The exponent σ enters the next step through the requirement $\beta + \sigma < 1$.

Definition 3.5 (Local endpoint maps)**.** A *local endpoint map* is a family $\Phi = (\Phi_x)_{x\in V}$ of functions $\Phi_x\colon \{0,1\}^E \times V \to [0,1]$ with $\sum_y \Phi_x(\omega, y) \le 1$, to be read as a randomised rule: given the configuration ω, the rule at x outputs y with probability $\Phi_x(\omega, y)$ and fails with the remaining probability. We require that there is $r_\Phi < \infty$ such that $\Phi_x(\omega, \cdot)$ depends only on the states of the edges with an endpoint in $B(x, r_\Phi)$, that $\Phi_x(\omega, y) = 0$ unless $y \in C_x(\omega)$, and that $\Phi_{gx}(g\omega, gy) = \Phi_x(\omega, y)$ for all $g \in \mathrm{Aut}(G)$. The map has *cap* $M \in \mathbb{N}$ if $\Phi_x(\omega, \cdot) \equiv 0$ unless $|C_x| \le M$ and C_x is M-good. Its *success probability*, *endpoint kernel* and *kernel norm* are

$$s := \mathbb{E}_{p_c} \sum_y \Phi_x(\omega, y)\,, \qquad K(x,y) := \frac{\mathbb{E}_{p_c}\Phi_x(\omega, y)}{s}\,, \qquad r := \|K\|\,;$$

by invariance s does not depend on x. If $s > 0$ and $r < 1$, the *efficiency ratio* of Φ is

$$\mathrm{eff}(\Phi) := \frac{\log(1/s) + \log M}{\log(1/r)}\,.$$

Thus $K(x,y)$ is the probability that the rule at x outputs y, given that it succeeds; caps refer to M-goodness for the parameter ε of Theorem 3.3, which is kept fixed. Under (A2) the endpoint kernel is doubly stochastic, so $r \le 1$ (Theorem 6.2). A small efficiency ratio means that the selected vertex is spread out very widely, as measured on $\ell^2(V)$, at a moderate price in success probability and cluster size. Steps of the lazy random walk along open edges give a first map with $r < \mathrm{e}^{-1}$ (Theorem 6.3); the point of the next proposition is to make the efficiency ratio small.

Proposition 3.6 (Step 1b: heavy tails yield efficient maps)**.** *Assume (A1) and (A2), and suppose that* $\mathbb{E}_{p_c}|C_o|^\alpha = \infty$ *for some* $\alpha \in (0, 1/2 - \varepsilon)$*. Then there are local endpoint maps with cap* $M \ge 2$ *and* $r < \mathrm{e}^{-1}$ *whose efficiency ratios are arbitrarily small.*

The proof in Section 6 improves a given map Φ in one *amplification step*, at a heavy scale n, that is, a scale with $\mathbb{P}_{p_c}(n \le |C_o| < 2n) \ge n^{-\beta}$ for some fixed $\beta < 1/2 - \varepsilon$. For a root x, let A be the cluster of x in a first configuration, and for a vertex z run ℓ steps of Φ from z, each in a fresh independent configuration. If $n \le |A| < 2n$ and the last step lands in A, we splice the pieces of the ℓ old clusters onto A by exchanging finitely many edge states between the configurations. Afterwards z lies in the cluster of x, which has fewer than $2n + \ell M$ vertices. The new map at x outputs z with probability proportional to the total weight of the splicings that produce the observed configuration and end at z. Its endpoint kernel is essentially the ℓ-th power of the old one composed with a stochastic kernel, so $\log(1/r)$ is multiplied by about ℓ. Since the old kernel is doubly stochastic, summing over z the probability that the chain from z succeeds and ends in A gives $s^\ell|A|$; this makes the new success probability of order $\mathbb{P}_{p_c}(n \le |C_o| < 2n) \ge n^{-\beta}$, up to a factor polynomial in ℓ and M. The cap is about $2n$. To control fluctuations, the splicing must be undone: once the positions of the ℓ seams among the bridges of the final cluster are known, everything else is determined, and Theorem 3.4 bounds the number of possible seam positions, weighted by the probabilities of choosing them, by about $M^{\sigma\ell}$. A second-moment estimate needs this number to be at most ns^ℓ, which is why the number of steps can be as large as about $\log n/(\log(1/s) + \sigma \log M)$. With this choice, each amplification step multiplies the ratio $(\log(1/s) + \sigma \log M)/\log(1/r)$ by a factor close to $\beta + \sigma < 1$.

Proposition 3.7 (Step 1c: efficient maps yield light tails)**.** *Assume (A1) and (A2), and suppose that there are local endpoint maps with cap* $M \ge 2$ *and* $r < \mathrm{e}^{-1}$ *whose efficiency ratios are arbitrarily small. Then* $\mathbb{E}_{p_c}|C_o|^\alpha < \infty$ *for every* $\alpha \in (0, 1/2)$*.*

The proof in Section 7 runs a fixed efficient map at parameter p slightly below p_c; its cap is used only through $|C_x| \le M$, not through goodness. Starting from the root of an

independent, size-biased cluster A, iterates of the endpoint kernel spread so quickly that after $\ell \approx \log \chi(p)/\log(1/r)$ steps the endpoint has left A, and the susceptibility in $V \setminus A$ at the endpoint is comparable to $\chi(p)$. On the other hand, a successful output lies within M open steps of its input, and moving along at most M open edges increases the susceptibility in $V \setminus A$ by a factor at most of order M (Theorem 7.1); after dividing by the success probability, one step of the map multiplies it by at most a factor $Z \le CM/s$, up to a contribution from the boundary of A. The boundary contribution is controlled by the quantity

$$\Delta(p) := \chi(p)^{-2} \sum_{b \sim o} \mathbb{E}_p[|C_o|\,|C_b|;\, C_o \neq C_b] = (1-p)\,\frac{\chi'(p)}{\chi(p)^2}\,,$$

where the identity holds for almost every p by Russo's formula. Comparing the two bounds gives $\Delta(p) \ge cZ^{-\ell}$, that is, $\chi' \ge c\chi^{2-\gamma}$ with $\gamma \approx \mathrm{eff}(\Phi)$, and so $\chi(p) \le C\delta^{-1/(1-\gamma)}$. A comparison of explorations at p and at p_c then bounds the critical tail: $\mathbb{P}_{p_c}(|C_o| \ge N) \le N^{-1/2+o(1)}$.

Proof of Theorem 3.2 given Theorems 3.6 and 3.7. Suppose that $\mathbb{E}_{p_c}|C_o|^{\alpha_0} = \infty$ for some $\alpha_0 \in (0, 1/2)$, and choose the parameter of Theorem 3.3 as some $\varepsilon \in (0, 1/2 - \alpha_0)$. By Theorem 3.6, the hypothesis of Theorem 3.7 holds, and Theorem 3.7 gives $\mathbb{E}_{p_c}|C_o|^{\alpha_0} < \infty$, a contradiction. □

Theorem 3.8 (Step 2: near-critical a priori bounds)**.** *Assume (A1) and (A2). For every $\eta > 0$ there is $C_\eta < \infty$ such that*

$$\chi(p) \le C_\eta\,(p_c - p)^{-1-\eta} \qquad \text{for all } p \in [0, p_c)\,, \tag{3.3}$$

$$\nabla_p \le C_\eta\,(p_c - p)^{-\eta} \qquad \text{for all } p \in [p_c/2, p_c)\,. \tag{3.4}$$

Both bounds are proved from Theorem 3.2. The susceptibility bound (3.3) (Section 7.3) follows from a differential inequality for $\mathbb{P}_p(|C_o| \ge n)$ in p, obtained from the decision-tree inequality of O'Donnell, Saks, Schramm and Servedio, applied to an algorithm that marks every vertex independently with probability $1/n$ and explores the clusters of the marked vertices. The triangle bound (3.4) (Section 8) follows from a differential inequality for ∇_p. By Russo's formula, ∇_p' is a sum over pairs of distinct clusters, weighted by the number of edges between them and by a kernel built from T_p. Pairs with a small cluster contribute at most a power of the size cutoff times ∇_p. For two large clusters, Step 1 bounds the probability that two adjacent distinct clusters are both large, and the number of edges between them is controlled by comparing two explorations of one cluster, with and without the other cluster removed. The centred scores of the two explorations differ by exactly $-p$ times this number, and each score is concentrated.

3.4 Part II: the corridor deformation (Steps 3 and 4)

We now define the deformation precisely. Throughout this subsection $p \in [p_c/2, p_c)$ and $k \ge 3$ are fixed, and $q := 1 - p$.

Definition 3.9 (Corridors and the deformed kernel)**.** A *raw path* is a sequence $w = (w_0, \dots, w_k)$ of vertices with $P(w_{i-1}, w_i) > 0$ for all i; it has *base* w_0, *end* w_k and probability $\mu(w) := \prod_{i=1}^{k} P(w_{i-1}, w_i)$. Delete the holding steps $w_{i-1} = w_i$ and erase the loops of the remaining path in chronological order. This gives a self-avoiding path γ_w from w_0 to w_k with $|\gamma_w|$ edges. Its *interior* I_w is the set of its vertices other than w_0 and w_k, and $\Lambda_w := V \setminus I_w$. The *blockers* Bl_w are the edges incident to I_w that are not edges of γ_w. The raw path is a *template* if $|\gamma_w| \ge 3$. Its *corridor event* Cor_w is the event that all edges of γ_w are open and all blockers are closed, and its *weight* is

$$c(w) := \mu(w)\, p^{-|\gamma_w|}\, q^{-|\mathrm{Bl}_w|}\,, \qquad \text{so that} \qquad c(w)\,\mathbb{P}_p(\mathrm{Cor}_w) = \mu(w)\,.$$

For a domain $\Lambda \subseteq V$ and $x, y \in \Lambda$, a template w is *counted* for (x, y) in Λ if all vertices of γ_w lie in Λ, every edge of γ_w is open and *pivotal* for $x \overset{\Lambda}{\longleftrightarrow} y$ (that is, $x \overset{\Lambda}{\longleftrightarrow} y$ holds and fails after closing this edge), and every blocker is closed or has an endpoint outside Λ. With the sum over all templates w of any base, define

$$N_{xy}^{\Lambda} := \sum_{w \text{ counted for } (x,y) \text{ in } \Lambda} c(w)\,, \qquad W_t^{\Lambda}(x,y) := \mathbb{1}\{x \overset{\Lambda}{\longleftrightarrow} y\}\, \mathrm{e}^{-tN_{xy}^{\Lambda}} \qquad (t \geq 0)\,,$$

and $W_t^{\Lambda}(x, y) := 0$ if $x \notin \Lambda$ or $y \notin \Lambda$. The *deformed susceptibilities* and the *deformed kernel* are

$$\chi_t^{\Lambda}(x) := \mathbb{E}_p \sum_{y \in V} W_t^{\Lambda}(x,y)\,, \qquad \chi_t := \chi_t^{V}(o)\,, \qquad T_t(x,y) := \mathbb{E}_p\, W_t^{V}(x,y)\,.$$

Finally, $\mathbf{E}_k^{\mathrm{rw}}[F(w)] := \sum_w \mu(w) F(w)$ denotes the average over k-step lazy-walk paths w with base $w_0 = o$.

At $t = 0$ nothing is penalised: $T_0 = T_p$ and $\chi_0 = \chi(p)$. Different raw paths with the same loop erasure are different templates, and the convention for blockers outside Λ lets us compare domains without changing the weights.

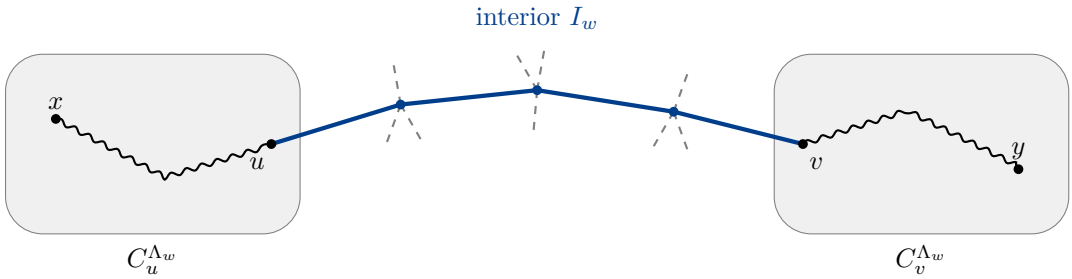


Figure 2: A corridor on the event Cor_w (schematic). The loop-erased path γ_w (thick) is open, and every other edge at its interior I_w is closed (dashed). Its edges are all pivotal for $x \leftrightarrow y$ if and only if x and y lie in the clusters $C_u^{\Lambda_w}$ and $C_v^{\Lambda_w}$ of its endpoints in $\Lambda_w = V \setminus I_w$, in either order, and these two clusters are distinct (Theorem 9.2).

Proposition 3.10 (Step 3: properties of the deformation)**.** *Assume (A1) and (A2). There is $C_0 \geq 1$, depending only on G, such that, with $c_k := \mathrm{e}^{C_0 k}$, the following holds for all $p \in [p_c/2, p_c)$, $k \geq 3$ and $t \geq 0$.*

(a) Basic bounds. *T_t is symmetric and invariant, $0 \leq T_t \leq T_p$ entrywise, and $\|T_t\| \leq \chi_t \leq \chi(p)$. The function $t \mapsto \chi_t$ is continuously differentiable on $[0, \infty)$.*

(b) The norm barely moves. $\|T_t\| \geq \|T_p\| - 2t\rho^k \|T_p\|^2$.

(c) The row sum decreases.

$$-\frac{\mathrm{d}}{\mathrm{d}t}\chi_t \geq \mathrm{e}^{-tc_k}\Big(\mathbf{E}_k^{\mathrm{rw}}\big[\mathbb{1}\{|\gamma_w| \geq 3\}\, \chi_t^{\Lambda_w}(w_0)\, \chi_t^{\Lambda_w}(w_k)\big] - C_0\, \chi(p)^2 \rho^k\, \nabla_p\Big)\,.$$

(d) Deletion. *For every domain Λ, every $A \subseteq V$ and every $z \in V$, $\chi_t^{\Lambda \setminus A}(z) \geq \chi_t^{\Lambda}(z) - \chi(p) \sum_{b \in \mathrm{N}[A]} T_p(z, b)$.*

(e) Transport along a path. *If Λ contains a self-avoiding path of length l from b to z, then $\chi_t^{\Lambda}(b) \geq \mathrm{e}^{-tlc_k} p^l \chi_t^{\Lambda}(z)$.*

Items (b) and (c) are the two halves of the comparison described in Section 3.2. Item (b) holds because a penalised connection factors through a corridor, so that $T_p - T_t \le 2t\,T_p P^k T_p$ entrywise. Item (c) is obtained by moving the root of the susceptibility to the base of the corridor with the mass-transport principle: the calibration $c(w)\mathbb{P}_p(\mathrm{Cor}_w) = \mu(w)$ then produces the average over raw paths, and the two exterior clusters contribute the product of the deformed susceptibilities at the two ends. The error term accounts for the requirement that the exterior clusters be distinct and is bounded with the van den Berg–Kesten inequality and the trace. The factor e^{-tc_k} is the price for the penalties of corridors that overlap the given one. Items (d) and (e) are the tools that make the main term of (c) usable: deleting a set A can only destroy the connections that pass near A, and an open path transports mass at the cost of its probability. Theorem 3.10 is proved in Section 9.

Proposition 3.11 (Step 4: escapes at both ends)**.** *Assume (A1) and (A2). There are $J, k_0 \in \mathbb{N}$ and $C_1, c_\star > 0$, depending only on G, with the following property. Let $p \in [p_c/2, p_c)$, $k \ge k_0$ and $R \ge 1$, and put $j := J + \lceil C_1 \log((k+1)R)\rceil$. Let $\mathcal{W}$ be the set of raw paths w with $|\gamma_w| \ge 3$ such that each of w_0 and w_k is joined by a self-avoiding path in Λ_w of length at most j to a vertex z with*

$$\sum_{b \in \mathrm{N}[I_w]} T_p(z,b) \le \frac{1}{4R}\,.$$

Then $\mathbf{E}_k^{\mathrm{rw}}[\mathbb{1}\{w \in \mathcal{W}\}] \ge c_\star$.

The vertices z are *escapes*: from them, deleting the corridor costs little in Theorem 3.10(d). The proof in Section 10 lets the raw path begin and end with halves of geodesic segments, and lets the escapes leave along the opposite halves and then continue as independent random walks of length m of order $C_1 \log((k+1)R)$. Avoidance of the raw path follows from the decay of P^l on $\ell^2(V)$. The construction is symmetric under reversal of the raw path, and the mass-transport principle turns this symmetry into an identity of probabilities, which transfers the avoidance estimate from one endpoint to the other. Finally, the walks smooth the connection kernel, $(P^m T_p)(a,b) \le \rho^m$, which makes the sum over $\mathrm{N}[I_w]$ small.

3.5 Proof of critical boundedness

We now prove Theorem 3.1 from the four steps and the sprinkling bound.

Proof of Theorem 3.1. Suppose, for a contradiction, that $\|T_{p_c}\| = \infty$. For $p \in [p_c/2, p_c)$ put

$$\delta := p_c - p\,, \qquad S := \log(1/\delta)\,, \qquad R := \frac{\chi(p)}{\|T_p\|} \ge 1\,.$$

All limits below are taken as $\delta \downarrow 0$, that is, $p \uparrow p_c$.

Stage 1: size of the parameters. By (3.1), which follows from Theorem 4.8, $\|T_p\| \ge (1-p_c)/(d\delta)$, so $\log\|T_p\| \ge S - C$. By Theorem 3.8, for every $\eta > 0$ we have $R \le C_\eta \delta^{-1-\eta} \cdot d\delta/(1-p_c) \le C'_\eta \delta^{-\eta}$ and $\nabla_p \le C_\eta \delta^{-\eta}$. Hence

$$\log R = o(S) \qquad \text{and} \qquad \log \nabla_p = o(S)\,. \tag{3.5}$$

Stage 2: choice of the constants. Fix the constants $J, k_0, C_1, c_\star$ of Theorem 3.11; they do not depend on p. Since $p \ge p_c/2$ and $j \le J + 1 + C_1 \log((k+1)R)$, we have

$$p^{2j} \ge c_2\,(kR)^{-C_2} \qquad \text{for all } k \ge 1 \text{ and } R \ge 1\,, \tag{3.6}$$

with $C_2 := 2C_1 \log(2/p_c)$ and $c_2 := (p_c/2)^{2J+2}\,2^{-C_2}$. Let $\lambda := \log(1/\rho) > 0$, fix $K_\star > 0$ with $\kappa_\star := \lambda K_\star > C_2 + 4$, and set

$$A_\delta := \log(2R) + \log(2 + \nabla_p) + \log S\,, \qquad k := \lceil K_\star A_\delta \rceil\,, \qquad D_\star := (kR)^{C_2+2}\,, \qquad t_\star := \frac{D_\star}{\|T_p\|}\,.$$

By (3.5), $k \to \infty$ and $k = o(S)$; in particular $k \geq \max\{k_0, 3\}$ and $k \leq S$ for small δ. We claim that

$$D_\star \rho^k \to 0\,, \qquad (kR)^{C_2} R^2 \nabla_p \rho^k \to 0\,, \qquad t_\star j c_k \to 0\,. \tag{3.7}$$

Indeed, $\rho^k \leq \mathrm{e}^{-\lambda K_\star A_\delta} = (2R)^{-\kappa_\star}(2+\nabla_p)^{-\kappa_\star} S^{-\kappa_\star}$. Since $R, \nabla_p \geq 1$ and $k \leq S$, this gives $D_\star \rho^k \leq S^{C_2+2-\kappa_\star} \to 0$ and $(kR)^{C_2} R^2 \nabla_p \rho^k \leq S^{C_2-\kappa_\star} \to 0$. Finally, $c_k = \mathrm{e}^{C_0 k}$ and $\log \|T_p\| \geq S - C$ give

$$\log(t_\star j c_k) \leq (C_2+2)(\log k + \log R) - S + C + \log j + C_0 k = -S + o(S)\,,$$

by (3.5), $k = o(S)$ and $j = O(\log(kR))$.

Stage 3: the norm stays large. Consider the deformation of Theorem 3.9 with these p and k, for $t \in [0, t_\star]$. By Theorem 3.10(b) and (3.7), for small δ,

$$\|T_t\| \geq \|T_p\|(1 - 2t_\star \rho^k \|T_p\|) = \|T_p\|(1 - 2D_\star \rho^k) \geq \tfrac{1}{2}\|T_p\|\,, \qquad \text{hence} \qquad \chi_t \geq \tfrac{1}{2}\|T_p\| \tag{3.8}$$

by Theorem 3.10(a).

Stage 4: the row sum decreases quadratically. Let $w \in \mathcal{W}$, the set of Theorem 3.11 for our p, k and R, and let z be the escape vertex of the endpoint w_0. The deformation is invariant, so $\chi_t^V(z) = \chi_t$. By Theorem 3.10(d) with $\Lambda = V$ and $A = I_w$, then $\chi(p) = R\|T_p\|$ and (3.8),

$$\chi_t^{\Lambda_w}(z) \geq \chi_t - \frac{\chi(p)}{4R} = \chi_t - \tfrac{1}{4}\|T_p\| \geq \tfrac{1}{2}\chi_t\,.$$

Theorem 3.10(e), applied along the escape path of length at most j in Λ_w, gives $\chi_t^{\Lambda_w}(w_0) \geq \mathrm{e}^{-t j c_k} p^j \chi_t/2$, and the same bound holds at w_k. Since $\mathrm{e}^{-2t_\star j c_k} \geq 1/2$ for small δ by (3.7), the bound (3.6) and $\mathbf{E}_k^{\mathrm{rw}}[\mathbb{1}\{w \in \mathcal{W}\}] \geq c_\star$ yield, for all $t \in [0, t_\star]$,

$$\mathbf{E}_k^{\mathrm{rw}}\big[\mathbb{1}\{|\gamma_w| \geq 3\}\, \chi_t^{\Lambda_w}(w_0)\, \chi_t^{\Lambda_w}(w_k)\big] \geq c_\star \cdot \tfrac{1}{2}\, p^{2j} \cdot \tfrac{1}{4}\chi_t^2 \geq \tfrac{1}{8} c_\star c_2\, (kR)^{-C_2} \chi_t^2\,.$$

The error term in Theorem 3.10(c) satisfies, by $\chi(p) = R\|T_p\| \leq 2R\chi_t$ and (3.7),

$$C_0\, \chi(p)^2 \rho^k \nabla_p \leq 4C_0\, R^2 \nabla_p\, \rho^k\, \chi_t^2 = o\big((kR)^{-C_2}\big)\, \chi_t^2\,.$$

Since also $\mathrm{e}^{-t c_k} \geq 1/2$ for $t \leq t_\star$, Theorem 3.10(c) gives, for small δ,

$$-\frac{\mathrm{d}}{\mathrm{d}t}\chi_t \geq c_3\, (kR)^{-C_2} \chi_t^2 \qquad \text{for all } t \in [0, t_\star]\,, \qquad c_3 := c_\star c_2/32\,. \tag{3.9}$$

Stage 5: contradiction. The function $t \mapsto \chi_t$ is positive and continuously differentiable, so (3.9) integrates to

$$\frac{1}{\chi_{t_\star}} \geq \frac{1}{\chi_0} + c_3\, (kR)^{-C_2} t_\star \geq c_3\, \frac{(kR)^2}{\|T_p\|}\,.$$

By (3.8), $1/\chi_{t_\star} \leq 2/\|T_p\|$. Hence $c_3 (kR)^2 \leq 2$, which fails for small δ because $k \to \infty$. This contradiction proves $\|T_{p_c}\| < \infty$. □

Remark 3.12 (Order of the choices). The constants $J, k_0, C_1, c_\star$ of the escape construction and C_0 of the deformation depend only on G; they are fixed before p is chosen. The length k and the final time $t_\star$ are then chosen as functions of p through R, ∇_p and S. The a priori bounds of Step 2 are used only through (3.5).

3.6 Roadmap

Figure 1 shows how the statements depend on each other. Part I (Sections 5 to 8) establishes the a priori estimates. It only concerns critical and near-critical percolation and does not involve the deformation. Part II (Sections 9 and 10) holds at every fixed $p \in [p_c/2, p_c)$ and uses only subcritical facts; the two parts meet only in the proof above.

4 Preliminaries

This section collects the tools used in the proof. Classical results are quoted; proofs are given for the variants that are less standard.

4.1 Kernels and operators

For a kernel K we use the operator norm

$$\|K\| := \sup\Big\{\Big|\sum_{x,y}\overline{g(x)}K(x,y)f(y)\Big| : f,g \text{ finitely supported, } \|f\|_2=\|g\|_2=1\Big\} \in [0,\infty]\,.$$

If it is finite, K extends to a bounded operator on $\ell^2(V)$ whose matrix is K; conversely, a bounded operator S has the matrix $S(x,y)=\langle \mathbb{1}_x, S\mathbb{1}_y\rangle$, where $\mathbb{1}_x$ is the indicator function of $\{x\}$. We identify the two. For a nonnegative kernel L and vertex sets A,B we write

$$L(A,B) := \sum_{a\in A}\sum_{b\in B} L(a,b)\,;$$

for instance, $\mathrm{Adj}(A,B)$ is the number of edges with one endpoint in A and the other in B.

Lemma 4.1 (Kernel bounds)**.** *Let K and L be kernels with $L\ge 0$.*

(a) Domination. *If $|K|\le L$ entrywise, then $\|K\|\le\|L\|$.*

(b) Schur test. *If all row sums and all column sums of L are at most b, then $\|L\|\le b$.*

(c) Monotone limits. *For $0<p_*\le 1$ and all x,y, $T_p(x,y)\uparrow T_{p_*}(x,y)$ as $p\uparrow p_*$, and $\|T_{p_*}\|=\sup_{p<p_*}\|T_p\|\in[0,\infty]$. Moreover $T_p\le T_{p'}$ entrywise for $p\le p'$; in particular T_p is bounded for every $p<p_{2\to2}$.*

Proof. For (a), $|\sum\overline{g(x)}K(x,y)f(y)|\le\sum|g(x)|L(x,y)|f(y)|\le\|L\|\|f\|_2\|g\|_2$. For (b), the Cauchy–Schwarz inequality with the weights $L(x,y)$ gives $\sum L(x,y)|g(x)||f(y)|\le(\sum L(x,y)|g(x)|^2)^{1/2}(\sum L(x,y)|f(y)|^2)^{1/2}\le b\|f\|_2\|g\|_2$.

For (c), monotonicity in p follows from the standard monotone coupling. The event $\{x\leftrightarrow y\}$ is the increasing union of the events $\{x\xleftrightarrow{B(x,n)}y\}$, $n\in\mathbb{N}$, whose probabilities are polynomials in p; hence $\lim_{p\uparrow p_*}T_p(x,y)=\sup_p\sup_n\mathbb{P}_p(x\xleftrightarrow{B(x,n)}y)=\sup_n\mathbb{P}_{p_*}(x\xleftrightarrow{B(x,n)}y)=T_{p_*}(x,y)$. If $\|T_p\|\le M$ for all $p<p_*$, then for finitely supported f,g the finite sum $\sum\overline{g(x)}T_{p_*}(x,y)f(y)$ is the limit of the corresponding sums for T_p, so $\|T_{p_*}\|\le M$. The reverse inequality and the last assertion follow from monotonicity and (a). □

4.2 Unimodularity, mass transport and the trace

Equip $\mathrm{Aut}(G)$ with the topology of pointwise convergence. It is a locally compact group whose vertex stabilisers $\mathrm{Stab}(x):=\{g\in\mathrm{Aut}(G): gx=x\}$ are compact and open. The group is *unimodular* if its left Haar measure is also right-invariant. For a transitive graph this is equivalent to $|\mathrm{Stab}(x)y|=|\mathrm{Stab}(y)x|$ for all $x,y\in V$, where $\mathrm{Stab}(x)y$ is the orbit of y under $\mathrm{Stab}(x)$, and it is equivalent to the mass-transport principle below; see [BLPS99b, Section 3] and [LP16, Chapter 8]. For instance, the automorphism group of every Cayley graph is unimodular [LP16, Chapter 8]. A function $F\colon V\times V\to[0,\infty]$ is *diagonally invariant* if $F(gx,gy)=F(x,y)$ for all $g\in\mathrm{Aut}(G)$ and $x,y\in V$.

Lemma 4.2 (Mass-transport principle, [BLPS99b], [LP16, Chapter 8])**.** *Assume (A1) and (A2). For every diagonally invariant $F\colon V\times V\to[0,\infty]$,*

$$\sum_{y\in V}F(o,y)=\sum_{x\in V}F(x,o)\,.$$

We think of $F(x,y)$ as mass sent from x to y: the expected mass sent out by the root equals the expected mass it receives. The identity extends to diagonally invariant real or complex F with $\sum_y |F(o,y)| < \infty$, by splitting into positive and negative parts. Typically $F(x,y) = \mathbb{E}_p[m(x,y;\omega)]$ for a random mass m that is *equivariant*, that is, $m(gx,gy;g\omega) = m(x,y;\omega)$, where $g\omega$ is the image of the configuration under g; such F are diagonally invariant because $\mathbb{P}_p$ is invariant. Two consequences are used repeatedly. First, every nonnegative invariant kernel has equal row and column sums; in particular an invariant stochastic kernel is doubly stochastic and has norm at most one by Theorem 4.1. Second, the diagonal of an invariant operator defines a trace.

A bounded operator S on $\ell^2(V)$ is *invariant* if its matrix is an invariant kernel, equivalently if S commutes with the unitary operators $U_g f := f \circ g^{-1}$, $g \in \mathrm{Aut}(G)$. For such S we set

$$\operatorname{tr} S := S(o,o) = \langle \mathbb{1}_o, S\mathbb{1}_o \rangle .$$

Since $\mathbb{1}_o$ is a unit vector, $|\operatorname{tr} S| \le \|S\|$. For example, $\operatorname{tr}(T_p^3) = \nabla_p$ whenever T_p is bounded.

Lemma 4.3 (The trace)**.** *Assume (A1) and (A2), and let A and B be bounded invariant operators.*

(a) $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

(b) *If A is positive semidefinite and B is self-adjoint, then* $|\operatorname{tr}(BA)| \le \|B\| \operatorname{tr} A$.

(c) *If A is positive semidefinite, then* $A(x,x) = \operatorname{tr} A$ *and* $|A(x,y)| \le \operatorname{tr} A$ *for all* $x, y \in V$.

Proof. For (a), the series defining $\operatorname{tr}(AB)$ converges absolutely, since

$$\sum_y |A(o,y)B(y,o)| \le \|A^*\mathbb{1}_o\|_2 \, \|B\mathbb{1}_o\|_2 \le \|A\| \, \|B\| .$$

The function $F(x,y) := A(x,y)B(y,x)$ is diagonally invariant, and the mass-transport principle gives $\operatorname{tr}(AB) = \sum_y A(o,y)B(y,o) = \sum_x A(x,o)B(o,x) = \operatorname{tr}(BA)$.

For (b), the positive square root $A^{1/2}$ commutes with every U_g, because A does and the functional calculus preserves commutation; so $A^{1/2}$ is invariant. By (a), $\operatorname{tr}(BA) = \operatorname{tr}(A^{1/2}BA^{1/2})$. Since $-\|B\|A \le A^{1/2}BA^{1/2} \le \|B\|A$ in the order of quadratic forms, evaluating at $\mathbb{1}_o$ gives the claim.

For (c), $A(x,x) = A(o,o)$ by invariance and transitivity, and $|A(x,y)|^2 \le A(x,x)A(y,y)$ by the Cauchy–Schwarz inequality for the positive semidefinite form $(f,g) \mapsto \langle f, Ag \rangle$. □

4.3 The lazy random walk

Recall from Section 3.2 the lazy simple random walk $P = (I + \mathrm{Adj}\,/d)/2$ and $\rho = \|P\|$. The following is a version of the discrete Cheeger inequality; see [Dod84] for its background.

Lemma 4.4 (Spectral gap)**.** *Assume (A1). The kernel P is symmetric, invariant and positive semidefinite, and*

$$\rho = \|P\| \le 1 - \frac{\iota^2}{4d^2} < 1 .$$

Proof. Symmetry and invariance are clear. By the Schur test $\|\mathrm{Adj}\,/d\| \le 1$, so $P = (I + \mathrm{Adj}\,/d)/2$ is positive semidefinite. For finitely supported f,

$$\langle f, (I-P)f \rangle = \frac{1}{2d} \sum_{\{x,y\} \in E} |f(x) - f(y)|^2 .$$

Let $f \ge 0$. By the co-area formula and the isoperimetric inequality, $\sum_{\{x,y\}\in E} |f(x)^2 - f(y)^2| = \int_0^\infty |\partial_E \{f^2 > s\}| \, \mathrm{d}s \ge \iota \|f\|_2^2$. The Cauchy–Schwarz inequality and $\sum_{\{x,y\}\in E} (f(x) + f(y))^2 \le$

$2d\|f\|_2^2$ bound the left-hand side by $(2d)^{1/2}\|f\|_2(\sum_{\{x,y\}\in E}(f(x)-f(y))^2)^{1/2}$. Hence $\langle f,(I-P)f\rangle \geq \iota^2\|f\|_2^2/(4d^2)$, that is, $\langle f,Pf\rangle \leq (1-\iota^2/(4d^2))\|f\|_2^2$. Since P has nonnegative entries, $|\langle f,Pf\rangle| \leq \langle |f|,P|f|\rangle$ for general f, and for the self-adjoint operator P this bound on the quadratic form bounds the norm. □

4.4 Correlation inequalities

An event or a function of the configuration is *increasing* if it is preserved, respectively does not decrease, when edges are opened. Two increasing events $\mathcal{A},\mathcal{B}$ occur *disjointly*, written $\mathcal{A}\circ\mathcal{B}$, if there are disjoint finite sets of open edges that witness $\mathcal{A}$ and $\mathcal{B}$ respectively.

Lemma 4.5 (Harris, van den Berg–Kesten, tree-graph bound)**.** *Let $p\in[0,1]$.*

(a) *Increasing nonnegative functions of the configuration are positively correlated under $\mathbb{P}_p$.*

(b) *For increasing events $\mathcal{A}_1,\dots,\mathcal{A}_m$ whose occurrence is witnessed by finite sets of open edges, $\mathbb{P}_p(\mathcal{A}_1\circ\cdots\circ\mathcal{A}_m)\leq\prod_i\mathbb{P}_p(\mathcal{A}_i)$.*

(c) *For all $x,y,z\in V$ and every domain Λ,*

$$\mathbb{P}_p(x\overset{\Lambda}{\longleftrightarrow}y,\ x\overset{\Lambda}{\longleftrightarrow}z)\leq\sum_{w\in V}T_p(x,w)\,T_p(w,y)\,T_p(w,z)\,.$$

In particular $\mathbb{E}_p|C_x|^2\leq\chi(p)^3$.

Proof. Parts (a) and (b) are the Harris inequality [Har60] and the van den Berg–Kesten inequality [vdBK85]; see [Gri99, Sections 2.2 and 2.3]. The passage from finitely many edges to events with finite witnesses on G is by monotone approximation. For (c), on the event on the left choose self-avoiding open paths in Λ from x to y and from x to z. Their union contains an open tree spanning x,y,z, which has a vertex w joined to x, y and z by three edge-disjoint open paths; paths of length zero are allowed. A union bound over w and part (b) give the claim, using $T_p^\Lambda\leq T_p$. Summing over y,z gives the bound on the second moment. □

4.5 The subcritical regime

Lemma 4.6 (Subcritical susceptibility)**.** *Assume (A1). For every $p<p_c$ the susceptibility $\chi(p)$ is finite. The kernel T_p is then a bounded, invariant, symmetric and positive semidefinite operator with $\|T_p\|\leq\chi(p)$.*

Proof. Finiteness of $\chi(p)$ below p_c is the sharpness of the phase transition. It was proved on the hypercubic lattice by Aizenman and Barsky [AB87] and Menshikov [Men86], and on quasi-transitive graphs by Antunović and Veselić [AV08, Theorem 2]; on every locally finite, transitive, infinite graph it is [DCT16, Theorem 1.1], which also gives a short proof. The norm bound is the Schur test. Since $T_p(x,y)=\mathbb{E}_p\sum_C\mathbb{1}\{x,y\in C\}$, where the sum runs over the clusters, every finitely supported f satisfies $\sum_{x,y}\overline{f(x)}T_p(x,y)f(y)=\mathbb{E}_p\sum_C|\sum_{x\in C}f(x)|^2\geq 0$. □

The next lemma is Russo's formula in the form we need: in operator norm and for the infinite graph. Its proof is given in Section A.1.

Lemma 4.7 (Russo's formula)**.** *Assume (A1) and (A2). The map $p\mapsto T_p$ is locally absolutely continuous on $(0,p_c)$ in operator norm, and χ is locally absolutely continuous on $(0,p_c)$. For almost every $p\in(0,p_c)$,*

$$(1-p)\,T_p'(x,y)=\mathbb{E}_p\big[\mathrm{Adj}(C_x,C_y)\,\mathbb{1}\{C_x\neq C_y\}\big]\qquad\text{for all } x,y\in V\,, \tag{4.1}$$

$$(1-p)\,\chi'(p)=\sum_{b\sim o}\mathbb{E}_p\big[|C_o|\,|C_b|;\,C_o\neq C_b\big]=\chi(p)^2\,\Delta(p)\,, \tag{4.2}$$

where T_p' is the derivative in operator norm and $\Delta(p)$ is defined by the last identity.

The quantity $\Delta(p)$ is the normalised expected product of the sizes of two adjacent distinct clusters; it already appeared in Section 3.3. In (4.1), $\mathrm{Adj}(C_x, C_y)$ counts the closed edges between the two clusters, each of which is pivotal for $x \leftrightarrow y$.

Lemma 4.8 (Sprinkling, [Hut19, Lemma 2.4])**.** *For $p, a \in [0,1]$,*

$$T_{p+(1-p)a} \le \sum_{j\ge 0} a^j\, T_p(\mathrm{Adj}\; T_p)^j \qquad \textit{entrywise.} \tag{4.3}$$

Consequently, if $ad\|T_p\| < 1$ then $\|T_{p+(1-p)a}\| \le \|T_p\|/(1 - ad\|T_p\|)$, and if $ad\chi(p) < 1$ then $\chi(p + (1-p)a) \le \chi(p)/(1 - ad\chi(p))$.

Proof. Let ω_1 and ω_2 be independent configurations with laws $\mathbb{P}_p$ and $\mathbb{P}_a$. Their union $\omega_1 \vee \omega_2$ has law $\mathbb{P}_{p+(1-p)a}$. If x and y are joined by a self-avoiding path open in $\omega_1 \vee \omega_2$, let $e_1, \dots, e_j$ be the edges of the path, in order, that are closed in ω_1; they are open in ω_2 and pairwise distinct. Orient $e_i = (x_i, y_i)$ along the path. The pieces of the path between them are edge-disjoint and open in ω_1, joining x to x_1, y_1 to x_2, …, y_j to y. By a union bound over j and the oriented edges, independence of ω_2 and the van den Berg–Kesten inequality for ω_1,

$$T_{p+(1-p)a}(x,y) \le \sum_{j\ge 0} a^j \sum_{(x_i,y_i)_{i\le j}} T_p(x,x_1)\, T_p(y_1,x_2) \cdots T_p(y_j,y)\,,$$

which is (4.3). Since $\|\mathrm{Adj}\| \le d$ and the row sums of Adj equal d, the norm and the row sums of the right-hand side are at most $\sum_j a^j d^j \|T_p\|^{j+1}$ and $\sum_j a^j d^j \chi(p)^{j+1}$, respectively. Theorem 4.1(a) concludes. □

4.6 Criticality and two reductions

Lemma 4.9 (Critical finiteness)**.** *Assume (A1). Then $0 < p_c < 1$, and $\mathbb{P}_{p_c}$-almost surely every cluster is finite. In particular, for every domain Λ, all clusters C_x^Λ are finite $\mathbb{P}_{p_c}$-almost surely.*

Proof. There are at most d^n self-avoiding paths of length n from o, so $\theta(p) = 0$ for $p < 1/d$. On the other hand, a connected vertex set $A \ni o$ with $|A| = n$ can be encoded by a closed walk of length $2n-2$ from o around a spanning tree of A, so there are at most d^{2n} such sets. If $C_o = A$, then all $|\partial_E A| \ge \iota n$ edges of $\partial_E A$ are closed. Hence $\mathbb{P}_p(|C_o| < \infty) \le \sum_{n\ge 1} d^{2n}(1-p)^{\iota n} < 1$ for p close to one, and $p_c < 1$.

Every ball satisfies $|B(o, r+1)| \ge (1 + \iota/d)|B(o,r)|$, since each vertex of the outer vertex boundary of $B(o,r)$ is incident to at most d edges of $\partial_E B(o,r)$. Thus G has exponential growth, and by a theorem of Hutchcroft [Hut16, Theorem 1.2] there are almost surely no infinite clusters at p_c. Clusters in a domain are contained in clusters of G. □

Lemma 4.10 (The operator threshold precedes uniqueness)**.** *Assume (A1). If $\mathbb{P}_p(N_\infty = 1) = 1$, then T_p is unbounded. Consequently $p_{2\to 2} \le p_u$.*

Proof. If $\mathbb{P}_p(N_\infty = 1) = 1$, then some vertex lies in an infinite cluster with positive probability, so $\theta(p) > 0$ by transitivity. By uniqueness and the Harris inequality, $T_p(x,y) \ge \mathbb{P}_p(|C_x| = |C_y| = \infty) \ge \theta(p)^2$ for all x, y. For finite $F \subseteq V$ this gives $\langle \mathbb{1}_F, T_p \mathbb{1}_F \rangle \ge \theta(p)^2 |F|^2 = \theta(p)^2 |F|\, \|\mathbb{1}_F\|_2^2$, so T_p is unbounded. If we had $p_{2\to 2} > p_u$, the definition of p_u as an infimum would provide $p < p_{2\to 2}$ with $\mathbb{P}_p(N_\infty = 1) = 1$, and T_p would be both bounded (Theorem 4.1) and unbounded. □

Proposition 4.11 (The nonunimodular case, [Hut19, Theorem 2.9])**.** *Assume (A1). If $\mathrm{Aut}(G)$ is not unimodular, then $p_c < p_{2\to 2}$.*

Proof. Hutchcroft's theorem states that $p_c < p_{q\to q}$ for every $q \in (1,\infty)$ on every connected, locally finite graph whose automorphism group has a quasi-transitive nonunimodular subgroup. Under our assumption, $\mathrm{Aut}(G)$ itself is such a subgroup, and $q = 2$ gives the claim. □

The last result of this section turns boundedness of one connection operator above p_c into a simultaneous nonuniqueness statement. It follows the argument of [Ope26].

Proposition 4.12 (Simultaneous nonuniqueness)**.** *Assume (A1), and let $p_c < p_1 < p_2 < 1$ be such that T_{p_2} is bounded. In the standard monotone coupling, almost surely for every $p \in [p_1, p_2]$ there are infinitely many infinite clusters at parameter p.*

Proof. Stage 1: ergodicity. Every $\mathrm{Aut}(G)$-invariant event $\mathcal{E}$ of the labels $(U_e)_{e\in E}$ has probability zero or one. Indeed, for $r \in \mathbb{N}$ choose an event $\mathcal{A}_r$ depending only on the labels in the ball $B(o,r)$ such that $\mathbb{P}(\mathcal{E}\triangle\mathcal{A}_r) \to 0$ as $r \to \infty$, and an automorphism g_r with $\mathrm{dist}(o, g_r o) > 2r + 2$, which exists by transitivity. Then $\mathcal{A}_r$ and $g_r\mathcal{A}_r$ are independent, while $g_r\mathcal{E} = \mathcal{E}$, so $|\mathbb{P}(\mathcal{E}) - \mathbb{P}(\mathcal{E})^2| \le 4\mathbb{P}(\mathcal{E}\triangle\mathcal{A}_r) \to 0$.

Stage 2: the count. Let $N_{12} \in \{0, 1, 2, \dots\} \cup \{\infty\}$ be the number of p_2-clusters that contain an infinite p_1-cluster. It is invariant, hence almost surely constant. Since $\theta(p_1) > 0$, it is positive.

It is not one. Otherwise any two vertices in infinite p_1-clusters would lie in the same p_2-cluster, and the Harris inequality would give $T_{p_2}(x,y) \ge \theta(p_1)^2$ for all x, y; as in the proof of Theorem 4.10, T_{p_2} would be unbounded.

It is not an integer $n \ge 2$. Otherwise there would be vertices x, y such that, with positive probability, x and y lie in infinite p_1-clusters contained in distinct p_2-clusters; call this event $\mathcal{E}_{xy}$. Fix a path π from x to y, and replace the labels of the edges of π by independent labels uniform on $[0, p_1]$. The law of the modified labels is absolutely continuous with respect to the original law, since finitely many coordinates were replaced by variables with bounded densities. The modification only opens edges, at both parameters, and only finitely many. Hence clusters at either parameter can only merge: every cluster after the modification is the union of finitely many clusters before it. In particular every infinite p_1-cluster after the modification contains an infinite p_1-cluster from before. Consequently every counted p_2-cluster before the modification lies in a counted p_2-cluster after it, and every counted p_2-cluster after the modification contains a counted one from before. On $\mathcal{E}_{xy}$ the path π joins two distinct counted p_2-clusters, which therefore lie in the same counted cluster after the modification. So the modified labels have $N_{12} \le n - 1$ with positive probability, which contradicts absolute continuity and $N_{12} = n$ almost surely.

Stage 3: conclusion. Hence $N_{12} = \infty$ almost surely. On this event choose infinite p_1-clusters in infinitely many distinct counted p_2-clusters. For every $p \in [p_1, p_2]$ they lie in infinite p-clusters, which are distinct because they lie in distinct p_2-clusters. □

5 Bridges in critical clusters

In this section we prove Theorem 3.4. Only (A1) is used, and all configurations have parameter p_c. The idea is to detect long chains of bridges with random marks. Mark every vertex independently with a small probability h. If a vertex y of C_o is the only mark in C_o, then every bridge separating o from y is pivotal for the event that C_o contains a mark. An exploration-martingale argument shows that this event has, conditionally on its occurrence, at most about $\sqrt{\log(1/h)/h}$ open pivotal edges, with exponential tails (Theorem 5.1). Choosing h as a function of M yields a tail bound for the bridge radius of clusters of size at most M (Theorem 5.2). Finally, a cluster that is not M-good is turned into one with a large bridge radius by closing the few edges that attach a long piece (Section 5.3).

5.1 Pivotal edges of a marked cluster

Let Λ be a domain, $x \in \Lambda$ and $h \in (0, 1/2]$. Independently of the edges, mark every vertex with probability h, and write $\mathbb{P}$ and $\mathbb{E}$ for the joint law of the configuration at parameter p_c and the marks. Let

$$\mathcal{G} = \mathcal{G}(\Lambda, x, h) := \{C_x^\Lambda \text{ contains a mark}\}.$$

An edge is *pivotal* for $\mathcal{G}$ if changing its state, with the marks and all other edges fixed, changes $\mathbb{1}_\mathcal{G}$. Let $\mathrm{Piv} = \mathrm{Piv}(\Lambda, x, h)$ be the set of open edges that are pivotal for $\mathcal{G}$. Since $\mathcal{G}$ is increasing, an open pivotal edge forces $\mathcal{G}$ to occur.

Lemma 5.1 (Pivotal edges of the marked event)**.** *Assume (A1). There is a constant $C \geq 1$, depending only on d and p_c, such that with $\kappa_h := C\sqrt{\log(1/h)/h}$ the following holds for every domain Λ, every $x \in \Lambda$, every integer $i \geq 0$ and every $r \geq 0$:*

$$\mathbb{E}\Big[\binom{|\mathrm{Piv}|}{i}\mathbb{1}_\mathcal{G}\Big] \leq \kappa_h^i\,\mathbb{P}(\mathcal{G}), \qquad \mathbb{P}(\mathcal{G},\ |\mathrm{Piv}| \geq r) \leq 2\,\mathbb{P}(\mathcal{G})\,\mathrm{e}^{-r/(4\kappa_h)}. \tag{5.1}$$

Proof. Stage 1: an exploration and its score. Explore C_x^Λ edge by edge in an order that does not depend on the marks, testing each edge of Λ incident to a discovered vertex at most once, and check the mark of each newly discovered vertex immediately. Stop at the first mark or when the cluster is exhausted. By Theorem 4.9 the cluster is almost surely finite, so the procedure stops and decides $\mathcal{G}$. Let $\mathcal{T}$ be the number of tested edges and

$$Y := \sum_{e \text{ tested}} (\omega(e) - p_c).$$

Given the configuration, the discovery order of C_x^Λ is fixed, and the index $\mathtt{g}$ of the first marked vertex in this order is geometric with parameter h; moreover $\mathcal{G} = \{\mathtt{g} \leq |C_x^\Lambda|\}$ and $\mathcal{T} \leq d\min\{\mathtt{g}, |C_x^\Lambda|\}$. A geometric variable conditioned to be at most n is stochastically smaller than itself, so $\mathbb{E}[\mathtt{g}\mathbb{1}\{\mathtt{g} \leq n\}] \leq \mathbb{P}(\mathtt{g} \leq n)/h$. Averaging over the configuration gives

$$\mathbb{E}\mathcal{T} \leq d/h \qquad \text{and} \qquad \mathbb{E}[\mathcal{T} \,|\, \mathcal{G}] \leq d/h. \tag{5.2}$$

Stage 2: the score counts open pivotal edges. Whether an edge e is tested is determined by the marks and the states of the other edges, because the procedure decides to test e before seeing $\omega(e)$. Every pivotal edge for $\mathcal{G}$ is tested, since otherwise the outcome of the procedure would not depend on its state. Conditionally on the marks and on all edges other than e, either e is pivotal, and then $\mathbb{1}_\mathcal{G} = \omega(e)$ and e is tested, or $\mathbb{1}_\mathcal{G}$ and the testing indicator do not depend on $\omega(e)$. In both cases $\mathbb{E}[(\omega(e) - p_c)\mathbb{1}\{e \text{ tested}\}\mathbb{1}_\mathcal{G} \,|\, \cdot\,] = p_c(1-p_c)\mathbb{1}\{e \text{ pivotal}\}$. Summing over e, which is justified by $|Y| \leq \mathcal{T}$ and (5.2), and using that pivotality of e does not depend on $\omega(e)$, we get

$$\mathbb{E}[Y\mathbb{1}_\mathcal{G}] = p_c(1-p_c)\sum_e \mathbb{P}(e \text{ pivotal}) = (1-p_c)\,\mathbb{E}|\mathrm{Piv}| = (1-p_c)\,\mathbb{E}\big[|\mathrm{Piv}|\mathbb{1}_\mathcal{G}\big]. \tag{5.3}$$

Stage 3: the first moment. Each tested edge is a fresh Bernoulli variable given the history of the exploration, and $\log \mathbb{E}\mathrm{e}^{\vartheta(\omega(e)-p_c)} \leq \vartheta^2/8$ by Hoeffding's lemma. Hence $\exp(\vartheta Y_n - \vartheta^2\mathcal{T}_n/8)$, with Y_n and $\mathcal{T}_n$ the score and the number of tests after n tests, is a supermartingale, and Fatou's lemma gives $\mathbb{E}\exp(\vartheta Y - \vartheta^2\mathcal{T}/8) \leq 1$ for all $\vartheta \in \mathbb{R}$. Jensen's inequality conditionally on $\mathcal{G}$, together with $\mathbb{P}(\mathcal{G}) \geq \mathbb{P}(x \text{ is marked}) = h$ and (5.2), yields

$$\vartheta\,\mathbb{E}[Y \,|\, \mathcal{G}] \leq \log\frac{1}{\mathbb{P}(\mathcal{G})} + \frac{\vartheta^2}{8}\,\mathbb{E}[\mathcal{T} \,|\, \mathcal{G}] \leq \log\frac{1}{h} + \frac{\vartheta^2 d}{8h}.$$

With $\vartheta = \sqrt{8h\log(1/h)/d}$ and (5.3) this gives

$$\mathbb{E}[|\mathrm{Piv}|\,|\,\mathcal{G}] \le \frac{\sqrt{d/2}}{1-p_c}\sqrt{\frac{\log(1/h)}{h}}\,,$$

which is the case $i=1$ of (5.1) once $C \ge \sqrt{d/2}/(1-p_c)$. We also take $C \ge 1$, so that $\kappa_h \ge 1$ for $h \le 1/2$. The bound is uniform in Λ and x.

Stage 4: a recursion for binomial moments. Put $b_i(\Lambda,x) := \mathbb{E}[\binom{|\mathrm{Piv}|}{i}\mathbb{1}_{\mathcal{G}}]$, so $b_0(\Lambda,x) = \mathbb{P}(\mathcal{G})$. On $\mathcal{G}$, every open pivotal edge is a bridge of C_x^Λ separating x from all marks of C_x^Λ, so the open pivotal edges are linearly ordered from x towards the marks. For an oriented edge (a,b) with $a,b \in \Lambda$, let $C_{x,ab}^\Lambda$ be the cluster of x in Λ when the edge $\{a,b\}$ is declared closed. We claim that, for $i \ge 1$,

$$b_i(\Lambda,x) = p_c \sum_{(a,b)} \sum_{W} \mathbb{P}(C_{x,ab}^\Lambda = W)\,(1-h)^{|W|}\, b_{i-1}(\Lambda\setminus W, b)\,, \tag{5.4}$$

where the inner sum runs over finite sets $W \subseteq \Lambda$ with $x, a \in W$ and $b \notin W$. To see this, decompose a selected set of i open pivotal edges according to its element $e = \{a,b\}$ closest to x, oriented away from x. Then $W = C_{x,ab}^\Lambda$ contains no mark, e is open, and e is the only open edge between W and its complement in Λ. Hence $C_x^\Lambda = W \cup C_b^{\Lambda\setminus W}$, and an edge of $C_b^{\Lambda\setminus W}$ is pivotal for $\mathcal{G}$ if and only if it is pivotal for $\mathcal{G}(\Lambda\setminus W, b, h)$. So the other $i-1$ selected edges are open pivotal edges of the latter event, and conversely every such configuration arises. The event $\{C_{x,ab}^\Lambda = W\}$ depends only on the edges with an endpoint in W other than e; the marks of W, the state of e, and the edges and marks in $\Lambda\setminus W$ are independent of it and of each other. This gives (5.4). Every i-set is counted once: if there are n ordered pivotal edges, $\sum_{j=1}^{n}\binom{n-j}{i-1} = \binom{n}{i}$. All terms are nonnegative, so no integrability is needed.

Stage 5: conclusion. By Stage 3, $b_1 \le \kappa_h b_0$ for all domains and roots. If $b_{i-1} \le \kappa_h^{i-1} b_0$ for all domains and roots, then (5.4) and its case $i=1$ give $b_i(\Lambda,x) \le \kappa_h^{i-1} b_1(\Lambda,x) \le \kappa_h^i b_0(\Lambda,x)$. This proves the first bound in (5.1). Summing, $\mathbb{E}[(1+1/(2\kappa_h))^{|\mathrm{Piv}|}\mathbb{1}_{\mathcal{G}}] = \sum_i (2\kappa_h)^{-i} b_i \le 2\mathbb{P}(\mathcal{G})$. Since $\log(1+1/(2\kappa_h)) \ge 1/(4\kappa_h)$ for $\kappa_h \ge 1$, Markov's inequality gives the second bound. □

5.2 The bridge radius of a capped cluster

Recall that $\varepsilon \in (0,1/2)$ is fixed and $\sigma = 1/2+\varepsilon$. A cluster is regarded as a graph with its open edges. The *bridge radius* of C_o about o is $\mathrm{rad}_{\mathrm{br}}(C_o,o) := \max_{y\in C_o} d_{\mathrm{br}}^{C_o}(o,y)$, the largest number of bridges of C_o separating o from a vertex of C_o.

Lemma 5.2 (Bridge radius)**.** *Assume (A1). There is $c>0$ such that for all sufficiently large M and all integers $1 \le m \le M^{1-\sigma}$,*

$$\mathbb{P}_{p_c}\big(|C_o| \le M,\ \mathrm{rad}_{\mathrm{br}}(C_o,o) \ge mM^\sigma/2\big) \le \exp\Big(-\frac{cm^2M^{2\varepsilon}}{\log M}\Big)\,. \tag{5.5}$$

Proof. Let $\mathcal{R}$ be the event on the left. On $\mathcal{R}$ choose, by a fixed rule, a vertex $y \in C_o$ with $d_{\mathrm{br}}^{C_o}(o,y) \ge mM^\sigma/2$. Mark vertices as in Section 5.1 with $\Lambda = V$ and $x = o$. Conditionally on the configuration, y is the only mark in C_o with probability $h(1-h)^{|C_o|-1} \ge h(1-h)^M$. In that case every bridge of C_o separating o from y is open and pivotal for $\mathcal{G}$, so $|\mathrm{Piv}| \ge mM^\sigma/2$. By Theorem 5.1,

$$\mathbb{P}_{p_c}(\mathcal{R})\,h(1-h)^M \le \mathbb{P}(\mathcal{G}, |\mathrm{Piv}| \ge mM^\sigma/2) \le 2\mathrm{e}^{-mM^\sigma/(8\kappa_h)}\,.$$

Choose

$$X := \frac{m^2M^{2\varepsilon}}{\log M}\,, \qquad h := \frac{X}{D_0 M} = \frac{m^2M^{2\varepsilon-1}}{D_0\log M}\,,$$

with a constant $D_0 \geq 2$ fixed below. Since $m^2 \leq M^{2-2\sigma} = M^{1-2\varepsilon}$, we have $h \leq 1/(D_0 \log M) \leq 1/2$ and $h \geq M^{-2}$ for large M, so $\log(1/h) \leq 2 \log M$. Hence

$$\frac{mM^\sigma}{8\kappa_h} = \frac{mM^\sigma \sqrt{h}}{8C\sqrt{\log(1/h)}} \geq \frac{m^2 M^{2\varepsilon}}{8C\sqrt{2D_0}\,\log M} = \frac{X}{8C\sqrt{2D_0}}\,.$$

Using $-\log(1-h) \leq 2h$, the logarithm of $2/(h(1-h)^M)$ is at most $C' \log M + 2X/D_0$. Therefore

$$\log \mathbb{P}_{p_c}(\mathcal{R}) \leq C' \log M + \frac{2X}{D_0} - \frac{X}{8C\sqrt{2D_0}}\,.$$

Fix D_0 so large that $2/D_0 \leq 1/(16C\sqrt{2D_0})$. Since $X \geq M^{2\varepsilon}/\log M$, the term $C' \log M$ is at most a small multiple of X for large M, uniformly in m. This proves (5.5). □

5.3 Proof of Theorem 3.4

Proof of Theorem 3.4. Stage 1: from a bad piece to a large bridge radius. Let ω be a configuration with $|C_o| \leq M$ in which C_o is not M-good. Choose, by a fixed rule, a set $A \subseteq C_o$ and a component B of the open subgraph induced by $C_o \setminus A$ that violate Theorem 3.3, let $\mathcal{K} \neq \emptyset$ be the set of open edges between B and A, and put $m := |\mathcal{K}|$. Since $\operatorname{diam}_{\mathrm{br}}(B) \leq |B| - 1 < M$, a violation $mM^\sigma < \operatorname{diam}_{\mathrm{br}}(B)$ forces $m < M^{1-\sigma}$.

Every open edge leaving B ends in A, because B is a component of $C_o \setminus A$ and C_o is a cluster. Define ω' by closing all edges of $\mathcal{K}$, except one if $o \notin B$: in that case keep the first edge of B entered by a self-avoiding open path from o to B. The part of this path before that edge avoids B and uses none of the closed edges, so the new cluster C'_o of o contains B; if $o \in B$, then $C'_o = B$. In either case B is joined to the rest of C'_o by at most one edge. A self-avoiding path between two vertices of B therefore stays in B, so the bridges of C'_o separating two vertices of B are exactly the bridges of B separating them. Choose $y, y' \in B$ with $d^B_{\mathrm{br}}(y, y') = \operatorname{diam}_{\mathrm{br}}(B) > mM^\sigma$. By the triangle inequality for the bridge distance of C'_o, one of them has bridge distance more than $mM^\sigma/2$ from o. Moreover $|C'_o| \leq |C_o| \leq M$. Hence ω' belongs to the event of Theorem 5.2.

Stage 2: the cost of the modification. All events involved are determined by the edges with an endpoint in $B(o, M)$, because a cluster of at most M vertices containing o lies in $B(o, M-1)$. Conditionally on all other edges we compare probabilities on a finite product space. Fix m. Given ω', every changed edge is incident to $B \subseteq C'_o$, and C'_o has at most M vertices, so the changed edges form a subset of size at most m of a set of at most dM edges determined by ω'. Reopening them recovers ω. Hence each ω' has at most $\sum_{j \leq m} \binom{\lfloor dM \rfloor}{j} \leq (m+1)(dM)^m$ preimages, and $\mathbb{P}_{p_c}(\omega)/\mathbb{P}_{p_c}(\omega') \leq \max\{1, p_c/(1-p_c)\}^m$. With Theorem 5.2, the probability of a violation with a given m is at most

$$\exp\Bigl(C'm \log M - \frac{cm^2 M^{2\varepsilon}}{\log M}\Bigr) \leq \exp\Bigl(-\frac{cM^{2\varepsilon}}{2 \log M}\Bigr)$$

for M large, uniformly in $m \geq 1$, since $M^{2\varepsilon}/(\log M)^2 \to \infty$. Summing over the at most M values of m gives the bound $M \exp(-cM^{2\varepsilon}/(2 \log M)) \leq \exp(-M^\varepsilon)$ for large M.

Stage 3: locality. Explore C_o until it is exhausted or more than M vertices are found; this uses only edges with an endpoint in $B(o, M)$. In the first case the cluster with its open edges is known, and M-goodness can be checked by inspecting all subsets $A \subseteq C_o$. In the second case the event fails. □

6 Local endpoint maps and their amplification

In this section we prove Theorem 3.6. We assume (A1) and (A2). Let $\varepsilon \in (0, 1/2)$ be the parameter of Theorem 3.3, to which all caps and M-goodness refer (Theorems 3.3 and 3.5),

and assume that $\mathbb{E}_{p_c}|C_o|^\alpha = \infty$ for some $\alpha \in (0, 1/2 - \varepsilon)$. We fix $\beta \in (\alpha, 1/2 - \varepsilon)$, so that

$$\beta + \sigma < 1\,, \qquad \sigma = \tfrac{1}{2} + \varepsilon\,. \tag{6.1}$$

Configurations have parameter p_c unless stated otherwise.

The heavy tail provides infinitely many scales n at which clusters of size about n are frequent (Section 6.1). One amplification step takes a map Φ and such a scale n and constructs a new map, in four parts: a splicing experiment (Section 6.2), an inverse description of its outcome (Section 6.3), first and second moment bounds for the resulting density (Section 6.4), and a truncation (Section 6.5). Iterating the step proves Theorem 3.6.

6.1 Heavy scales and a first map

Lemma 6.1 (Heavy scales)**.** *There is an infinite set $\mathcal{N}$ of powers of two such that for every $n \in \mathcal{N}$*

$$\pi_n := \mathbb{P}_{p_c}(n \le |C_o| < 2n) \ge n^{-\beta} \qquad \text{and} \qquad \mathbb{P}_{p_c}(n/2 \le |C_o| < 2n) \le (1 + 2^\beta)\,\pi_n\,. \tag{6.2}$$

Proof. Put $\pi'_j := \mathbb{P}_{p_c}(2^j \le |C_o| < 2^{j+1})$ for $j \ge 0$. Since $|C_o|$ is almost surely finite (Theorem 4.9), $\mathbb{E}_{p_c}|C_o|^\alpha \le \sum_j 2^{\alpha(j+1)}\pi'_j$. If the sequence $2^{\beta j}\pi'_j$ were bounded, the right-hand side would be finite because $\alpha < \beta$, contrary to our assumption. So the sequence $2^{\beta j}\pi'_j$ is unbounded, and it has infinitely many record indices j, that is, $2^{\beta j}\pi'_j > 2^{\beta i}\pi'_i$ for all $i < j$. Keep the records with value at least one. At such a record, $\pi'_j \ge 2^{-\beta j}$ and $\pi'_{j-1} < 2^\beta \pi'_j$, which is (6.2) for $n = 2^j$. □

The next lemma collects the basic properties of local endpoint maps. For a map Φ and $p \in [0,1]$ we call $F_p(x,y) := \mathbb{E}_p \Phi_x(\omega, y)$ its *unconditional endpoint kernel* at parameter p, and $s_p := \sum_y F_p(x,y)$ its success probability at p; thus $s = s_{p_c}$ and $K = F_{p_c}/s$.

Lemma 6.2 (Basic properties)**.** *Let Φ be a local endpoint map with $s > 0$. Then K is invariant and doubly stochastic, and $r = \|K\| \le 1$. For every $p \in [0,1]$ the kernel F_p is invariant, $F_p(x,y) = 0$ if $\operatorname{dist}(x,y) > r_\Phi + 1$, its row and column sums equal s_p, and $\|F_p - F_{p_c}\| \to 0$ as $p \to p_c$.*

Proof. Invariance of F_p follows from the covariance of Φ and the invariance of $\mathbb{P}_p$. If $\operatorname{dist}(x,y) > r_\Phi + 1$, closing all edges without an endpoint in $B(x, r_\Phi)$ disconnects x from y without changing $\Phi_x(\omega, \cdot)$, so $\Phi_x(\omega, y) = 0$ for all ω. By the mass-transport principle the column sums of the nonnegative invariant kernels F_p and $|F_p - F_{p_c}|$ equal their row sums; in particular K is doubly stochastic, and the Schur test gives $r \le 1$. The entries of F_p are polynomials in p, since Φ_x depends on finitely many edges. Hence the row sums of $|F_p - F_{p_c}|$, which have at most $|B(o, r_\Phi + 1)|$ nonzero terms, tend to zero, and the Schur test gives $\|F_p - F_{p_c}\| \to 0$. □

Lemma 6.3 (A first map)**.** *For every M_1 there is a local endpoint map with cap $M \ge M_1$ and $r < \mathrm{e}^{-1}$.*

Proof. Fix $\ell_0 \in \mathbb{N}$ with $2\rho^{\ell_0} < \mathrm{e}^{-1}$. For a path w of ℓ_0 steps of the lazy walk, let $e(w)$ be the number of distinct edges it traverses; holding steps traverse no edge. Call w *open* if all edges it traverses are open. For $M \ge M_1$ put

$$\Phi_x(\omega, y) := \mathbb{1}\{|C_x| \le M,\ C_x \text{ is } M\text{-good}\} \sum_{w\colon x \to y} \mu(w)\, p_c^{\ell_0 - e(w)}\, \mathbb{1}\{w \text{ is open}\}\,,$$

where the sum runs over ℓ_0-step paths from x to y. This is a local endpoint map with cap M: the sum over y is at most $\sum_w \mu(w) = 1$, an open path joins x to y, and by Theorem 3.4 the

indicator depends only on the edges near x. Without the indicator, the unconditional endpoint kernel would be $\sum_w \mu(w) p_c^{\ell_0 - e(w)} p_c^{e(w)} = p_c^{\ell_0} P^{\ell_0}(x,y)$. Hence $F_{p_c} \le p_c^{\ell_0} P^{\ell_0}$ entrywise and

$$s \ge p_c^{\ell_0} - \mathbb{P}_{p_c}(|C_o| > M) - \mathbb{P}_{p_c}(|C_o| \le M,\ C_o \text{ not } M\text{-good}) \ge p_c^{\ell_0}/2$$

for M large, by Theorems 3.4 and 4.9. Thus $r = \|F_{p_c}\|/s \le p_c^{\ell_0}\rho^{\ell_0}/(p_c^{\ell_0}/2) = 2\rho^{\ell_0} < \mathrm{e}^{-1}$. □

6.2 The splicing experiment

For the rest of this section fix an old map Φ with cap $M \ge 2$, success probability s, endpoint kernel K and $r = \|K\|$; fix $\ell \in \mathbb{N}$ and $n \in \mathcal{N}$ with $\ell M \le n/2$, and put $\nu := \ell M$. For vertices x and z we run the following experiment.

(1) Sample independent configurations $\omega^{(0)}, \omega^{(1)}, \dots, \omega^{(\ell)}$ with law $\mathbb{P}_{p_c}$. We call $\omega^{(0)}$ the *distinguished* configuration.

(2) Starting from $y_\ell := z$, for $i = \ell, \ell-1, \dots, 1$ apply the rule Φ_{y_i} to $\omega^{(i)}$, using randomness that is independent across the steps and of all other randomness; if it outputs a vertex, call it y_{i-1}, and if it fails, the experiment fails.

(3) The experiment *succeeds* if all ℓ rules succeed, the cluster $A := C_x(\omega^{(0)})$ satisfies $n \le |A| < 2n$, and $y_0 \in A$.

A successful experiment has *weight* $1/n$. Let $C_i := C_{y_i}(\omega^{(i)})$ be the cluster used in the i-th step; it contains y_{i-1}, has at most M vertices and is M-good.

On success, the following *switching procedure* splices pieces of the clusters C_i onto A. It maintains a set S, initially $S = A$, and a set of eligible indices, initially $\{1, \dots, \ell\}$. While $z \notin S$, it performs a round:

(i) let i be the largest eligible index with $C_i \cap S \ne \emptyset$, and declare all indices $\le i$ ineligible;

(ii) let B_i be the component of y_i in the subgraph of $\omega^{(i)}$-open edges induced by $V \setminus S$, and let $\mathcal{K}_i$ be the set of $\omega^{(i)}$-open edges between B_i and S;

(iii) choose the *seam* $e_i \in \mathcal{K}_i$ uniformly at random, independently of everything else;

(iv) exchange the states of $\omega^{(0)}$ and $\omega^{(i)}$ on the *mask* $\mathcal{M}_i$, which consists of e_i and of all edges that have an endpoint in B_i and no endpoint in S;

(v) replace S by $S \cup B_i$.

The *output* consists of the configurations $\omega'^{(0)}, \dots, \omega'^{(\ell)}$ after all exchanges. The random objects of a successful run, namely the input configurations, the chain $y_0, \dots, y_{\ell-1}$ and the seams, form its *trajectory* τ.

Lemma 6.4 (Geometry of the switching procedure)**.** *For a successful experiment every round is well defined, and the procedure stops after at most ℓ rounds. Let $i_1 < \cdots < i_t$ be the selected indices. Before each round $S = C_x(\omega^{(0)})$ for the current $\omega^{(0)}$, and after the round $C_x(\omega^{(0)}) = S \cup B_i$, with e_i the only open edge between S and B_i. The piece B_{i_1} is attached to A by e_{i_1}, and each later piece B_{i_j} is attached to $B_{i_{j-1}}$ by e_{i_j} and to no other earlier piece. Consequently, in the final distinguished configuration $\omega'^{(0)}$,*

$$C_x(\omega'^{(0)}) = A \cup B_{i_1} \cup \cdots \cup B_{i_t} \ni z\,, \qquad |C_x(\omega'^{(0)})| < 2n + \nu\,, \tag{6.3}$$

the seams are bridges of $C_x(\omega'^{(0)})$ separating x from z, in the order $e_{i_1}, \dots, e_{i_t}$ from x, and the open edges of $\omega'^{(0)}$ inside each B_i are those of $\omega^{(i)}$. Every edge whose state in the distinguished configuration is changed has an endpoint in the hanging part *$H := B_{i_1} \cup \cdots \cup B_{i_t}$, which is the vertex set of the component of $C_x(\omega'^{(0)})$ beyond the first seam e_{i_1}.*

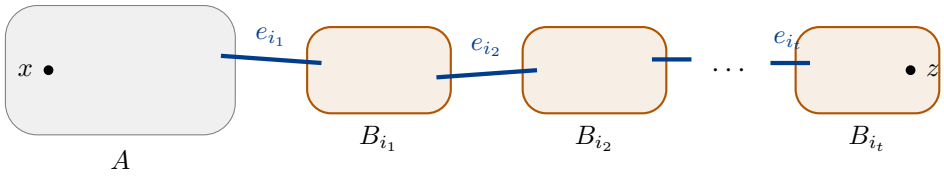


Figure 3: The cluster of x after the switching procedure (schematic). Each inserted piece B_{i_j} is attached to its predecessor by a single open seam e_{i_j}, so the seams are bridges separating x from z. Cutting the seams recovers A and all pieces (Theorems 6.4 and 6.5).

Proof. Initially $C_1 \ni y_0 \in A$, so an index meets S. Suppose index i is selected. Then $y_i \notin S$: for $i < \ell$ this holds because $C_{i+1} \ni y_i$ and $i+1$ was eligible but not selected, and for $i = \ell$ because $y_\ell = z \notin S$. Since C_i is connected, meets S and contains y_i, the piece $B_i \subseteq C_i \setminus S$ has at least one open edge to S, so $\mathcal{K}_i \neq \emptyset$.

Before the exchange all edges of the current $\omega^{(0)}$ leaving S are closed. The exchange imports the $\omega^{(i)}$-states of all edges inside B_i and of all edges from B_i to $V \setminus (S \cup B_i)$; the latter are closed in $\omega^{(i)}$, because B_i is a component of the $\omega^{(i)}$-open subgraph on $V \setminus S$. It also opens e_i. All other edges between B_i and S keep their closed states, and no edge inside S changes. Hence the new cluster of x is $S \cup B_i$, with e_i its only open edge between S and B_i. Edges inside B_i are not touched again, since later masks consist of edges with no endpoint in the enlarged S, apart from their seams.

Every eligible index larger than the selected i missed the old S. Hence the next selected cluster meets the enlarged set only in B_i, so the next piece is attached to B_i and has no open edge to earlier pieces or to A. If $i < \ell$ and $z \notin S \cup B_i$, then $C_{i+1} \ni y_i \in B_i$ provides an eligible index for the next round; if $i = \ell$, then $z = y_\ell \in B_\ell$ and the procedure stops. Since the selected indices increase, there are at most ℓ rounds. The pieces form a chain attached by single seams, which gives the claims about the seams and (6.3), using $|A| < 2n$ and $|B_i| \leq M$. Every mask edge has an endpoint in its piece, which proves the last claim. □

6.3 Undoing the splicing

Fix x and z. For a configuration ω' with $z \in C_x(\omega')$, list the bridges of $C_x(\omega')$ that separate x from z in their order from z towards x. A *code* is a sequence $c = (c_\ell, \dots, c_1)$ with entries in $\{*\} \cup \{1, \dots, M\}$. A successful run with output distinguished configuration $\omega'^{(0)}$ has the following code: $c_i = *$ if i is not selected, and for the selected indices, read from z towards x, the entry c_{i_j} is the number of separating bridges after the previous seam (or after z), up to and including e_{i_j}. We call c_{i_j} the *gap* of e_{i_j}. Recall that $\sigma = 1/2 + \varepsilon$, and put

$$\Theta_M := 1 + \sum_{m=1}^{M} \min\Big\{1, \frac{2M^\sigma}{m}\Big\} \leq CM^\sigma \log M\,.$$

Lemma 6.5 (Inverse codes)**.** *Fix x, z and output configurations $\omega'^{(0)}, \dots, \omega'^{(\ell)}$.*

(a) *A code determines the input configurations, the sets A and B_i, the seams, the masks and the contact sets $\mathcal{K}_i$ of every successful run with this output and this code.*

(b) *In every successful run each gap satisfies $1/|\mathcal{K}_i| \leq \min\{1, 2M^\sigma/c_i\}$.*

(c) *Summed over the codes c of successful runs with this output, $\sum_c \prod_{i:\, c_i \neq *} |\mathcal{K}_i|^{-1} \leq \Theta_M^\ell$.*

Proof. (a): The code locates the seams among the separating bridges. By Theorem 6.4, cutting them decomposes $C_x(\omega'^{(0)})$ into A (the part containing x) and the pieces $B_{i_1}, \dots, B_{i_t}$

in their order. This determines the set S before each round, hence every mask, and undoing the exchanges in reverse order recovers the input configurations. The contact sets are then determined by the input configurations and the pieces.

(b): The separating bridges counted in the gap of e_i, other than e_i itself, lie inside B_i, between the point where the next seam (or z) attaches and the endpoint of e_i. Since B_i is joined to the rest of the cluster by at most two edges, both of them bridges, a self-avoiding path between two vertices of B_i stays in B_i; so these edges are bridges of B_i separating two of its vertices. By Theorem 6.4 the open edges inside B_i are those of $\omega^{(i)}$. Hence $c_i - 1 \le \mathrm{diam}_{\mathrm{br}}(B_i)$, the bridge diameter of B_i in $\omega^{(i)}$. Now B_i is a component of the $\omega^{(i)}$-open subgraph induced by $C_i \setminus S$, joined to $C_i \cap S$ by the $|\mathcal{K}_i| \ge 1$ open edges of $\mathcal{K}_i$. Since C_i is M-good, $\mathrm{diam}_{\mathrm{br}}(B_i) \le |\mathcal{K}_i| M^\sigma$. If $c_i = 1$ the claim is $|\mathcal{K}_i| \ge 1$; otherwise $c_i - 1 \ge c_i/2$ gives $1/|\mathcal{K}_i| \le M^\sigma/(c_i - 1) \le 2M^\sigma/c_i$. Also $c_i - 1 \le |B_i| - 1$, so $c_i \le M$ and codes indeed take values in $\{*\} \cup \{1, \dots, M\}$.

(c): By (b), the product for a code of a successful run is at most $\prod_{i:\, c_i \neq *} \min\{1, 2M^\sigma/c_i\}$. Summing this over all sequences $c \in (\{*\} \cup \{1, \dots, M\})^\ell$ gives Θ_M^ℓ. The bound on Θ_M is the harmonic sum. □

All configurations of a successful run are determined, as far as they matter, by the edges in a fixed finite ball around x: the cluster A is explored within distance $2n$, the chain moves at most ν steps from $z \in C_x(\omega'^{(0)})$, each rule reads edges within distance r_Φ of its vertex, and every mask lies within distance one of a piece. Conditioning on all other edges, we may therefore treat the trajectories as elements of a finite product space. On it, the exchanges preserve the total number of open edges among the $\ell + 1$ configurations, so input and output configurations have the same probability under the product measure. This is the basis of the following density description.

Lemma 6.6 (The load)**.** *For $x, z \in V$ and a configuration ω let*

$$L_x(z, \omega) := \frac{1}{n} \mathbb{E}\Big[\sum_{\tau} \prod_{i=1}^{\ell} \Phi_{y_i}(\omega^{(i)}, y_{i-1}) \prod_{j=1}^{t} |\mathcal{K}_{i_j}|^{-1}\Big],$$

where the expectation is over independent auxiliary configurations $\omega'^{(1)}, \dots, \omega'^{(\ell)}$ with law $\mathbb{P}_{p_c}$, and the sum is over the trajectories τ of successful runs whose output is $(\omega, \omega'^{(1)}, \dots, \omega'^{(\ell)})$; the factors are evaluated on these trajectories. Then, for every event $\mathcal{B}$,

$$\frac{1}{n} \mathbb{P}(\textit{the experiment succeeds and } \omega'^{(0)} \in \mathcal{B}) = \mathbb{E}_{p_c}\big[L_x(z, \omega) \mathbb{1}_{\mathcal{B}}(\omega)\big]. \tag{6.4}$$

Moreover $0 \le L_x(z, \omega) \le \Theta_M^\ell / n$; the function $L_x(z, \cdot)$ depends only on the edges in a finite ball around x; $L_{gx}(gz, g\omega) = L_x(z, \omega)$ for $g \in \mathrm{Aut}(G)$; and $L_x(z, \omega) = 0$ unless $z \in C_x(\omega)$ and $|C_x(\omega)| < 2n + \nu$.

Proof. A trajectory has probability $\mathbb{P}_{p_c}(\omega^{(0)}) \cdots \mathbb{P}_{p_c}(\omega^{(\ell)}) \prod_i \Phi_{y_i}(\omega^{(i)}, y_{i-1}) \prod_j |\mathcal{K}_{i_j}|^{-1}$ on the finite product space. The product of the configuration probabilities equals that of the output configurations. Summing over trajectories with a given output gives (6.4). For the upper bound, group the trajectories with a given output by their codes. By Theorem 6.5(a), the code determines the input configurations and the seams, so for fixed code the sum over trajectories is a sum over chains, and $\sum_{\text{chains}} \prod_i \Phi_{y_i}(\omega^{(i)}, y_{i-1}) \le 1$ because these are the probabilities of the chains given the input configurations. Theorem 6.5(c) bounds the remaining sum over codes by Θ_M^ℓ. Locality and covariance hold because every ingredient is defined locally and covariantly, and the support property is (6.3). □

6.4 First and second moment of the load

Let $L(\omega) \coloneqq \sum_z L_x(z,\omega)$ be the *total load* and $\bar{L} \coloneqq \mathbb{E}_{p_c} L$; by invariance these do not depend on x in law. Write $\pi \coloneqq \pi_n$, and let

$$T^{(n)}(x,y) \coloneqq \mathbb{P}_{p_c}\big(y \in C_x,\ n \le |C_x| < 2n\big)\,,$$
$$\chi^{(n)} \coloneqq \sum_y T^{(n)}(o,y) = \mathbb{E}_{p_c}\big[|C_o|;\, n \le |C_o| < 2n\big]\,.$$

The kernel $T^{(n)}$ is symmetric, because $y \in C_x$ implies $C_x = C_y$, and $n\pi \le \chi^{(n)} \le 2n\pi$.

Lemma 6.7 (Mean load)**.** *The* mean load $\bar{L}(x,z) \coloneqq \mathbb{E}_{p_c} L_x(z,\omega)$ *is a kernel with row sums* $\bar{L}$*, and*

$$\bar{L}(x,z) = \frac{s^\ell}{n} \sum_y T^{(n)}(x,y)\, K^\ell(z,y)\,, \qquad \pi s^\ell \le \bar{L} \le 2\pi s^\ell\,, \qquad \|\bar{L}(\cdot,\cdot)\| \le \bar{L}\, r^\ell\,. \tag{6.5}$$

Proof. By (6.4), $\bar{L}(x,z)$ is $1/n$ times the success probability of the experiment. The ℓ steps of the chain use independent configurations, each succeeding with probability s and moving according to K, so the chain succeeds and ends at y with probability $s^\ell K^\ell(z,y)$; independently, $y \in A$ and $n \le |A| < 2n$ with probability $T^{(n)}(x,y)$. Since K^ℓ is doubly stochastic, $\bar{L} = \sum_z \bar{L}(x,z) = s^\ell \chi^{(n)}/n \in [\pi s^\ell, 2\pi s^\ell]$. Finally $\bar{L}(\cdot,\cdot) = (s^\ell/n)\, T^{(n)} (K^\ell)^\top$, and the Schur test gives $\|T^{(n)}\| \le \chi^{(n)}$, so $\|\bar{L}(\cdot,\cdot)\| \le (s^\ell/n)\chi^{(n)} r^\ell = \bar{L} r^\ell$. □

The mean kernel already has the desired small norm. To turn the load into the output probabilities of a new map we must divide by a bound on L, and we need that most of the mean is carried by configurations on which L is not much larger than its mean. This is what the second moment controls.

Lemma 6.8 (Second moment of the load)**.** *There is a polynomial* $Q(\ell, M) \ge 1$*, with coefficients depending only on* G*, such that*

$$\mathbb{E}_{p_c} L^2 \le Q(\ell, M)\, \pi s^{2\ell} \qquad \text{whenever } n \in \mathcal{N},\ \ell M \le n/2 \text{ and } n \ge (\Theta_M/s)^\ell\,. \tag{6.6}$$

Proof. Expand $L^2 = \sum_{z_1,z_2} L_x(z_1,\omega) L_x(z_2,\omega)$, using two independent families of auxiliary output configurations. Each summand is a weighted sum over pairs (τ^1,τ^2) of trajectories of successful runs with the common distinguished output ω and endpoints z_1, z_2. We split the pairs into three classes. For a trajectory with at least one round, let H^1 or H^2 be its hanging part (Theorem 6.4); it contains the endpoint and has at most ν vertices. Two hanging parts in the same cluster are the far sides of bridges, so they are nested or disjoint.

Class 1: one of the trajectories has no round. Let $L^\emptyset(\omega)$ be the part of $L(\omega)$ coming from trajectories without rounds. For them the output equals the input, so $A = C_x(\omega)$, $n \le |A| < 2n$, $z \in A$, and the chain from z ends in A. Hence

$$L^\emptyset(\omega) \le \frac{s^\ell}{n}\, \mathbb{1}\{n \le |C_x(\omega)| < 2n\} \sum_{z \in C_x(\omega)} \sum_{y \in C_x(\omega)} K^\ell(z,y) \le \frac{s^\ell}{n}\, 2n = 2s^\ell\,,$$

using that the column sums of K^ℓ are one. The pairs of class 1 contribute at most $2\mathbb{E}_{p_c}[L^\emptyset L] \le 4s^\ell \bar{L} \le 8\pi s^{2\ell}$.

Class 2: the hanging parts are nested or joined by an edge of G*.* Fix τ^1, with hanging part $H^1 \ni z_1$. We claim that at most $N_\nu \coloneqq \nu + \nu^2 + d\nu^3$ endpoints z_2 admit a trajectory τ^2 in this class. If $H^2 \subseteq H^1$, then $z_2 \in H^1$: at most ν choices. If $H^1 \subsetneq H^2$, the first seam of τ^2 is a bridge separating x from z_1, and the bridges between it and z_1 lie in H^2; so it is among the last ν such bridges, it determines H^2, and $z_2 \in H^2$: at most ν^2 choices. If H^1 and H^2 are

disjoint and joined by an edge, pick a vertex of H^2 adjacent to H^1 (at most $d\nu$ choices); the first seam of τ^2 is among the last ν bridges separating x from it, and $z_2 \in H^2$: at most $d\nu^3$ choices. By Theorem 6.6, $L_x(z_2,\omega) \le \Theta_M^\ell/n \le s^\ell$ for each such z_2. Hence class 2 contributes at most $\mathbb{E}_{p_c}[L] N_\nu s^\ell \le 2N_\nu \pi s^{2\ell}$.

Class 3: the hanging parts are disjoint and not joined by any edge of G. Undo both trajectories: by Theorem 6.4 the changes of the distinguished configuration made by τ^a concern edges with an endpoint in H^a, and these two sets of edges are disjoint, so the undoing operations commute. Let ω_{00} be the jointly restored distinguished configuration. We claim:

(a) $C_x(\omega_{00}) = A_{00} \coloneqq C_x(\omega) \setminus (H^1 \cup H^2)$, and $n - \nu \le |A_{00}| < 2n$;

(b) running the switching procedure of τ^1 from A_{00} instead of from $A^1 \coloneqq C_x(\omega) \setminus H^1$, with the jointly restored configurations and the same chain and seams, reproduces the selected indices, pieces, contact sets and masks of τ^1; the same holds for τ^2. In particular, the first selected cluster of each chain meets A_{00}.

For (a): undoing τ^1 alone restores the input of τ^1, whose cluster of x is A^1 by Theorem 6.4; its edges to H^1 are closed and remain so, because undoing τ^2 changes only edges with an endpoint in H^2, and no edge joins H^1 to H^2. The same holds with the roles exchanged. Edges with both endpoints in A_{00} are not changed, and A_{00} is connected, because removing the far sides of two bridges leaves the near side connected. Hence $C_x(\omega_{00}) = A_{00}$. Finally $|A^1| \in [n, 2n)$ and $|H^2| \le \nu$.

For (b) we argue by induction over the rounds of τ^1, writing S for the current set of the original run and S_0 for that of the new run; initially $S_0 = S \setminus H^2$, and $H^2 \subseteq S$ throughout, since $H^2 \subseteq A^1$. Let i be the index selected in the original run and $e_i = \{a_i, b_i\}$ its seam with $a_i \in S$ and $b_i \in B_i \subseteq H^1$. Since no edge joins H^1 to H^2, we have $a_i \notin H^2$, so $C_i \ni a_i$ meets S_0, while every larger eligible index misses $S \supseteq S_0$. Hence the same index is selected. The new piece is the component of y_i in the $\omega^{(i)}$-open subgraph on $V \setminus S_0 = (V \setminus S) \cup H^2$; an open path leaving B_i would have to enter H^2 through an edge from $B_i \subseteq H^1$, which does not exist, so the piece is again B_i. For the same reason the contact sets and the masks of the two runs coincide, and the states exchanged on the mask agree, because undoing τ^2 changed no edge with an endpoint in H^1. Afterwards $S_0 \cup B_i = (S \cup B_i) \setminus H^2$. The stopping rules agree because $z_1 \in H^1$ is not in H^2. The first selected cluster meets $S_0 = A_{00}$ by the argument just given.

By (b), a pair (τ^1, τ^2) of class 3 is determined by the jointly restored configurations together with the two chains and the two seam sequences: rerunning both procedures from A_{00} and applying their disjoint exchanges recovers the output. The exchanges preserve the total number of open edges, so the weight of the pair equals the probability of the restored configurations, times the chain probabilities and the factors $\prod|\mathcal{K}|^{-1}$ of the two procedures run from A_{00}. For fixed restored configurations and chains, the selected indices, the pieces and the contact sets of each procedure do not depend on the seam choices, because every round adds the whole piece to the current set; only the masks and the output depend on the seams. Hence the factors $\prod|\mathcal{K}|^{-1}$ are the probabilities of the uniform seam choices of each run, their sum over the seam sequences of one run is one, and the sum over pairs of seam sequences is at most the product of these two sums. Dropping all conditions except (a) and the fact that each chain succeeds and meets A_{00}, we obtain that class 3 contributes at most

$$\frac{1}{n^2} \sum_{z_1, z_2} \mathbb{E}_{p_c}\left[\mathbb{1}\{n - \nu \le |C_x| < 2n\} \operatorname{hit}_{z_1}(C_x) \operatorname{hit}_{z_2}(C_x)\right],$$

where, for a fixed set D, $\operatorname{hit}_z(D)$ is the probability that a chain of ℓ independent steps of Φ from z succeeds and that one of its clusters C_i meets D. To bound it, let every vertex z send unit mass to each vertex of $C_1 \cup \cdots \cup C_\ell$ when its chain succeeds. A successful chain has probability s^ℓ and its clusters have at most ν vertices in total, so each vertex sends expected

mass at most νs^ℓ, and by the mass-transport principle each vertex receives at most νs^ℓ. A union bound over the vertices of D gives

$$\sum_z \mathrm{hit}_z(D) \le \nu s^\ell |D| \,.$$

Since $\nu \le n/2$, class 3 therefore contributes at most

$$\frac{\nu^2 s^{2\ell}}{n^2}\,\mathbb{E}_{p_c}\big[|C_x|^2;\, n/2 \le |C_x| < 2n\big] \le 4\nu^2 s^{2\ell}\,\mathbb{P}_{p_c}(n/2 \le |C_o| < 2n) \le 4(1+2^\beta)\nu^2 \pi s^{2\ell}\,,$$

by (6.2). Adding the three classes gives (6.6) with $Q(\ell, M) := 8 + 2N_\nu + 10\nu^2$, since $4(1+2^\beta) \le 10$. □

6.5 One amplification step and the iteration

By Theorem 3.4 there is M_ε such that the bound of Theorem 3.4 holds for all $M \ge M_\varepsilon$.

Lemma 6.9 (One amplification step)**.** *Let Φ be a local endpoint map with cap $M \ge 2$, success probability s and kernel norm r. Let $\ell \in \mathbb{N}$ and $n \in \mathcal{N}$ satisfy*

$$\ell M \le n/2\,, \qquad n \ge (\Theta_M/s)^\ell\,, \qquad 2n \ge M_\varepsilon\,, \qquad 16\,Q(\ell,M)\,n^\beta \mathrm{e}^{-n^\varepsilon} \le 1\,. \tag{6.7}$$

Then there is a local endpoint map Φ' with cap $M' := 2n + \ell M$, success probability $s' \ge \pi_n/(8Q(\ell,M))$ and kernel norm $r' \le 2r^\ell$.

Proof. Put $b := 4Q(\ell,M)s^\ell$. By Theorems 6.7 and 6.8, $\mathbb{E}_{p_c}[L; L > b] \le \mathbb{E}_{p_c} L^2/b \le \pi s^\ell/4 \le \bar{L}/4$. The load vanishes unless $|C_x| < 2n + \nu \le M'$, so by Theorem 3.4, $M' \ge M_\varepsilon$ and $\pi_n \ge n^{-\beta}$,

$$\mathbb{E}_{p_c}\big[L;\, L \le b,\ C_x \text{ not } M'\text{-good}\big] \le b\,\mathrm{e}^{-(M')^\varepsilon} \le \bar{L}\cdot\frac{4Q(\ell,M)}{\pi_n}\,\mathrm{e}^{-n^\varepsilon} \le \frac{\bar{L}}{4}\,,$$

by the last condition in (6.7). Let $\mathcal{R}$ be the event that $L \le b$ and C_x is M'-good, and define

$$\Phi'_x(\omega, z) := \mathbb{1}_{\mathcal{R}}(\omega)\,\frac{L_x(z,\omega)}{b}\,.$$

By Theorem 6.6 this is a local endpoint map with cap M': on $\mathcal{R}$ the outputs have total probability $L/b \le 1$, every output lies in C_x, and $|C_x| < M'$ with C_x being M'-good. Its unconditional endpoint kernel $F'_{p_c}(x,z) = \mathbb{E}_{p_c}[L_x(z,\omega); \mathcal{R}]/b$ satisfies $0 \le bF'_{p_c}(x,z) \le \bar{L}(x,z)$, and its row sums are $s' = \mathbb{E}_{p_c}[L;\mathcal{R}]/b \ge \bar{L}/(2b) \ge \pi_n/(8Q(\ell,M))$. Hence $r' = \|F'_{p_c}\|/s' \le \|\bar{L}(\cdot,\cdot)\|/(\bar{L}/2) \le 2r^\ell$ by Theorem 4.1(a) and (6.5). □

The new map Φ' consists of fixed functions $\Phi'_x(\omega,\cdot)$ of the configuration. Like every local endpoint map, it can be evaluated on configurations of any parameter; this is how the maps are used in Section 7.

Proof of Theorem 3.6. We measure a map Φ with cap M and success probability s by

$$\mathrm{cost}(\Phi) := \log(1/s) + (\sigma+\xi)\log M\,,$$

where $\xi > 0$ is fixed so small that, by (6.1),

$$c_\xi := \frac{\beta + \xi + (\sigma+\xi)(1+\xi)}{1-\xi} < 1\,.$$

Fix the constant C in $\Theta_M \le CM^\sigma \log M$, and choose $M_1 \ge M_\varepsilon$ so large that $\log(2C\log M) \le \xi \log M$ for $M \ge M_1$. By Theorem 6.3 there is a map Φ_0 with cap $M \ge M_1$ and kernel norm $r < \mathrm{e}^{-1}$.

One step. Let Φ have cap $M \ge M_1$, success probability s and kernel norm $r < \mathrm{e}^{-1}$, and put $a := \lceil CM^\sigma \log M/s \rceil$. Then $a \ge \max\{2, \Theta_M/s\}$ and $\log a \le \log(1/s) + \sigma \log M + \log(2C\log M) \le \mathrm{cost}(\Phi)$. For $n \in \mathcal{N}$ put $\ell := \lfloor \log n/\log a \rfloor$, so that $a^\ell \le n < a^{\ell+1}$. As $n \to \infty$ through $\mathcal{N}$, we have $\ell \to \infty$ and $\ell \le \log n/\log 2$; hence all conditions in (6.7) hold for large n, using that $Q(\ell, M)$ is polynomial in ℓ and $n \ge a^\ell \ge (\Theta_M/s)^\ell$. Theorem 6.9 gives a map Φ' with cap M', success probability s' and kernel norm r'. By $\pi_n \ge n^{-\beta}$ and $\log n < (\ell+1)\,\mathrm{cost}(\Phi)$,

$$\begin{aligned} \log(1/s') &\le \beta \log n + \log(8Q(\ell, M)) \le \beta(\ell+1)\,\mathrm{cost}(\Phi) + O_M(\log(\ell+1))\,, \\ \log M' &= \log(2n + \ell M) \le \log n + 1 \le (\ell+1)\,\mathrm{cost}(\Phi) + 1\,, \\ \log(1/r') &\ge \ell \log(1/r) - \log 2\,. \end{aligned}$$

Since $\mathrm{cost}(\Phi) > 0$ and $\log(1/r) > 1$ are fixed, for n large enough this gives $\log(1/s') \le \ell(\beta+\xi)\,\mathrm{cost}(\Phi)$, $\log M' \le \ell(1+\xi)\,\mathrm{cost}(\Phi)$ and $\log(1/r') \ge \ell(1-\xi)\log(1/r) \ge 2\log(1/r)$, and therefore

$$\frac{\mathrm{cost}(\Phi')}{\log(1/r')} \le \frac{\ell\,\mathrm{cost}(\Phi)\,(\beta + \xi + (\sigma+\xi)(1+\xi))}{\ell(1-\xi)\log(1/r)} = c_\xi \frac{\mathrm{cost}(\Phi)}{\log(1/r)}\,.$$

Moreover Φ' has cap $M' \ge 2n \ge M_1$ and $r' < \mathrm{e}^{-2}$.

Iteration. Starting from Φ_0 and applying the step repeatedly, we obtain maps $\Phi_0, \Phi_1, \dots$ with kernel norms $r_j < \mathrm{e}^{-1}$ and

$$\frac{\mathrm{cost}(\Phi_j)}{\log(1/r_j)} \le c_\xi^j \frac{\mathrm{cost}(\Phi_0)}{\log(1/r_0)} \longrightarrow 0\,.$$

Since $\log(1/s) + \log M \le \max\{1, (\sigma+\xi)^{-1}\}\,\mathrm{cost}(\Phi)$ for every map, the efficiency ratio $\mathrm{eff}(\Phi_j)$ tends to zero. □

7 Critical moments and the near-critical susceptibility

In this section we prove Theorem 3.7, which completes the proof of Theorem 3.2, and then the susceptibility bound (3.3) of Theorem 3.8. We assume (A1) and (A2) throughout. For a domain Λ, $p < p_c$ and $v \in V$ we write

$$\chi^\Lambda(v) := \mathbb{E}_p |C_v^\Lambda|\,,$$

with $\chi^\Lambda(v) = 0$ for $v \notin \Lambda$; the parameter p will be clear from the context.

The strategy for Theorem 3.7 was described after its statement in Section 3.3. We run a fixed efficient map slightly below p_c and compare two bounds on the normalised susceptibility in the complement of an independent cluster A, evaluated at the endpoint of ℓ steps of the map. The upper bound (Section 7.1) says that one step can only multiply this quantity by a bounded factor, up to a boundary contribution controlled by $\Delta(p)$. The lower bound says that after $\ell \approx \log \chi(p)/\log(1/r)$ steps the endpoint has escaped from A. Together they bound $\Delta(p)$ from below, which by Russo's formula controls the growth of χ (Section 7.2).

7.1 Two comparison lemmas

For a domain Λ, $v \in \Lambda$ and $M \in \mathbb{N}_0$, let $B_{\mathrm{int}}^\Lambda(v, M)$ be the set of vertices joined to v by an open path in Λ of length at most M.

Lemma 7.1 (Maximal susceptibility in an intrinsic ball)**.** *For every domain Λ, $v \in \Lambda$, $M \in \mathbb{N}_0$ and $p \in (0, p_c)$,*

$$\mathbb{E}_p \max_{z \in B^\Lambda_{\mathrm{int}}(v,M)} \chi^\Lambda(z) \le \Big(1 + \frac{1-p}{p}\, M\Big) \chi^\Lambda(v)\,. \tag{7.1}$$

Proof. For $M = 0$ there is nothing to prove. Let first Λ be finite, and let $Y = Y(\omega)$ be a maximiser of χ^Λ over $B^\Lambda_{\mathrm{int}}(v, M)$, chosen by a fixed rule.

A resampling identity. For $w \in \Lambda$, let $W := C^\Lambda_w(\omega)$, and let $\tilde\omega$ be obtained from ω by resampling, independently, all edges with an endpoint in W. Given W, the other edges keep their product law, because the event $\{C^\Lambda_w = W\}$ depends only on the edges with an endpoint in W. Hence $\tilde\omega$ has law $\mathbb{P}_p$ and is independent of W, and $\mathbb{P}(Y(\tilde\omega) \in W) = \sum_z \mathbb{P}_p(Y = z)\, T^\Lambda_p(w, z)$. Summing over w and using the symmetry of T^Λ_p gives

$$\sum_{w \in \Lambda} \mathbb{P}(Y(\tilde\omega) \in C^\Lambda_w(\omega)) = \mathbb{E}_p\, \chi^\Lambda(Y) = \mathbb{E}_p \max_{z \in B^\Lambda_{\mathrm{int}}(v,M)} \chi^\Lambda(z)\,. \tag{7.2}$$

Entering the cluster of w. If $v \notin W$ and $Y(\tilde\omega) \in W$, take a self-avoiding $\tilde\omega$-open path in Λ of length at most M from v to $Y(\tilde\omega)$, and its first edge $\{b, a\}$ with $b \notin W$ and $a \in W$. This edge is closed in ω, since it belongs to the boundary of the ω-cluster W. The part of the path before it uses only edges without an endpoint in W, which are the same in ω and $\tilde\omega$, and it has length at most $M - 1$. Let $N_w(\omega)$ be the number of edges $\{b, a\}$ with $a \in W$, $b \in \Lambda \setminus W$, such that b is joined to v by an ω-open path in $\Lambda \setminus W$ of length at most $M - 1$; put $N_w := 0$ if $v \in W$. Then

$$\mathbb{P}\big(Y(\tilde\omega) \in W\big) \le T^\Lambda_p(v, w) + \mathbb{E}_p N_w\,.$$

Counting the entering edges. Let e be counted by $N_w(\omega)$, and let ω^e be ω with e opened. In ω^e, the edge e is the only open edge between W and its complement, so $v \xleftrightarrow{\Lambda} w$ and e is a bridge separating v from w. Every bridge separating v from w that comes before e lies on the path of length at most $M - 1$ from v to b. Hence e is among the first M bridges separating v from w in ω^e. Since ω is recovered from ω^e by closing e, each configuration ω' with $v \xleftrightarrow{\Lambda} w$ arises from at most M pairs (ω, e), and $\mathbb{P}_p(\omega) = ((1-p)/p)\, \mathbb{P}_p(\omega^e)$. Therefore $\mathbb{E}_p N_w \le ((1-p)/p)\, M\, T^\Lambda_p(v, w)$. Summing over w and using (7.2) proves (7.1) for finite Λ.

Infinite domains. Exhaust Λ by finite domains containing v. The susceptibilities increase to χ^Λ, the intrinsic balls increase to $B^\Lambda_{\mathrm{int}}(v, M) \subseteq B(v, M)$, and the maxima increase to the maximum. Monotone convergence concludes. □

Recall from Theorem 6.2 the unconditional endpoint kernel F_p of a map and its success probability s_p at parameter p, and put $K_p := F_p / s_p$.

Lemma 7.2 (A map in a deleted domain)**.** *Let Φ be a local endpoint map with cap $M \ge 2$, success probability $s > 0$ and kernel norm $r < 1$. There is $p_\Phi \in [p_c/2, p_c)$ such that $s_p \ge s/2$ and $\|K_p\| \le e r$ for $p \in [p_\Phi, p_c)$. Moreover, there are $Z \in [2, CM/s]$, with C depending only on G, and $C_\Phi < \infty$ such that for all $p \in [p_\Phi, p_c)$, $A \subseteq V$ and $v \in V$,*

$$(K_p F_A)(v) \le Z\, F_A(v) + C_\Phi\, g_A(v)\,, \tag{7.3}$$

where

$$F_A(v) := \frac{\chi^{V \setminus A}(v)}{\chi(p)} \qquad \text{and} \qquad g_A(v) := \sum_{\substack{a \in A \\ \mathrm{dist}(v,a) \le M}} \sum_{b \sim a} F_A(b)\,.$$

The function F_A is the susceptibility in the complement of A, normalised by the full susceptibility; it vanishes on A. The term $g_A(v)$ collects its values at the outer boundary of A near v.

Proof. By Theorem 6.2, $s_p \to s$ and $\|F_p - F_{p_c}\| \to 0$ as $p \to p_c$, so $\|K_p\| \to \|K\| = r$; this gives p_Φ. Fix $p \in [p_\Phi, p_c)$, A and v. Whenever the rule at v outputs y, the cluster C_v has at most M vertices, so a self-avoiding open path π of length at most M joins v to y; choose one by a fixed rule. We bound $\sum_y F_p(v,y)F_A(y)$ by splitting according to π.

If π avoids A, then $v \notin A$ and $y \in B_{\mathrm{int}}^{V\setminus A}(v,M)$, so by Theorem 7.1 this part is at most $\mathbb{E}_p \max_{z \in B_{\mathrm{int}}^{V\setminus A}(v,M)} F_A(z) \le (1+(1-p)M/p)\, F_A(v)$. If $y \in A$, then $F_A(y) = 0$. In the remaining case π meets A and $y \notin A$. Let $\{a,b\}$ be the last edge of π with $a \in A$ and $b \notin A$; then $\mathrm{dist}(v,a) \le M$, and the segment of π from b to y lies in $V \setminus A$ and has length $l \le M$. By the Harris inequality applied to the event that this segment is open,

$$\chi^{V\setminus A}(b) \ge \mathbb{E}_p\big[|C_y^{V\setminus A}|;\ \text{the segment is open}\big] \ge p^l \chi^{V\setminus A}(y)\,,$$

so $F_A(y) \le p^{-M} F_A(b)$, and this part is at most $p^{-M} g_A(v)$. Altogether $\sum_y F_p(v,y)F_A(y) \le (1+(1-p)M/p)F_A(v) + p^{-M} g_A(v)$. Dividing by $s_p \ge s/2$ and using $p \ge p_c/2$ gives (7.3) with $Z := \max\{2, (2/s)(1+2M/p_c)\}$ and $C_\Phi := (2/s)(2/p_c)^M$. □

7.2 Proof of Theorem 3.7

Proof of Theorem 3.7. Stage 1: choice of a map. Let $\gamma \in (0,1)$. Let C_Z be a constant, depending only on G, with $\log Z \le \log(1/s) + \log M + C_Z$ for the constant Z of Theorem 7.2. If a map Φ with $r < \mathrm{e}^{-1}$ satisfies $\mathrm{eff}(\Phi) \le (\log 2)/2$, then $\log(1/r) \ge (\log 2)/\mathrm{eff}(\Phi) \ge 2$, because $\log(1/s) + \log M \ge \log 2$, and

$$\frac{\log(1/s) + \log M + C_Z}{\log(1/r) - 1} = \frac{\mathrm{eff}(\Phi)\log(1/r) + C_Z}{\log(1/r) - 1} \le \Big(2 + \frac{2C_Z}{\log 2}\Big)\,\mathrm{eff}(\Phi)\,.$$

By hypothesis there are such maps with arbitrarily small efficiency ratio. We fix one, Φ, with cap $M \ge 2$, success probability s and $r < \mathrm{e}^{-1}$, for which the left-hand side is at most γ. Let p_Φ, Z and C_Φ be as in Theorem 7.2, and note that $\log Z/(\log(1/r) - 1) \le \gamma$.

Stage 2: an upper bound through the boundary. Fix $p \in [p_\Phi, p_c)$ and write $\chi = \chi(p)$. Let $A := C_o(\hat\omega)$ for an independent configuration $\hat\omega$ with law $\mathbb{P}_p$, and size-bias it: for functions G of a vertex set put $\widehat{\mathbb{E}}[\mathsf{G}(A)] := \mathbb{E}_p[|C_o|\,\mathsf{G}(C_o)]/\chi$. For each vertex a, exposing the cluster C_a leaves the edges without an endpoint in C_a fresh, so

$$\widehat{\mathbb{E}}\Big[\mathbb{1}\{a \in A\}\sum_{b\sim a} F_A(b)\Big] = \frac{1}{\chi^2}\sum_{b\sim a}\mathbb{E}_p\big[|C_a|\mathbb{1}\{o \in C_a\}\,|C_b|\mathbb{1}\{C_b \ne C_a\}\big] \le \Delta(p)\,,$$

where we used that $|C_b^{V\setminus C_a}| = |C_b|\mathbb{1}\{C_b \ne C_a\}$ and, by transitivity, that $\sum_{b\sim a}\mathbb{E}_p[|C_a||C_b|; C_a \ne C_b] = \chi^2\Delta(p)$ for every a. Hence $\widehat{\mathbb{E}}\, g_A(v) \le |B(o,M)|\,\Delta(p)$ for every v. Iterating (7.3), using that K_p has nonnegative entries,

$$K_p^\ell F_A \le Z^\ell F_A + C_\Phi \sum_{i=0}^{\ell-1} Z^{\ell-1-i} K_p^i g_A\,.$$

Evaluate at $o \in A$, where $F_A(o) = 0$, and take $\widehat{\mathbb{E}}$; the kernel K_p is deterministic and stochastic, and $Z \ge 2$. This gives

$$\widehat{\mathbb{E}}\,(K_p^\ell F_A)(o) \le C_\Phi |B(o,M)|\,\Delta(p)\sum_{i=0}^{\ell-1} Z^{\ell-1-i} \le 2C_\Phi |B(o,M)|\,Z^{\ell-1}\Delta(p)\,. \tag{7.4}$$

Stage 3: a lower bound from spreading. Exposing $A = C_o$ as before,

$$\widehat{\mathbb{E}}\,(K_p^\ell F_A)(o) = \frac{1}{\chi^2}\sum_v K_p^\ell(o,v)\,\mathbb{E}_p\big[|C_o||C_v|;\ C_o \ne C_v\big]\,.$$

Without the restriction $C_o \neq C_v$, the Harris inequality gives $\mathbb{E}_p[|C_o||C_v|] \geq \chi^2$, and K_p^ℓ is stochastic, so the sum would be at least one. The removed part equals, by the mass-transport principle applied to the mass $|C_z| \sum_{v \in C_z} K_p^\ell(z,v)$ sent from z to every vertex of C_z,

$$\sum_v K_p^\ell(o,v)\, \mathbb{E}_p\big[|C_o|^2;\, v \in C_o\big] = \mathbb{E}_p\big[|C_o|\, \langle \mathbb{1}_{C_o}, K_p^\ell \mathbb{1}_{C_o} \rangle\big] \leq \|K_p\|^\ell\, \mathbb{E}_p|C_o|^2 \leq (\mathrm{er})^\ell \chi^3\,,$$

by Theorem 4.5(c). Choose $\ell := \lceil \log(2\chi)/(\log(1/r)-1) \rceil$, so that $(\mathrm{er})^\ell \chi = \mathrm{e}^{-(\log(1/r)-1)\ell}\chi \leq 1/2$. Then $\widehat{\mathbb{E}}\,(K_p^\ell F_A)(o) \geq 1/2$.

Stage 4: growth of the susceptibility. Since $\ell - 1 \leq \log(2\chi)/(\log(1/r) - 1)$ and $\log Z \leq \gamma(\log(1/r) - 1)$, we have $Z^{\ell-1} \leq (2\chi)^\gamma$. Comparing with (7.4), we obtain $\Delta(p) \geq c_\Phi \chi^{-\gamma}$ for all $p \in [p_\Phi, p_c)$, with $c_\Phi > 0$ depending on the fixed map. By Theorem 4.7, $(1-p)\chi' = \chi^2 \Delta(p) \geq c_\Phi \chi^{2-\gamma}$ for almost every such p. Hence the locally absolutely continuous function $\chi^{-(1-\gamma)}$ has derivative at most $-(1-\gamma)c_\Phi$ almost everywhere on $[p_\Phi, p_c)$. Integrating from p to $p' < p_c$ and letting $p' \uparrow p_c$ gives $\chi(p)^{-(1-\gamma)} \geq (1-\gamma)c_\Phi(p_c - p)$. Since χ is bounded on $[0, p_\Phi]$, we conclude that for every $\gamma \in (0,1)$ there is C_γ with

$$\chi(p) \leq C_\gamma\, (p_c - p)^{-1/(1-\gamma)} \qquad \text{for all } p \in [0, p_c)\,. \tag{7.5}$$

Stage 5: the critical tail. Let $N \in \mathbb{N}$ be large and $p := p_c - N^{-1/2}$. Explore C_o in a fixed order, testing each edge at most once, until the cluster is exhausted or N vertices have been found. This uses at most dN tests, and the record of the exploration decides the event $\{|C_o| \geq N\}$. Let $\mathcal{L}_N$ be the likelihood ratio of the record under $\mathbb{P}_{p_c}$ with respect to $\mathbb{P}_p$; each test contributes a factor p_c/p if the edge is open and $(1-p_c)/(1-p)$ if it is closed. For $b > 1$, the conditional b-th moment of one such factor under $\mathbb{P}_p$ is

$$\varphi_b(p) := p\Big(\frac{p_c}{p}\Big)^b + (1-p)\Big(\frac{1-p_c}{1-p}\Big)^b \leq 1 + C_b(p_c - p)^2\,,$$

because φ_b is smooth near the interior point p_c, with $\varphi_b(p_c) = 1$ and $\varphi_b'(p_c) = 0$. Conditioning successively on the at most dN tests gives $\mathbb{E}_p \mathcal{L}_N^b \leq (1 + C_b/N)^{dN} \leq \mathrm{e}^{dC_b}$. By Hölder's and Markov's inequalities,

$$\mathbb{P}_{p_c}(|C_o| \geq N) = \mathbb{E}_p[\mathcal{L}_N \mathbb{1}\{|C_o| \geq N\}] \leq \mathrm{e}^{dC_b/b} \Big(\frac{\chi(p)}{N}\Big)^{1-1/b}\,.$$

By (7.5), $\chi(p) \leq C_\gamma N^{1/(2(1-\gamma))}$, so $\mathbb{P}_{p_c}(|C_o| \geq N) \leq C_{\gamma,b} N^{-\alpha'}$ with $\alpha' := (1-1/b)(1-1/(2(1-\gamma)))$. Given $\alpha \in (0, 1/2)$, choose γ small and b large such that $\alpha' > \alpha$. Then

$$\mathbb{E}_{p_c}|C_o|^\alpha = \sum_{N \geq 0} ((N+1)^\alpha - N^\alpha)\mathbb{P}_{p_c}(|C_o| > N) \leq 1 + C \sum_{N \geq 1} N^{\alpha - 1 - \alpha'} < \infty\,. \qquad \square$$

7.3 The near-critical susceptibility

We now prove the susceptibility bound (3.3). Its source is the following covariance inequality for functions determined by an algorithm, due to O'Donnell, Saks, Schramm and Servedio [OSSS05, Section 3.3, Equation (6)]; we include a proof in Section A.2.

Lemma 7.3 (Decision-tree covariance inequality)**.** *Let $X = (X_i)_{i \in I}$ be a finite family of independent random variables, each taking finitely many values, let $g = g(X)$ take values in $[0,1]$, and let an algorithm, possibly using independent randomness, determine $g(X)$ by querying the coordinates of X one at a time, never the same coordinate twice. Let* rev_i*, the* revealment *of i, be the probability that coordinate i is queried, and let $X^{(i)}$ be X with X_i replaced by an independent copy. Then, for every real function f,*

$$|\mathrm{Cov}(f(X), g(X))| \leq \sum_{i \in I} \mathrm{rev}_i\; \mathbb{E}|f(X) - f(X^{(i)})|\,.$$

Proof of (3.3). Fix $\alpha \in (0,1/2)$ and put $C_\alpha := \mathbb{E}_{p_c}|C_o|^\alpha$, which is finite by Theorem 3.2. Let Λ be a finite domain, $v \in \Lambda$, $n \geq 2$ and $p \in (0,p_c]$. Let $f := \mathbb{1}\{|C_v^\Lambda| \geq n\}$ and $\Psi(p) := \mathbb{P}_p(|C_v^\Lambda| \geq n)$. Independently, mark every vertex of Λ with probability $1/n$, and let g be the indicator that C_v^Λ contains a mark. An algorithm that reveals all marks and then explores the clusters in Λ of all marked vertices determines g. It queries an edge only if one of its endpoints lies in the cluster of a marked vertex, which for a given vertex x has probability

$$\mathbb{E}_p[1-(1-1/n)^{|C_x^\Lambda|}] \leq \mathbb{E}_p \min\{1, |C_x^\Lambda|/n\} \leq n^{-\alpha}\mathbb{E}_p|C_x^\Lambda|^\alpha \leq C_\alpha n^{-\alpha}\,,$$

by monotonicity in p and in the domain. Thus every edge has revealment at most $2C_\alpha n^{-\alpha}$. The marks do not influence f, and for an edge e the increasing indicator f satisfies $\mathbb{E}|f(X)-f(X^{(e)})| = 2p(1-p)\mathbb{P}_p(e \text{ is pivotal for } f)$. By Theorem 7.3 and Russo's formula $\Psi'(p) = \sum_e \mathbb{P}_p(e \text{ is pivotal for } f)$ on the finite domain,

$$\operatorname{Cov}(f,g) \leq 4p(1-p)\,C_\alpha n^{-\alpha}\Psi'(p) \leq C_\alpha n^{-\alpha}\Psi'(p)\,.$$

On the other hand, conditionally on the configuration, $g=1$ with probability at least $1-(1-1/n)^n$ on $\{f=1\}$, and $\mathbb{E}g \leq C_\alpha n^{-\alpha}$, so $\operatorname{Cov}(f,g) \geq \Psi(p)(1-(1-1/n)^n - C_\alpha n^{-\alpha}) \geq \Psi(p)/2$ for $n \geq n_\alpha$. Hence $\Psi'(p) \geq c_\alpha n^\alpha \Psi(p)$ on $(0,p_c]$, and integrating from p to p_c,

$$\mathbb{P}_p(|C_v^\Lambda| \geq n) \leq \Psi(p_c)\,\mathrm{e}^{-c_\alpha(p_c-p)n^\alpha} \leq C_\alpha n^{-\alpha}\mathrm{e}^{-c_\alpha(p_c-p)n^\alpha}\,,$$

where $\Psi(p_c) \leq \mathbb{P}_{p_c}(|C_o| \geq n) \leq C_\alpha n^{-\alpha}$. Letting $\Lambda \uparrow V$, which is legitimate because $\{|C_v| \geq n\}$ has a finite witness, we obtain for $\delta = p_c - p$ and all $n \geq 1$

$$\mathbb{P}_p(|C_o| \geq n) \leq C'_\alpha n^{-\alpha}\mathrm{e}^{-c_\alpha \delta n^\alpha}\,, \qquad C'_\alpha := \max\{C_\alpha,\, n_\alpha^\alpha \mathrm{e}^{c_\alpha n_\alpha^\alpha}\}\,,$$

where the second term of the maximum covers the finitely many $n < n_\alpha$. The function $y \mapsto y^{-\alpha}\mathrm{e}^{-c_\alpha \delta y^\alpha}$ is decreasing on $(0,\infty)$, and the substitution $y = (c_\alpha\delta)^{-1/\alpha} z$ gives

$$\chi(p) = \sum_{n\geq 1} \mathbb{P}_p(|C_o| \geq n) \leq C'_\alpha \int_0^\infty y^{-\alpha}\mathrm{e}^{-c_\alpha\delta y^\alpha}\,\mathrm{d}y = C''_\alpha\,\delta^{-(1-\alpha)/\alpha}$$

with $C''_\alpha := C'_\alpha c_\alpha^{-(1-\alpha)/\alpha}\int_0^\infty z^{-\alpha}\mathrm{e}^{-z^\alpha}\,\mathrm{d}z$, which is finite because $\alpha<1$. For $\alpha = 1/(2+\eta)$ we have $(1-\alpha)/\alpha = 1+\eta$, which proves (3.3). □

8 The triangle diagram

In this section we prove the triangle bound (3.4) of Theorem 3.8. We assume (A1) and (A2), fix $p \in [p_c/2, p_c)$ and write $T = T_p$ and $\delta = p_c - p$. By Theorem 4.6 the operator T is bounded and positive semidefinite, and $\nabla_p = \operatorname{tr}(T^3) \geq T(o,o)^3 = 1$.

By Russo's formula, the derivative of ∇_p is a sum over pairs of distinct clusters, weighted by the number of edges between them and by a kernel built from T. We split this sum according to the size of the smaller cluster. If it is small, the weight is bounded by its size times ∇_p. If both clusters are large, we use that two distinct large clusters have few edges between them: comparing an exploration of one cluster with an exploration of the same cluster after deleting the other, the difference of their centred scores is exactly minus p times the number of these (closed) edges, and both scores are concentrated (Theorem 8.1). Together with the a priori bound on $\chi(p)$ this yields a differential inequality for ∇_p whose solution grows more slowly than every power of $1/\delta$.

Sums over pairs of clusters. For a nonnegative function F of ordered pairs of distinct finite vertex sets, invariant under automorphisms, put

$$\mathcal{S}(F) := \mathbb{E}_p\Big[\frac{1}{|C_o|}\sum_{B\neq C_o} F(C_o,B)\Big]\,,$$

where the sum runs over the clusters B other than C_o. This is a positive linear functional, so it satisfies the Cauchy–Schwarz inequality $\mathcal{S}(F_1F_2)\le \mathcal{S}(F_1^2)^{1/2}\mathcal{S}(F_2^2)^{1/2}$. For a nonnegative invariant kernel J, recall $\mathsf{J}(A,B)=\sum_{a\in A,b\in B}\mathsf{J}(a,b)$. The mass-transport principle gives the *rerooting formula*

$$\mathcal{S}\big(\mathsf{J}(A,B)\,F(A,B)\big) = \mathbb{E}_p\Big[\sum_v \mathsf{J}(o,v)\,F(C_o,C_v)\,\mathbb{1}\{C_o\neq C_v\}\Big]\ : \tag{8.1}$$

send from x to each $y\in C_x$ the mass $|C_x|^{-1}\sum_v \mathsf{J}(y,v)F(C_x,C_v)\mathbb{1}\{C_x\neq C_v\}$; the expected mass sent out by o is the left-hand side, and the expected mass received is the right-hand side. Sending $F(A,B)/(|A||B|)$ from every vertex of A to every vertex of B shows in the same way that $\mathcal{S}(F)=\mathcal{S}(F^\top)$, where $F^\top(A,B):=F(B,A)$.

Lemma 8.1 (Edges between two large clusters)**.** *Let $\alpha\in(0,1/2)$ and $\varepsilon_1>0$ with $\kappa := 1/2+\varepsilon_1-2\alpha<0$. There is $C<\infty$ such that for all $p\in[p_c/2,p_c)$ and $n_0\ge 1$,*

$$\mathcal{S}\big(\mathrm{Adj}(A,B)^2\,\mathbb{1}\{\min(|A|,|B|)\ge n_0\}\big)\le C\,n_0^{\kappa}\,. \tag{8.2}$$

Proof. Fix an edge $\{x,v\}$ and a real number $n\ge 1$, and let $\mathcal{H}_n := \{C_x\neq C_v,\ n\le \min(|C_x|,|C_v|)<2n\}$.

Stage 1: probability of two large clusters. On $\mathcal{H}_n$ the events $\{|C_x|\ge n\}$ and $\{|C_v|\ge n\}$ have disjoint finite witnesses, namely trees in the two distinct clusters. By the van den Berg–Kesten inequality, monotonicity in p and Markov's inequality,

$$\mathbb{P}_p(\mathcal{H}_n)\le \mathbb{P}_{p_c}(|C_o|\ge n)^2\le Cn^{-2\alpha}\,,$$

using Theorem 3.2.

Stage 2: two explorations. Explore C_v in the full graph, testing every edge incident to a discovered vertex exactly once (including edges between two discovered vertices), and stop after $N:=\lceil 2dn\rceil$ tests if the exploration has not finished. Let S_1 be the sum of $\omega(e)-p$ over the tested edges. Every test is a fresh Bernoulli variable given the past, so by the Azuma–Hoeffding inequality $\mathbb{P}_p(|S_1|>n^{1/2+\varepsilon_1})\le 2\exp(-2n^{1+2\varepsilon_1}/N)$, without any conditioning on the sizes. Second, on the same configuration, expose $A:=C_x$ together with the states of its internal and boundary edges; if $v\notin A$, explore the cluster of v in $V\setminus A$ in the same way, with the same cap N, and let S_2 be its score ($S_2:=0$ if $v\in A$). Given the exposure, the tested edges are fresh, so the same tail bound holds for S_2.

On $\mathcal{H}_n\cap\{|C_v|\le|C_x|\}$, the cluster $B:=C_v$ has fewer than $2n$ vertices and fewer than N incident edges, so both explorations finish. The first tests all edges incident to B, the second those that have no endpoint in A. The difference consists of the edges between A and B, all of which are closed, so

$$S_1-S_2=-p\ \mathrm{Adj}(A,B)\,.$$

Hence, on this event, $\mathrm{Adj}(A,B)>2n^{1/2+\varepsilon_1}/p$ forces $|S_1|>n^{1/2+\varepsilon_1}$ or $|S_2|>n^{1/2+\varepsilon_1}$. Using $p\ge p_c/2$ and $N\le 3dn$, and exchanging the roles of x and v for the other size order, we obtain

$$\mathbb{P}_p(\mathcal{H}_n,\ \mathrm{Adj}(C_x,C_v)>4n^{1/2+\varepsilon_1}/p_c)\le C\mathrm{e}^{-cn^{2\varepsilon_1}}\,.$$

The two scores are concentrated separately; no independence between them is used.

Stage 3: summation. Since $\mathrm{Adj}(C_x, C_v) \le d\min(|C_x|, |C_v|) < 2dn$ on $\mathcal{H}_n$, Stages 1 and 2 give $\mathbb{E}_p[\mathrm{Adj}(C_x, C_v)\mathbb{1}_{\mathcal{H}_n}] \le Cn^{1/2+\varepsilon_1-2\alpha} + Cne^{-cn^{2\varepsilon_1}} \le Cn^{\kappa}$. By the rerooting formula (8.1) with $\mathsf{J} = \mathrm{Adj}$, the left-hand side of (8.2) equals $\mathbb{E}_p[\sum_{v\sim o} \mathrm{Adj}(C_o, C_v)\mathbb{1}\{\min(|C_o|, |C_v|) \ge n_0,\ C_o \ne C_v\}]$. Splitting the minimum into the ranges $[2^i n_0, 2^{i+1} n_0)$, $i \ge 0$, bounds it by $d\sum_{i\ge 0} C(2^i n_0)^{\kappa} \le Cn_0^{\kappa}$, because $\kappa < 0$. □

Proof of (3.4). Stage 1: the derivative. Put $\mathsf{J} := T^2$, a bounded, invariant, symmetric kernel. By Theorem 4.7 the map $p \mapsto T_p$ is locally absolutely continuous in operator norm; the trace is cyclic by Theorem 4.3(a) and satisfies $|\mathrm{tr}\, S| \le \|S\|$ (Section 4.2). Hence $\nabla_p = \mathrm{tr}(T^3)$ is locally absolutely continuous on $(0, p_c)$, with $\nabla_p' = 3\,\mathrm{tr}(T^2 T')$ almost everywhere. By (4.1) and the rerooting formula with $F = \mathrm{Adj}$,

$$(1-p)\nabla_p' = 3\sum_v \mathsf{J}(o,v)\,\mathbb{E}_p[\mathrm{Adj}(C_v, C_o)\mathbb{1}\{C_v \ne C_o\}] = 3\,\mathcal{S}\big(\mathsf{J}(A,B)\,\mathrm{Adj}(A,B)\big)\,. \tag{8.3}$$

Stage 2: a square bound. By (8.1), expanding the second factor $\mathsf{J}(C_o, C_v)$,

$$\mathcal{S}\big(\mathsf{J}(A,B)^2\big) = \sum_{v,x',y'} \mathsf{J}(o,v)\,\mathsf{J}(x',y')\,\mathbb{P}_p\big(o \leftrightarrow x',\ v \leftrightarrow y',\ C_o \ne C_v\big)\,.$$

The two connections occur disjointly, so the van den Berg–Kesten inequality bounds the probability by $T(o,x')T(v,y')$. By symmetry of J and T, the resulting sum is $\mathrm{tr}(T\mathsf{J}T\mathsf{J}) = \mathrm{tr}(T^6)$, and Theorem 4.3(b) with the positive operator T^3 gives

$$\mathcal{S}\big(\mathsf{J}(A,B)^2\big) \le \mathrm{tr}(T^6) \le \|T\|^3\,\nabla_p\,. \tag{8.4}$$

Stage 3: splitting by size. Fix $\alpha \in (0, 1/2)$, $\varepsilon_1 > 0$ with $\kappa < 0$ as in Theorem 8.1, and a size $n_0 \ge 1$. For the pairs in which $A = C_o$ has fewer than n_0 vertices, reroot the factor Adj with (8.1) and expose A. If $v \notin A$, the cluster C_v is the cluster of v in $V \setminus A$, which is fresh, so

$$\mathbb{E}_p\big[\mathsf{J}(A, C_v)\,|\,A\big] \le \sum_{y\in A}\sum_z \mathsf{J}(y,z)\,T(v,z) = \sum_{y\in A} T^3(y,v) \le |A|\,\nabla_p\,,$$

by Theorem 4.3(c) for the positive operator T^3. Hence, using $|A|\mathbb{1}\{|A| < n_0\} \le n_0^{1-\alpha}|A|^{\alpha}$ and Theorem 3.2,

$$\mathcal{S}\big(\mathsf{J}(A,B)\,\mathrm{Adj}(A,B)\mathbb{1}\{|A| < n_0\}\big) \le d\,\nabla_p\,\mathbb{E}_p\big[|C_o|; |C_o| < n_0\big] \le C\,n_0^{1-\alpha}\,\nabla_p\,.$$

Since $\mathsf{J}(A,B)\,\mathrm{Adj}(A,B)$ is symmetric in A and B and $\mathcal{S}(F) = \mathcal{S}(F^{\top})$, the same bound holds for the pairs with $|B| < n_0$. For the remaining pairs, the Cauchy–Schwarz inequality, (8.2) and (8.4) give

$$\mathcal{S}\big(\mathsf{J}(A,B)\,\mathrm{Adj}(A,B)\mathbb{1}\{\min(|A|,|B|) \ge n_0\}\big) \le C\,n_0^{\kappa/2}\,\|T\|^{3/2}\,\nabla_p^{1/2}\,.$$

Stage 4: the differential inequality. The factor $(1-p)^{-1}$ in (8.3) is bounded. Since $\nabla_p \ge 1$, the function $\mathsf{b} := \nabla_p^{1/2}$ is locally absolutely continuous, and by Stages 1 and 3, for almost every $p \in [p_c/2, p_c)$,

$$\mathsf{b}' \le C\,n_0^{1-\alpha}\,\mathsf{b} + C\,n_0^{\kappa/2}\|T\|^{3/2}\,. \tag{8.5}$$

Stage 5: choice of exponents. Let $\eta > 0$, and fix $\varepsilon_2 \in (0, \min\{1, \eta\})$. Choose $\alpha < 1/2$ and $\varepsilon_1, \eta' > 0$ such that $\kappa < 0$ and

$$A_\star := (2-\varepsilon_2)(1-\alpha) < 1\,, \qquad B_\star := (2-\varepsilon_2)\big(\tfrac14 + \tfrac{\varepsilon_1}{2} - \alpha\big) + \tfrac32(1+\eta') < 1 + \tfrac{\eta}{2}\,.$$

This is possible by continuity, because at $\alpha = 1/2$ and $\varepsilon_1 = \eta' = 0$ the two expressions equal $1 - \varepsilon_2/2$ and $1 + \varepsilon_2/4$. Take $n_0 := \delta^{-(2-\varepsilon_2)} \geq 1$. By (3.3), $\|T\| \leq \chi(p) \leq C\delta^{-1-\eta'}$, and since $\kappa/2 = 1/4 + \varepsilon_1/2 - \alpha$, inequality (8.5) becomes

$$\mathsf{b}'(p) \leq C\,(p_c - p)^{-A_\star}\,\mathsf{b}(p) + C\,(p_c - p)^{-B_\star}\,.$$

Since $A_\star < 1$, the coefficient of b is integrable on $[p_c/2, p_c)$, and the integrating factor is bounded above and below. Hence

$$\mathsf{b}(p) \leq C\Big(\mathsf{b}(p_c/2) + \int_{p_c/2}^{p} (p_c - p')^{-B_\star}\,\mathrm{d}p'\Big) \leq C_\eta\,\delta^{-\eta/2}\,,$$

where the last step uses $B_\star < 1 + \eta/2$. Squaring gives $\nabla_p \leq C_\eta \delta^{-\eta}$. □

9 The corridor deformation

In this section we prove Theorem 3.10. We assume (A1) and (A2) and fix $p \in [p_c/2, p_c)$, $q = 1 - p$ and $k \geq 3$; only the deformation time $t \geq 0$ varies. We use the notation of Theorem 3.9 and write $T := T_p$ and $\chi := \chi(p)$. All constants depend only on G; in particular they are uniform in $p \in [p_c/2, p_c)$ and in k.

The section follows the order of Theorem 3.10. After some bookkeeping (Theorem 9.1) we identify the corridor event and its probability (Theorem 9.2), which gives the norm bound (b) at once. The rate bound (c) needs three ingredients: the deletion estimate (d), a lower bound for the mass of two distinct clusters (Theorem 9.3), and the factorisation of the penalty of a connection through a corridor into the penalties of its two legs (Theorem 9.4). Item (e) is proved last.

9.1 Bookkeeping

Lemma 9.1 (Basic bounds and monotonicity)**.** *Let* $D_1 := (2/p_c)(1-p_c)^{-d}$, $C_0 := \max\{\log D_1 + 1,\, 1 + 2d/p_c\}$ *and* $c_k := \mathrm{e}^{C_0 k}$. *Then the following holds.*

(i) *For every edge* e, $\sum_w c(w) \leq c_k/k$, *where the sum runs over all templates* w, *of any base, such that* e *is an edge of* γ_w.

(ii) *If* $x \overset{\Lambda}{\longleftrightarrow} y$, *then* $N_{xy}^{\Lambda} \leq c_k |C_x^{\Lambda}|$.

(iii) *For fixed* t, $W_t^{\Lambda}(x,y)$ *does not decrease when edges are opened or when* Λ *is enlarged.*

Moreover, Theorem 3.10(a) *holds, and for every domain* Λ, *all* $t \geq 0$ *and all* $x, y \in V$

$$-\frac{\mathrm{d}}{\mathrm{d}t}\chi_t^{\Lambda}(x) = \mathbb{E}_p \sum_y N_{xy}^{\Lambda} W_t^{\Lambda}(x,y)\,, \qquad -\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}_p W_t^{\Lambda}(x,y) = \mathbb{E}_p\big[N_{xy}^{\Lambda} W_t^{\Lambda}(x,y)\big]\,.$$

Proof. (i): A template has $|\gamma_w| \leq k$ and $|\mathrm{Bl}_w| \leq dk$, so $p^{-|\gamma_w|} q^{-|\mathrm{Bl}_w|} \leq D_1^k$, because $p \geq p_c/2$ and $q \geq 1 - p_c$. Every edge of γ_w is traversed by w. Summed over all bases, the probability that a lazy walk traverses $e = \{a, b\}$ at its i-th step is $\sum_u (P^{i-1}(u,a)P(a,b) + P^{i-1}(u,b)P(b,a)) = 1/d$, by symmetry of P. A union bound over the k steps gives $\sum_w c(w) \leq D_1^k k/d$, which is at most $D_1^k \mathrm{e}^k/k \leq \mathrm{e}^{C_0 k}/k$, since $k^2 \leq \mathrm{e}^k$.

(ii): An edge that is pivotal for $x \overset{\Lambda}{\longleftrightarrow} y$ lies on every self-avoiding open path from x to y in Λ. Fix one such path; it has fewer than $|C_x^{\Lambda}|$ edges, and every counted template contains one of them. Apply (i).

(iii): Let the configuration or the domain grow, and suppose x and y were already connected before; otherwise there is nothing to prove. Every self-avoiding open path from x to y before

the change is still one afterwards. Hence an edge that is pivotal afterwards lies on all old paths, so it was open and pivotal before, and the loop-erased path of a template counted afterwards was contained in the old domain. A blocker that is closed or outside the domain afterwards was so before. Thus every template counted afterwards was counted before, with the same weight, and N_{xy}^{Λ} cannot increase.

For Theorem 3.10(a): the counting condition is symmetric in x and y, and all definitions are covariant, so T_t is symmetric and invariant; $0 \le W_t^V \le \mathbb{1}\{x \leftrightarrow y\}$ gives $0 \le T_t \le T$ and $\chi_t \le \chi$; and the Schur test gives $\|T_t\| \le \chi_t$, the row sum. By (ii), $\sum_y N_{xy}^{\Lambda} W_t^{\Lambda}(x,y) \le c_k |C_x|^2$, whose expectation is at most $c_k \chi^3$ by Theorem 4.5(c); for a single entry the bound $N_{xy}^{\Lambda} W_t^{\Lambda}(x,y) \le c_k |C_x^{\Lambda}|$ suffices. Dominated convergence therefore justifies differentiating $W_t^{\Lambda}(x,y) = \mathbb{1}\{x \overset{\Lambda}{\longleftrightarrow} y\} \mathrm{e}^{-tN_{xy}^{\Lambda}}$ under the expectation and the sum, and shows that the derivatives are continuous in t. □

9.2 The corridor event

Recall that for a template w with base $u = w_0$ and end $v = w_k$, the interior I_w consists of the vertices of γ_w other than u and v, and $\Lambda_w = V \setminus I_w$.

Lemma 9.2 (The corridor event)**.** *Let w be a template with base u and end v.*

(a) *The event Cor_w depends only on the edges incident to I_w; in particular it is independent of the states of the edges with both endpoints in Λ_w.*

(b) $c(w)\,\mathbb{P}_p(\mathrm{Cor}_w) = \mu(w)$.

(c) *On Cor_w, the template w is counted for (x,y) in V if and only if $C_u^{\Lambda_w} \ne C_v^{\Lambda_w}$ and either $x \in C_u^{\Lambda_w}$ and $y \in C_v^{\Lambda_w}$, or $x \in C_v^{\Lambda_w}$ and $y \in C_u^{\Lambda_w}$.*

Proof. Every edge of γ_w has an endpoint in I_w because $|\gamma_w| \ge 3$, and the blockers are the other edges incident to I_w; this gives (a). The event Cor_w prescribes $|\gamma_w|$ open and $|\mathrm{Bl}_w|$ closed edges, which gives (b).

For (c), note that on Cor_w every interior vertex has exactly two open edges, the corridor edges (Figure 2). Suppose w is counted for (x,y). If $x \in I_w$, a self-avoiding open path starting at x uses only one of the two corridor edges at x, so the other one is not pivotal; hence $x \notin I_w$, and likewise $y \notin I_w$. A self-avoiding open path from x to y uses all corridor edges, so it runs through the whole corridor, from u to v or from v to u, and its two remaining parts lie in Λ_w. If $C_u^{\Lambda_w} = C_v^{\Lambda_w}$, an open path from u to v in Λ_w would bypass the corridor, so its edges would not be pivotal. This proves one direction. Conversely, if the two exterior clusters are distinct and contain x and y as stated, then the corridor is the only open link between them, so all its edges are open and pivotal for $x \leftrightarrow y$, and all blockers are closed. □

Proof of Theorem 3.10(b). By Theorem 9.1 with $\Lambda = V$, $-\partial_t T_t(x,y) = \mathbb{E}_p[N_{xy}^V W_t^V(x,y)] \le \mathbb{E}_p N_{xy}^V = \sum_w c(w)\,\mathbb{P}_p(w \text{ is counted for } (x,y))$. By Theorem 9.2, a counted template requires Cor_w and one of two exterior events, which are independent of Cor_w. In each exterior event the connections from x and from y lie in distinct clusters, so they occur disjointly, and the van den Berg–Kesten inequality and $T^{\Lambda_w} \le T$ bound its probability by $T(x,u)T(v,y)$, respectively $T(x,v)T(u,y)$. With $c(w)\mathbb{P}_p(\mathrm{Cor}_w) = \mu(w)$, adding the raw paths that are not templates, and summing over raw paths with base u and end v, which gives $P^k(u,v)$, we obtain

$$-\partial_t T_t(x,y) \le \sum_{u,v} P^k(u,v)\big(T(x,u)T(v,y) + T(x,v)T(u,y)\big) = 2\,(TP^kT)(x,y)\,,$$

by symmetry of P^k. Integrating, $0 \le T - T_t \le 2t\,TP^kT$ entrywise. By Theorem 4.1(a), $\|T - T_t\| \le 2t\|TP^kT\| \le 2t\rho^k\|T\|^2$, and the triangle inequality gives (b). No positivity of T_t is used. □

9.3 Deletion and the rate bound

Proof of Theorem 3.10(d). Fix a configuration and y, and suppose that $W_t^{\Lambda}(z,y) > W_t^{\Lambda\setminus A}(z,y)$; recall that $W_t^{\Lambda\setminus A} \le W_t^{\Lambda}$ by Theorem 9.1(iii). We claim that some self-avoiding open path from z to y in Λ visits $\mathrm{N}[A]$. If $z \xleftrightarrow{\Lambda\setminus A} y$ fails, every open path from z to y in Λ visits A. Otherwise some template w is counted for (z,y) in $\Lambda\setminus A$ but not in Λ. Its loop-erased path lies in $\Lambda\setminus A$ and is open, so either an edge e of γ_w is not pivotal in Λ, or a blocker f is open with both endpoints in Λ. In the first case an open path from z to y in Λ avoids e; it must visit A, since e is pivotal in $\Lambda\setminus A$. In the second case f has an endpoint in A, because it is not closed and must leave $\Lambda\setminus A$; its other endpoint is an interior vertex $i \notin A$ of γ_w, so $i \in \mathrm{N}[A]$, and every open path from z to y in $\Lambda\setminus A$ uses all edges of γ_w and hence visits i. This proves the claim; it covers the case of a terminal in A.

The loss $W_t^{\Lambda}(z,y) - W_t^{\Lambda\setminus A}(z,y)$ is at most one. Splitting a path that visits $b \in \mathrm{N}[A]$ at b gives two edge-disjoint open paths, so by the van den Berg–Kesten inequality

$$\mathbb{E}_p\bigl[W_t^{\Lambda}(z,y) - W_t^{\Lambda\setminus A}(z,y)\bigr] \le \sum_{b\in\mathrm{N}[A]} T(z,b)\,T(b,y)\,.$$

Summing over y proves (d). A single path event bounds the whole loss; no sum over penalty weights appears. □

Lemma 9.3 (Two distinct clusters)**.** *For every domain Λ and $u,v \in \Lambda$, with $X_t^{\Lambda}(x) \coloneqq \sum_y W_t^{\Lambda}(x,y)$,*

$$\mathbb{E}_p\bigl[X_t^{\Lambda}(u)\,X_t^{\Lambda}(v);\, C_u^{\Lambda} \ne C_v^{\Lambda}\bigr] \ge \chi_t^{\Lambda}(u)\,\chi_t^{\Lambda}(v) - C_0\,\chi^2\,T^3(u,v)\,. \tag{9.1}$$

Proof. Expose $A \coloneqq C_u^{\Lambda}$ together with the states of its internal and boundary edges. This determines $X_t^{\Lambda}(u)$: every template counted for a pair in A has its loop-erased path in A, and its blockers are internal edges of A, boundary edges of A or edges leaving Λ. The edges with both endpoints in $\Lambda\setminus A$ remain independent with parameter p. For two terminals in a cluster disjoint from A, the weighted connections in Λ and in $\Lambda\setminus A$ coincide: the open paths are the same, and the only blockers that change status are closed edges to A. Hence, with $\chi_t^{\Lambda\setminus A}(v) = 0$ for $v \in A$,

$$\mathbb{E}_p\bigl[X_t^{\Lambda}(u)\,X_t^{\Lambda}(v);\, C_u^{\Lambda} \ne C_v^{\Lambda}\bigr] = \mathbb{E}_p\bigl[X_t^{\Lambda}(u)\,\chi_t^{\Lambda\setminus A}(v)\bigr]\,.$$

Apply (d) inside the expectation, and use $X_t^{\Lambda}(u) \le |A|$. For adjacent a and z, the Harris inequality gives $pT(v,z) \le T(v,a)$, so $\sum_{z\in\mathrm{N}[A]} T(v,z) \le (1+d/p)\sum_{a\in A} T(v,a)$. By Theorem 4.5(c), summed over a third terminal, $\mathbb{E}_p[|A|\mathbb{1}\{a\in A\}] = \sum_y \mathbb{P}_p(u \xleftrightarrow{\Lambda} a, u \xleftrightarrow{\Lambda} y) \le \chi\,T^2(u,a)$. The error is therefore at most $(1+d/p)\chi^2\sum_a T^2(u,a)T(a,v)$, which proves (9.1) because $1+d/p \le 1+2d/p_c \le C_0$. □

Lemma 9.4 (Two legs)**.** *Let w be a template with base u and end v. On Cor_w, if $x \in C_u^{\Lambda_w}$, $y \in C_v^{\Lambda_w}$ and $C_u^{\Lambda_w} \ne C_v^{\Lambda_w}$, then*

$$N_{xy}^{V} \le c_k + N_{xu}^{\Lambda_w} + N_{vy}^{\Lambda_w}\,.$$

Proof. By Theorem 9.1(i), the templates whose loop-erased paths share an edge with γ_w have total weight at most $|\gamma_w|c_k/k \le c_k$. Let w' be any other template counted for (x,y). Its loop-erased path is open and avoids the edges of γ_w, so it does not visit I_w, since all open edges at I_w belong to γ_w. Being connected, it lies in one of the two exterior clusters. Suppose it lies in $C_u^{\Lambda_w}$. Each of its edges is pivotal for $x \xleftrightarrow{\Lambda_w} u$: otherwise an open path from x to u in Λ_w avoiding it, followed by the corridor and an open path from v to y in Λ_w, would connect x to y without it. Its blockers are closed, hence closed or outside Λ_w. So w' is counted for (x,u) in Λ_w, with the same weight. The case of $C_v^{\Lambda_w}$ is symmetric. □

Proof of Theorem 3.10(c). Write $\mathcal{C}_w(x,y)$ for the event that w is counted for (x,y) in V. By Theorem 9.1,

$$-\frac{\mathrm{d}}{\mathrm{d}t}\chi_t = \sum_y \sum_w c(w)\,\mathbb{E}_p\big[\mathbb{1}_{\mathcal{C}_w(o,y)} W_t^V(o,y)\big]\,.$$

The mass sent from a terminal x to a base u, $m(x,u) \coloneqq \sum_y \sum_{w\colon w_0=u} c(w)\,\mathbb{E}_p[\mathbb{1}_{\mathcal{C}_w(x,y)} W_t^V(x,y)]$, is diagonally invariant and nonnegative. By the mass-transport principle we may move the root from the terminal to the base:

$$-\frac{\mathrm{d}}{\mathrm{d}t}\chi_t = \sum_{w\colon w_0=o} c(w) \sum_{x,y} \mathbb{E}_p\big[\mathbb{1}_{\mathcal{C}_w(x,y)}\, W_t^V(x,y)\big]\,.$$

Keep only the orientation $x \in C_u^{\Lambda_w}$, $y \in C_v^{\Lambda_w}$ of Theorem 9.2. On this event and on Cor_w, Theorem 9.4 gives $W_t^V(x,y) \geq \mathrm{e}^{-tc_k} W_t^{\Lambda_w}(x,u)\, W_t^{\Lambda_w}(v,y)$. By the independence in Theorem 9.2(a) and $c(w)\mathbb{P}_p(\mathrm{Cor}_w) = \mu(w)$, summing over x and y yields

$$-\frac{\mathrm{d}}{\mathrm{d}t}\chi_t \geq \mathrm{e}^{-tc_k}\,\mathbf{E}_k^{\mathrm{rw}}\Big[\mathbb{1}\{|\gamma_w| \geq 3\}\,\mathbb{E}_p\big[X_t^{\Lambda_w}(w_0)\, X_t^{\Lambda_w}(w_k);\, C_{w_0}^{\Lambda_w} \neq C_{w_k}^{\Lambda_w}\big]\Big]\,.$$

Apply Theorem 9.3 with $\Lambda = \Lambda_w$. For the nonnegative error term we may add the raw paths that are not templates, and

$$\mathbf{E}_k^{\mathrm{rw}}\big[T^3(o,w_k)\big] = \sum_v P^k(o,v)\,T^3(v,o) = \mathrm{tr}(P^k T^3) \leq \rho^k\ \mathrm{tr}(T^3) = \rho^k\,\nabla_p\,,$$

by Theorem 4.3(b) for the positive operator T^3 and the self-adjoint operator P^k. This proves (c). No commutation between P and T is needed. □

9.4 Transport along a path

Proof of Theorem 3.10(e). Let π be a self-avoiding path in Λ of length l from b to z, and let $\mathcal{F}$ be the event that all its edges are open. On $\mathcal{F}$, every connection $z \overset{\Lambda}{\longleftrightarrow} y$ gives $b \overset{\Lambda}{\longleftrightarrow} y$. The templates counted for (b,y) whose loop-erased paths use an edge of π have total weight at most lc_k, by Theorem 9.1(i). Every other template counted for (b,y) is counted for (z,y): each of its edges e is pivotal for $z \overset{\Lambda}{\longleftrightarrow} y$, since an open path from z to y avoiding e, preceded by π, would connect b to y without e; its blocker condition does not depend on the terminals. Hence $W_t^\Lambda(b,y) \geq \mathrm{e}^{-tlc_k} W_t^\Lambda(z,y)$ on $\mathcal{F}$. Both $\mathbb{1}_{\mathcal{F}}$ and $W_t^\Lambda(z,y)$ are increasing by Theorem 9.1(iii), so the Harris inequality and $\mathbb{P}_p(\mathcal{F}) = p^l$ give $\mathbb{E}_p W_t^\Lambda(b,y) \geq \mathrm{e}^{-tlc_k} p^l\, \mathbb{E}_p W_t^\Lambda(z,y)$. Summing over y proves (e). □

10 Escapes at both ends of a random path

In this section we prove Theorem 3.11. We assume (A1) and (A2), fix $p \in [p_c/2, p_c)$ and write $T = T_p$. The construction is the one of [Ope26], with one simplification available in the transitive case: the geodesic segments below are chosen among all geodesic segments, rather than within the orbit of a fixed one.

We must find, for a positive fraction of lazy-walk paths w, short paths from both endpoints that avoid the interior of the corridor and end at vertices from which T sees little of the neighbourhood of the corridor. Two observations make this possible. First, after a few steps an independent random walk averages the connection kernel, and this averaging does not depend on $\|T\|$ (Theorem 10.1). Second, if the raw path starts and ends with halves of geodesic segments, an escape can leave along the other half of the segment and is then unlikely to meet the raw path, by the decay of P^l on $\ell^2(V)$. A symmetry of the construction under reversal, obtained from the mass-transport principle, transfers the estimate from one endpoint to the other.

Lemma 10.1 (Smoothing). *For all $a, b \in V$ and $l \geq 0$, $0 \leq (P^l T)(a,b) \leq \rho^l$.*

Proof. Since T is positive semidefinite (Theorem 4.6), the Cauchy–Schwarz inequality for the form $(f,g) \mapsto \langle f, Tg\rangle$ gives $(P^l T)(a,b)^2 = \langle P^l \mathbb{1}_a, T\mathbb{1}_b\rangle^2 \leq (P^l T P^l)(a,a)\, T(b,b)$, and $T(b,b) = 1$. The operator $P^l T P^l$ is invariant and positive semidefinite, so by Theorem 4.3,

$$(P^l T P^l)(a,a) = \operatorname{tr}(P^l T P^l) = \operatorname{tr}(P^{2l} T) \leq \|P^{2l}\| \operatorname{tr}(T) = \rho^{2l}\,,$$

since $\operatorname{tr}(T) = T(o,o) = 1$. □

Geodesic segments. For $J \in \mathbb{N}$ let S_J be the set of geodesic paths $g = (g_{-J}, \dots, g_J)$ in G, that is, of sequences of vertices with $\operatorname{dist}(g_i, g_{i'}) = |i - i'|$. Since G is infinite, connected and locally finite, there are vertices at distance $2J$, so $\mathsf{S}_J \neq \emptyset$; it is invariant under automorphisms. For $x \in V$ and $|i| \leq J$, the number $n_i := \#\{g \in \mathsf{S}_J \colon g_i = x\}$ is finite, because such g lie in $B(x, 2J)$, and positive and independent of x by transitivity. The mass-transport principle applied to $\#\{g \in \mathsf{S}_J \colon g_i = x,\ g_{i'} = y\}$ shows that $n_i = n_{i'}$ for all i, i'; we write $n_\star$ for this number. Consequently, for $|i|, |i'| \leq J$ the *position kernel*

$$\Pi_{i \to i'}(x,y) := \frac{\#\{g \in \mathsf{S}_J \colon g_i = x,\ g_{i'} = y\}}{n_\star}$$

is doubly stochastic, so $\|\Pi_{i \to i'}\| \leq 1$.

The labelled experiment. Let $k > 2J$ and $m \in \mathbb{N}$. A *labelled object* rooted at u is a tuple $\Xi = (g, \pi, g', \zeta, \zeta')$ consisting of $g \in \mathsf{S}_J$ with $g_0 = u$; a lazy-walk path $\pi = (\pi_0, \dots, \pi_{k-2J})$ with $\pi_0 = g_J$; a segment $g' \in \mathsf{S}_J$ with $g'_J = \pi_{k-2J}$; and lazy-walk paths ζ and ζ' of length m starting at g_{-J} and g'_{-J}. Under the probability measure

$$\mathbb{Q}_u(\Xi) := \frac{\mu(\pi)\, \mu(\zeta)\, \mu(\zeta')}{n_\star^2} \tag{10.1}$$

the segment g is uniform, π is a lazy walk from g_J, the segment g' is uniform given its position g'_J, and ζ, ζ' are independent lazy walks. The object determines (see Figure 4):

- the *raw path*, a k-step lazy-walk path with base u and end $v := g'_0$,

$$w(\Xi) := (g_0, g_1, \dots, g_J, \pi_1, \dots, \pi_{k-2J}, g'_{J-1}, \dots, g'_0)\,;$$

- the *escapes* from u and from v, with endpoints $Z := \zeta_m$ and $Z' := \zeta'_m$,

$$\psi := (g_0, g_{-1}, \dots, g_{-J}, \zeta_1, \dots, \zeta_m)\,, \qquad \psi' := (g'_0, g'_{-1}, \dots, g'_{-J}, \zeta'_1, \dots, \zeta'_m)\,.$$

Reversal. The *reversal* of Ξ is $\Xi^\dagger := (g', \overleftarrow{\pi}, g, \zeta', \zeta)$, where $\overleftarrow{\pi}$ is π traversed backwards; it is a labelled object rooted at v, its raw path is $w(\Xi)$ traversed backwards, and its two escapes are those of Ξ in exchanged order. By symmetry of P, $\mu(\overleftarrow{\pi}) = \mu(\pi)$, so $\mathbb{Q}_v(\Xi^\dagger) = \mathbb{Q}_u(\Xi)$. We claim that

$$\mathbb{E}_{\mathbb{Q}_o}[F] = \mathbb{E}_{\mathbb{Q}_o}[F \circ \dagger] \tag{10.2}$$

for every nonnegative function F of labelled objects that is invariant under automorphisms. Indeed, the mass $M_F(x,y) := \sum \mathbb{Q}_x(\Xi) F(\Xi)$, summed over the objects rooted at x whose raw path ends at y, is diagonally invariant. The mass sent out by o is $\mathbb{E}_{\mathbb{Q}_o} F$. Since $\Xi \mapsto \Xi^\dagger$ is a bijection between the objects rooted at x ending at o and the objects rooted at o ending at x, the mass received by o is $\mathbb{E}_{\mathbb{Q}_o}[F \circ \dagger]$. The mass-transport principle gives (10.2).

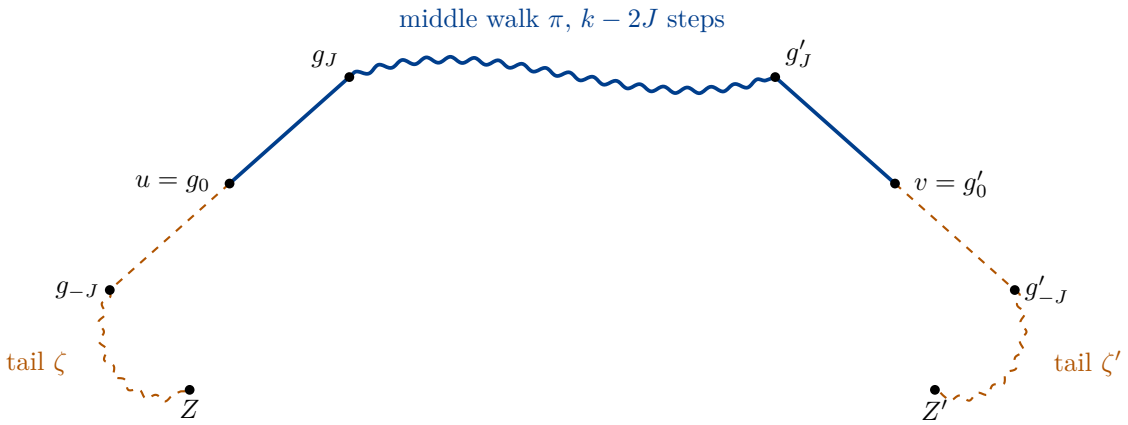


Figure 4: The labelled experiment (schematic). The raw path (solid) consists of the positive half of a geodesic segment g through u, a middle walk π, and the positive half of a geodesic segment g' through v, traversed backwards. The escapes (dashed) follow the negative halves of the same segments and continue as independent walks ζ, ζ' of length m. Reversing the middle walk and exchanging the roles of (g, ζ) and (g', ζ') gives the experiment rooted at v.

Comparison with the lazy walk. For a raw path w with base o, we claim

$$\mathbb{Q}_o(w(\Xi) = w) \le (2d)^{2J} \mu(w)\,. \tag{10.3}$$

Indeed, $w(\Xi) = w$ fixes π, the positive half of g and the reversed positive half of g'. Summing (10.1) over the at most $n_\star$ compatible segments g, the at most $n_\star$ compatible segments g', and all tails gives at most $\mu(\pi)$. The $2J$ steps of w along the two half-segments are steps between distinct neighbours, each of probability $1/(2d)$, so $\mu(w) = (2d)^{-2J}\mu(\pi)$.

Proof of Theorem 3.11. Let $\lambda := \log(1/\rho) > 0$. Throughout, Ξ has law $\mathbb{Q}_o$, so $u = o$.

Stage 1: the escape from u avoids the raw path. We bound the probability that ψ visits a vertex of $w(\Xi)$ other than u; revisits of u are allowed. Condition on g. The deterministic part $(g_0, \dots, g_{-J})$ of ψ meets the first block $(g_0, \dots, g_J)$ of the raw path only in u, because g is a geodesic. We use $P^r(a,b) = \langle \mathbb{1}_a, P^r \mathbb{1}_b \rangle \le \rho^r$ throughout.

- The middle walk visits a given vertex g_{-i}, $0 \le i \le J$, at time r with probability at most ρ^r, and with probability zero for $r < J$.
- The tail ζ visits a given vertex g_i, $0 \le i \le J$, at time l with probability at most ρ^l, and with probability zero for $l < J$.
- Since ζ and π are independent given g, the probability that $\zeta_l = \pi_r$ is

$$\sum_x P^l(g_{-J}, x)\, P^r(g_J, x) = P^{l+r}(g_{-J}, g_J) \le \rho^{l+r}\,,$$

  and it is zero for $l + r < 2J$.

These collisions have total probability at most

$$E_J := 2(J+1) \sum_{r \ge J} \rho^r + \sum_{s \ge 2J} (s+1)\rho^s\,,$$

which tends to zero as $J \to \infty$; the time sums run to infinity, so the bound is uniform in k and m. It remains to consider collisions with the last block $(g'_J, \dots, g'_0)$. Given g, the endpoint of the middle walk has law $\varphi := P^{k-2J}(g_J, \cdot)$ with $\|\varphi\|_2 \le \rho^{k-2J}$, and g'_i has law $\varphi \Pi_{J \to i}$, whose ℓ^2 norm is also at most ρ^{k-2J}. The escape ψ is independent of (π, g') given g.

A deterministic position of ψ is hit by g_i' with probability at most $\|\varphi\Pi_{J\to i}\|_2 \le \rho^{k-2J}$, and the tail position ζ_l, whose law has ℓ^2 norm at most ρ^l, with probability at most ρ^{l+k-2J} by the Cauchy–Schwarz inequality. A union bound over $0 \le i \le J$, the $J+1$ deterministic positions and $l \ge 0$ bounds these collisions by $F_J\rho^{k-2J}$ with $F_J := (J+1)(J+1+(1-\rho)^{-1})$. In the same way, $\mathbb{Q}_o(\mathrm{dist}(u,v)<3) \le |B(o,2)|^{1/2}\rho^{k-2J}$.

Stage 2: the escape from v. The event that ψ' visits a vertex of $w(\Xi)$ other than v is the event that the escape from the root of $\Xi^\dagger$ visits a vertex of the raw path of $\Xi^\dagger$ other than the root. By (10.2) it has the same probability as the event of Stage 1.

Stage 3: choice of J and k_0. Fix J with $2E_J \le 1/8$, and then L_0 with $(2F_J + |B(o,2)|^{1/2})\rho^{L_0} \le 1/8$. For $k - 2J \ge L_0$ and every m, with $\mathbb{Q}_o$-probability at least $3/4$, both escapes avoid every vertex of the raw path other than their starting points, and $\mathrm{dist}(u,v) \ge 3$. The escapes need not avoid each other.

Stage 4: the endpoints of the escapes. Fix C_1 with $\lambda C_1 > 1$, and let $m := \lceil C_1 \log((k+1)R)\rceil$. Given g, π, g', and hence the raw path, the tails are independent lazy walks, and $|\mathrm{N}[I_w]| \le (d+1)(k+1)$. By Theorem 10.1,

$$\mathbb{E}_{\mathbb{Q}_o}\Big[\sum_{b\in\mathrm{N}[I_{w(\Xi)}]} T(Z,b)\;\Big|\; g,\pi,g'\Big] = \sum_{b\in\mathrm{N}[I_{w(\Xi)}]} (P^mT)(g_{-J},b) \le (d+1)(k+1)\rho^m\,,$$

and the same holds for Z'. By Markov's inequality and $\rho^m \le ((k+1)R)^{-\lambda C_1}$, the probability that one of the two sums exceeds $1/(4R)$ is at most $8(d+1)((k+1)R)^{1-\lambda C_1}$. Fix $k_0 \ge 2J + L_0$ such that this is at most $1/4$ for all $k \ge k_0$ and $R \ge 1$.

Stage 5: conclusion. For $k \ge k_0$, with $\mathbb{Q}_o$-probability at least $1/2$ the events of Stages 3 and 4 occur together. Then $|\gamma_w| \ge \mathrm{dist}(u,v) \ge 3$ for $w = w(\Xi)$, and the interior I_w consists of vertices of the raw path other than u and v. So ψ avoids I_w, and erasing its holding steps and loops yields a self-avoiding path in Λ_w from u to Z of length at most $J + m \le j$; similarly for v. Hence $w(\Xi) \in \mathcal{W}$ on this event, and (10.3) gives

$$\mathbf{E}_k^{\mathrm{rw}}[\mathbb{1}\{w \in \mathcal{W}\}] \ge (2d)^{-2J}\,\mathbb{Q}_o(w(\Xi)\in\mathcal{W}) \ge \tfrac{1}{2}(2d)^{-2J} =: c_\star\,.$$

The constants J, k_0, C_1 and $c_\star$ depend only on G. □

11 Consequences: proof of Theorem 2.3

We deduce Theorem 2.3 from Theorem 2.1 and known conditional results. The two parts of Theorem 2.1 play different roles. Boundedness at p_c gives the triangle condition, the classical hypothesis of mean-field theory. The strict inequality $p_c < p_{2\to2}$ is the hypothesis of Hutchcroft's results on near-critical clusters [Hut22] and on exponential decay [Hut20a], which we apply directly. Since G is transitive and simple, no transfer between graphs is needed, in contrast to [Ope26]. Assume (A1).

The triangle condition. By Theorem 2.1, $T := T_{p_c}$ is a bounded operator on $\ell^2(V)$. By Tonelli's theorem the triangle sum equals the diagonal entry of the product, so $\nabla_{p_c} = \langle \mathbb{1}_o, T^3\mathbb{1}_o\rangle \le \|T^3\| \le \|T\|^3 < \infty$, which is (2.2).

Susceptibility and cluster volume. Theorem 2.1 gives $p_c < p_{2\to2}$, so the results of [Hut22], which are stated for connected, locally finite, quasi-transitive graphs with $p_c < p_{2\to2}$, apply to G. By [Hut22, Corollary 1.4] with $k=1$, there is $\delta_0 > 0$ with $c(p_c-p)^{-1} \le \mathbb{E}_p[|C_o|; |C_o| < \infty] \le C(p_c-p)^{-1}$ for $p \in (p_c-\delta_0, p_c)$; below p_c all clusters are finite, so this is the first part of (2.3). By [Hut22, Theorem 1.1] at $p = p_c$, $\mathbb{P}_{p_c}(n \le |C_o| < \infty) \asymp n^{-1/2}$; since critical clusters are finite (Theorem 4.9), this is the first part of (2.4).

Percolation probability. For the upper bound we use [Hut22, Lemma 2.1], whose hypothesis is the triangle condition (2.2). Applied with the subgraph G itself, it gives $\delta_0 > 0$ and C such that the intrinsic radius R_o of C_o satisfies $\mathbb{P}_p(R_o \geq r) \leq C \max\{1/r,\, p - p_c\}$ for all $r \geq 1$ and $p \in [0, p_c + \delta_0)$. Since G is locally finite, C_o is infinite if and only if R_o is infinite; letting $r \to \infty$ gives $\theta(p) \leq C(p - p_c)$ for $p \in (p_c, p_c + \delta_0)$. For the lower bound we use [DCT16, Theorem 1.1], which applies to every locally finite transitive infinite graph. In its parametrisation an edge is open with probability $1 - \mathrm{e}^{-b}$ for a parameter $b > 0$, and it gives $\theta \geq (b - b_c)/b$ for $b > b_c$, where b_c is the critical value of b. With $b = -\log(1-p)$ and $b_c = -\log(1-p_c)$ we have $b - b_c = \log(1 + (p - p_c)/(1-p)) \geq (p - p_c)/(1 - p_c)$, and b is bounded for $p \leq (1 + p_c)/2$. Hence $\theta(p) \geq c(p - p_c)$ near p_c, which completes the proof of (2.3).

Intrinsic and extrinsic radius. By [Hut22, Theorem 1.2] at $p = p_c$, both $\mathbb{P}_{p_c}(n \leq \mathrm{rad}_{\mathrm{int}}(C_o) < \infty)$ and $\mathbb{P}_{p_c}(n \leq \mathrm{rad}_{\mathrm{ext}}(C_o) < \infty)$ are of order n^{-1}. Since critical clusters are finite, this is the second part of (2.4). The extrinsic bound is also [Hut20a, Theorem 3.1].

Exponential decay. By [Hut20a, Theorem 2.2], $p_{2\to 2} \leq p_{\mathrm{exp}}$ on every connected, locally finite graph, and together with Theorem 2.1 this gives $p_c < p_{2\to 2} \leq p_{\mathrm{exp}}$. More precisely, [Hut20a, Remark 2.3] gives, for $0 < p < p_{2\to 2}$,

$$T_p(x,y) \leq 2\|T_p\| \exp\Bigl(-\frac{\mathrm{dist}(x,y)}{\mathrm{e}\|T_p\|}\Bigr) \qquad \text{for all } x, y \in V\,,$$

where $\|T_p\| < \infty$ by Theorem 4.1(c). For $p = 0$ the off-diagonal connection probabilities vanish. This proves (2.5).

A Proofs of two auxiliary results

A.1 Russo's formula in operator norm

Proof of Theorem 4.7. Finite domains. Let Λ be a finite domain. For an increasing event depending on finitely many edges, Russo's formula [Gri99, Section 2.4] expresses the derivative in p as the expected number of pivotal edges, and since pivotality of an edge does not depend on its state, $(1-p)\,\mathbb{P}_p(e \text{ is pivotal}) = \mathbb{P}_p(e \text{ is closed and pivotal})$. An edge $e = \{a, b\}$ is closed and pivotal for $x \overset{\Lambda}{\longleftrightarrow} y$ exactly when, for one of its two orientations, $x \in C_a^\Lambda$, $y \in C_b^\Lambda$ and $C_a^\Lambda \neq C_b^\Lambda$. Hence

$$(1-p)\frac{\mathrm{d}}{\mathrm{d}p} T_p^\Lambda(x,y) = \sum_a \sum_{b \sim a} \mathbb{P}_p\bigl(x \in C_a^\Lambda,\ y \in C_b^\Lambda,\ C_a^\Lambda \neq C_b^\Lambda\bigr)\,, \tag{A.1}$$

where both a and b range over Λ. The right-hand side equals $\mathbb{E}_p[\mathrm{Adj}(C_x^\Lambda, C_y^\Lambda)\mathbb{1}\{C_x^\Lambda \neq C_y^\Lambda\}]$.

Passage to G. Fix $\bar p < p_c$. For $p \leq \bar p$, each summand in (A.1) is at most $\mathbb{P}_p(\{x \leftrightarrow a\} \circ \{b \leftrightarrow y\}) \leq T_{\bar p}(x,a) T_{\bar p}(b,y)$ by the van den Berg–Kesten inequality. The bound is summable over (a,b), with sum $(T_{\bar p}\,\mathrm{Adj}\,T_{\bar p})(x,y)$, and also over y, with sum $d\chi(\bar p)^2$. Let Λ increase to V through finite domains. Below p_c all clusters are almost surely finite, so for almost every configuration the clusters C_a^Λ, C_b^Λ eventually coincide with C_a, C_b, and the events in (A.1) converge to their analogues on G. Integrating (A.1) in p and using dominated convergence, we obtain for $0 < p' < p \leq \bar p$

$$T_p(x,y) - T_{p'}(x,y) = \int_{p'}^{p} D_{\tilde p}(x,y)\,\mathrm{d}\tilde p\,, \qquad D_{\tilde p}(x,y) \coloneqq \frac{1}{1-\tilde p}\mathbb{E}_{\tilde p}[\mathrm{Adj}(C_x, C_y)\mathbb{1}\{C_x \neq C_y\}]\,, \tag{A.2}$$

with $0 \leq D_{\tilde p} \leq \bar D \coloneqq (1-\bar p)^{-1} T_{\bar p}\,\mathrm{Adj}\,T_{\bar p}$ entrywise.

Operator norm. The kernels $D_{\tilde{p}}$ and $\bar{D}$ are invariant and symmetric, and the row $\bar{D}(o,\cdot)$ is summable. By (A.2), every coordinate of the row $T_{\tilde{p}}(o,\cdot)$ is absolutely continuous with derivative $D_{\tilde{p}}(o,\cdot)$ almost everywhere. Let F_n be finite sets increasing to V with $\sum_{y\notin F_n}\bar{D}(o,y)\le 1/n$. For almost every $\tilde{p}$, all coordinates are differentiable at $\tilde{p}$ with the stated derivative, since V is countable, and then

$$\limsup_{h\to 0}\sum_y\Big|\frac{T_{\tilde{p}+h}(o,y)-T_{\tilde{p}}(o,y)}{h}-D_{\tilde{p}}(o,y)\Big|\le\limsup_{h\to 0}\sum_{y\in F_n}\Big|\cdots\Big|+2\sum_{y\notin F_n}\bar{D}(o,y)\le\frac{2}{n}$$

for every n. So the row is differentiable in $\ell^1(V)$ almost everywhere, and absolutely continuous in $\ell^1(V)$, since $\|T_p(o,\cdot)-T_{p'}(o,\cdot)\|_1\le\int_{p'}^{p}\|D_{\tilde{p}}(o,\cdot)\|_1\,\mathrm{d}\tilde{p}$. For an invariant symmetric kernel, all row and column sums of its absolute value equal the ℓ^1 norm of the row of o, so by the Schur test the operator norm is at most this ℓ^1 norm. Applied to the difference quotients and to the increments, this gives local absolute continuity in operator norm and (4.1).

The susceptibility. Summing (A.2) over y with Tonelli's theorem shows that χ is locally absolutely continuous with $(1-p)\chi'(p)=\sum_y\mathbb{E}_p[\mathrm{Adj}(C_o,C_y)\mathbb{1}\{C_o\neq C_y\}]$ almost everywhere. Grouping the vertices y by their clusters, the right-hand side equals $\mathbb{E}_p\sum_{a\in C_o}\mathsf{m}(a)$, where $\mathsf{m}(a):=\sum_{b\sim a}|C_b|\mathbb{1}\{C_b\neq C_a\}$. Let each vertex x send the mass $\mathsf{m}(a)$ to every $a\in C_x$; by the mass-transport principle, $\mathbb{E}_p\sum_{a\in C_o}\mathsf{m}(a)=\mathbb{E}_p[|C_o|\,\mathsf{m}(o)]$, which is (4.2). □

A.2 The decision-tree covariance inequality

Theorem 7.3 is a form of [OSSS05, Section 3.3, Equation (6)]; there the algorithm computes the bounded function, as here. We give the proof for completeness, for random variables taking finitely many values, which is the case needed in Section 7.3.

Proof of Theorem 7.3. Let Y be an independent copy of X, independent of the randomness of the algorithm. Run the algorithm on X, let $i_1,\dots,i_{\mathsf{T}}$ be the queried coordinates in the order of the queries, and let Z^t be X with the coordinates $i_1,\dots,i_t$ replaced by those of Y, so that $Z^0=X$. The *transcript* consists of the randomness of the algorithm, the queried indices and the queried values of X.

Given the transcript up to some time, the coordinates of X that have not been queried keep their original independent laws, because the decision to query a coordinate depends only on the transcript so far. The coordinates of Y are independent of everything else. Hence, given the full transcript, Z^{T} has the law of X. Since the transcript determines $g(X)$,

$$\mathrm{Cov}(f(X),g(X))=\mathbb{E}\big[(f(X)-f(Z^{\mathsf{T}}))\,g(X)\big]\,.$$

Write $f(X)-f(Z^{\mathsf{T}})$ as the telescoping sum of $f(Z^{t-1})-f(Z^t)$ and use $0\le g\le 1$: $|\mathrm{Cov}(f(X),g(X))|\le\mathbb{E}\sum_{t\le\mathsf{T}}|f(Z^{t-1})-f(Z^t)|$. Given the transcript before the t-th query, and given that this query concerns coordinate i, the pair (Z^{t-1},Z^t) has the law of $(X,X^{(i)})$ by the same reasoning. Since every coordinate is queried at most once, taking expectations gives $\mathbb{E}\sum_{t\le\mathsf{T}}|f(Z^{t-1})-f(Z^t)|=\sum_i\mathrm{rev}_i\,\mathbb{E}|f(X)-f(X^{(i)})|$. □

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